How to Read Piping & Instrumentation Diagram for Beginners

Intended for Engineers who are new in plant or facilities and final year engineering and OHS students.

By Cahyo Hardo, B.Eng., M.OHS.

Bahasa version had been shared in Indonesian Association of Oil and Gas Safety and Engineering Experts forum

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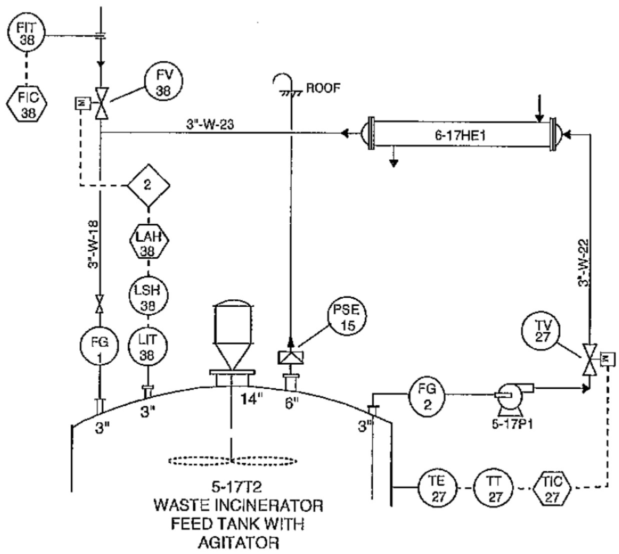
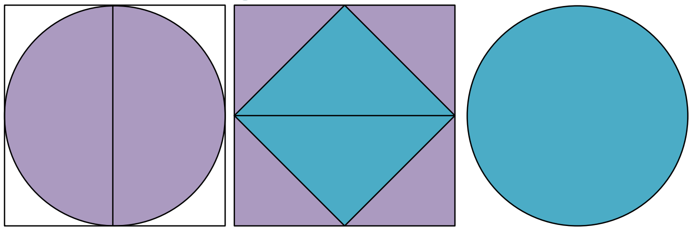


Indeed, with hardship there is ease..
(QS. Al Insyirah : 6)

How to Read Piping & Instrumentation Diagram for Beginners



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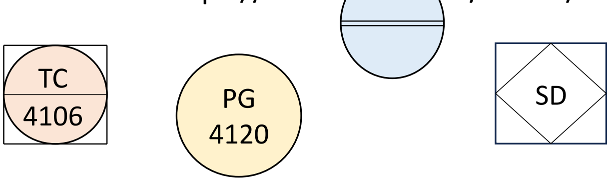


Introduction

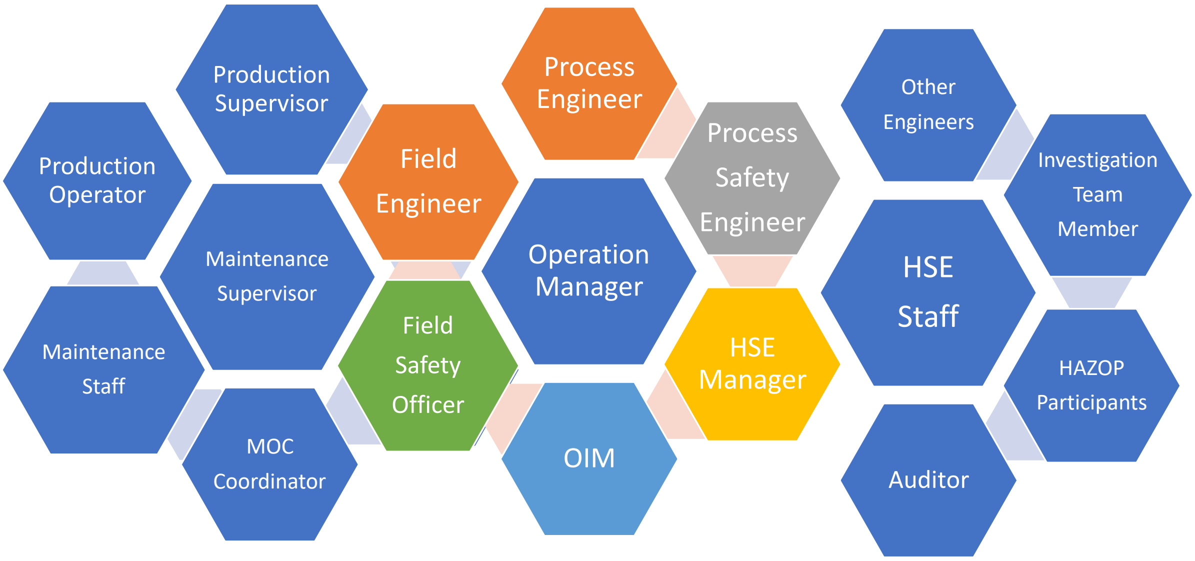
- For the upstream oil and gas industry.
- Primary target audience: Final-year engineering and OHS students, and engineers new to Plant/facilities
- Uses terminology from Chemical Engineering and Process Engineering.
- Challenge: Explaining P&IDs in a relatively short time.

Stages (ideal)

1. Understand Process Flow Diagrams.
2. Understand how common unit operations in a plant or production facility work.U
3. nderstand safety aspects in P&IDs.
4. Ready to read P&IDs properly.

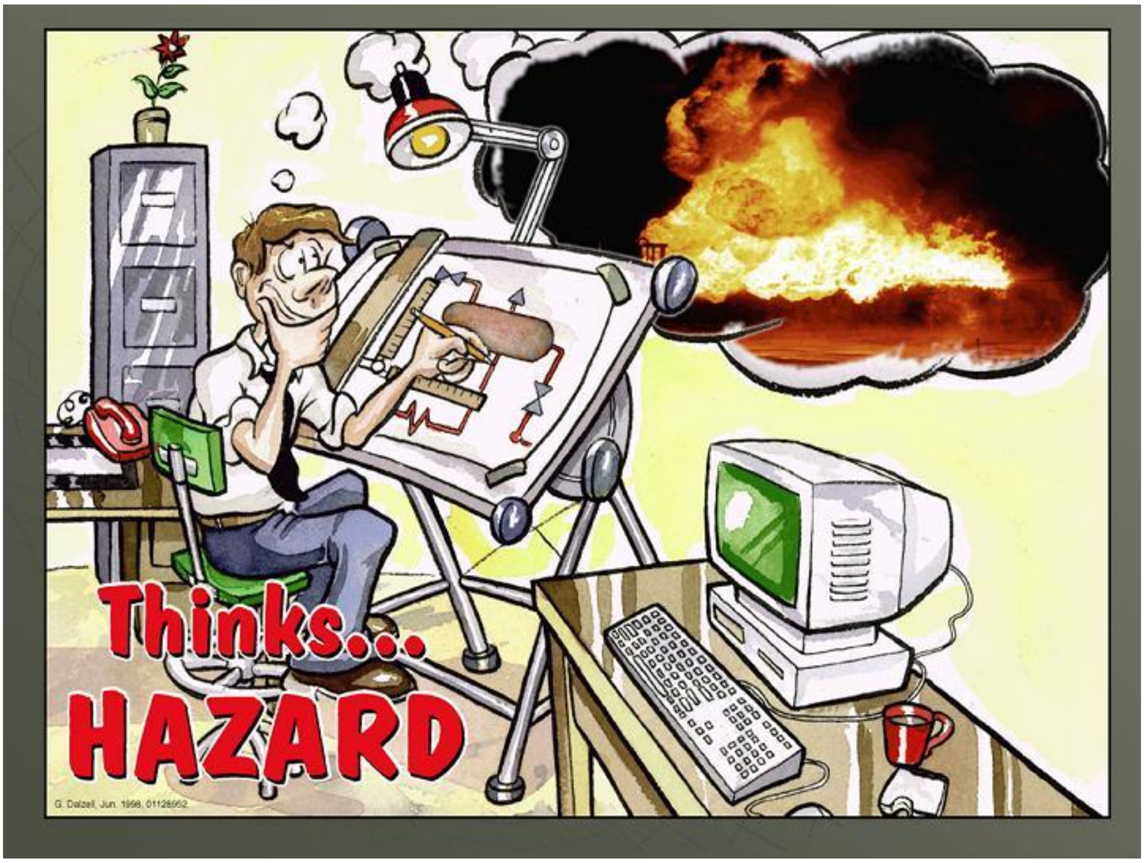
Essential supplies

Being able to read P&IDs is a good foundation for:

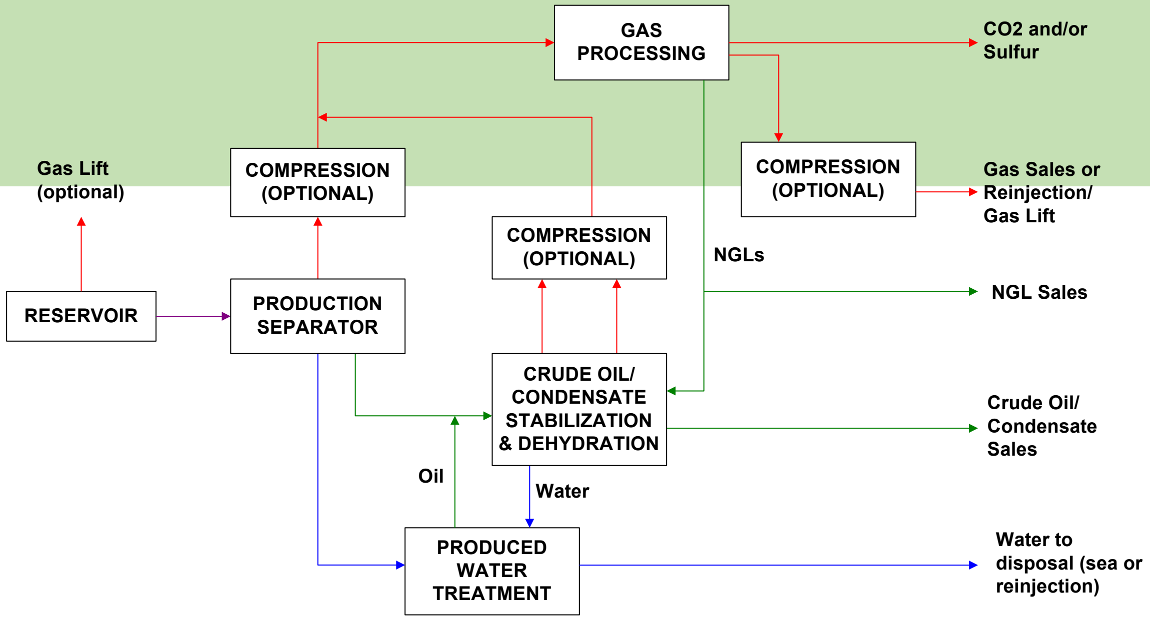


Uses of P&ID

1. Explained plant operating manual
2. Plant protection design guidelines.
3. Tools for troubleshooting operation.
4. Associate document in permit to work.
5. Reference in SOP or SOP development.
6. Supporting tools for new operator training.
7. Reference for audit.
8. Tools for engineer for plant optimisation.
9. One of main documents in Facility MOC.
10. HAZOP main document.
11. Supporting tools for investigation.
12. Supporting for emergency situation.
13. Part of documents in technical discussion.
14. One of interview tools.


Thinks...
HAZARD
G. Dalzell, Jun. 1998, 01128662

Process Flow Diagram



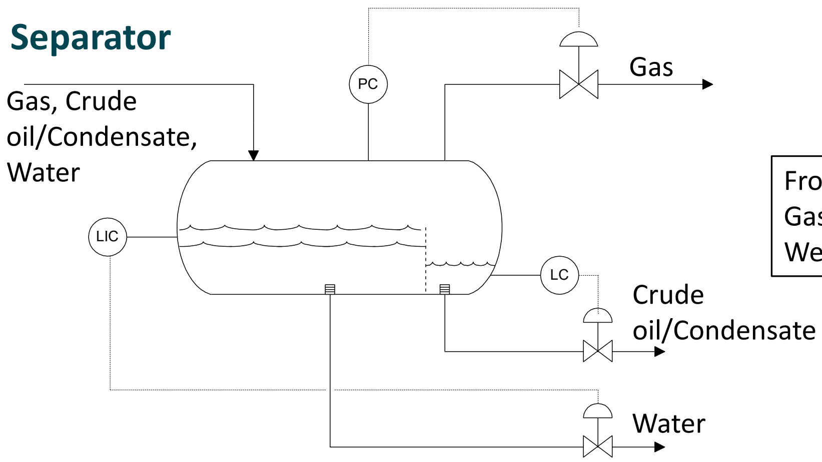
Schematic View of a Typical Integrated Surface Production Facility

How Equipment Works

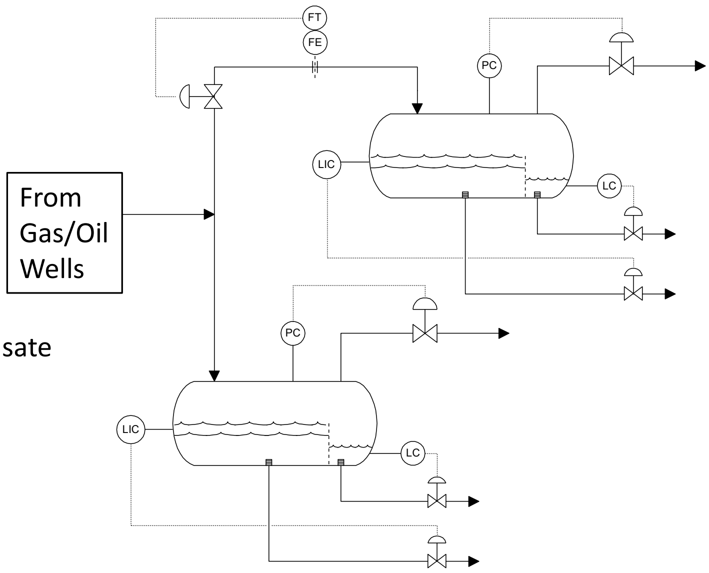
Separator

- Used to separate gas, oil/condensate, and water by exploiting their density differences.
- Inside the separator:
  - Gas will always be at the top and exit at the top of the separator.
  - The oil/condensate layer will always be above the water layer.
  - The oil/condensate will be discharged from the separator at the end or middle of the separator, depending on the liquid level control method.
- To ensure proper separation, the following must be observed:
  - The pressure inside the separator must be maintained using a pressure control valve (which maintains the gas outlet pressure) or indirectly controlled by a suction compressor.
  - The liquid level must be maintained using a level control valve.

Separator

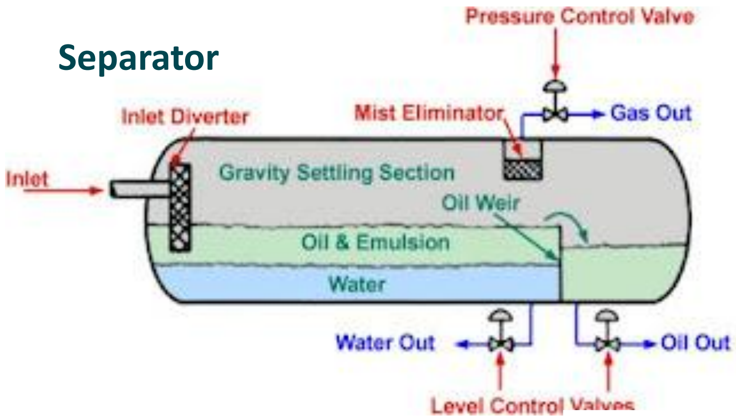


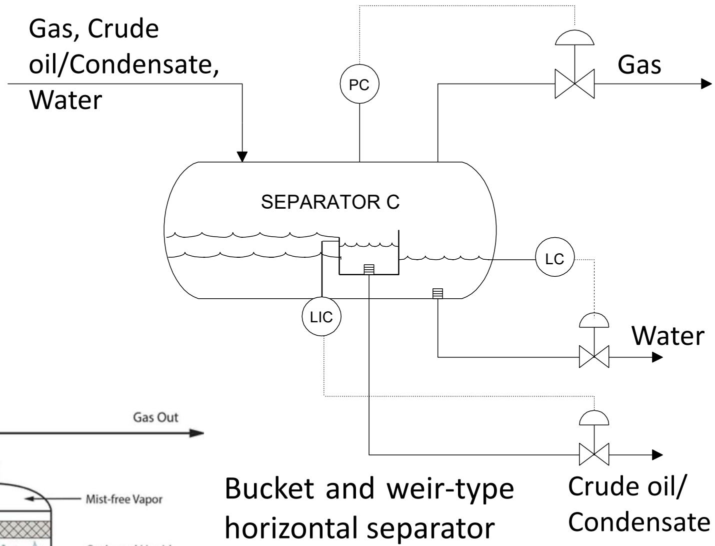
Separator Unit with its control system



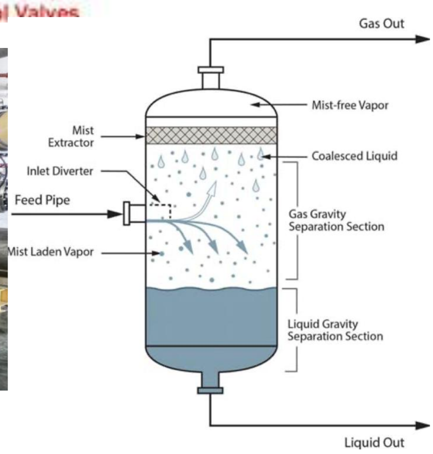
Separator operation with flow control due to capacity limitations

Separator



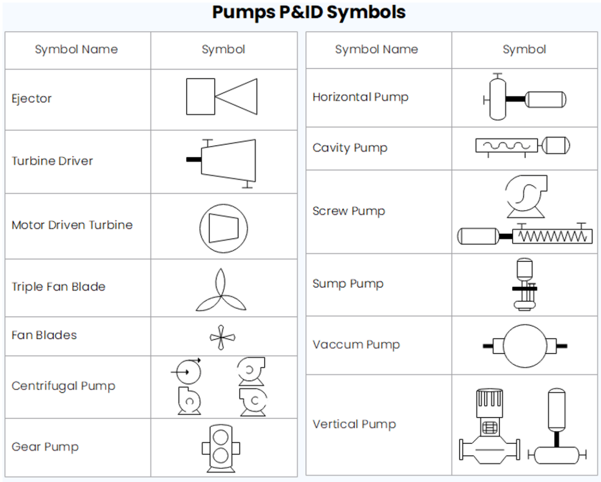


Bucket and weir-type horizontal separator



How Equipment Works

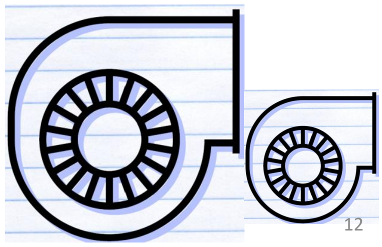
Centrifugal Pump



Pumps P&ID Symbols

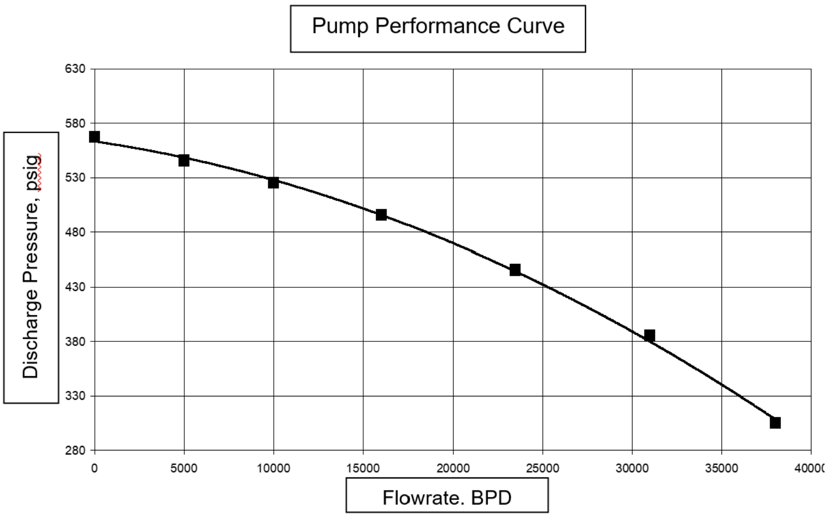
Symbol Name	Symbol	Symbol Name	Symbol
Ejector		Horizontal Pump	
Turbine Driver		Cavity Pump	
Motor Driven Turbine		Screw Pump	
Triple Fan Blade		Sump Pump	
Fan Blades		Vaccum Pump	
Centrifugal Pump		Vertical Pump	
Gear Pump			

- A device for transporting liquid fluids (water, oil, condensate)
- The flow rate follows the pump's performance curve—a parabolic curve.
- The lower the pressure at the pump outlet, the greater the pump flow rate.
- To increase capacity, additional pumps are used, operated in parallel or in series.
- To increase discharge pressure, additional pumps are used in series.
- Requires:
  - Sufficient inlet pressure to prevent cavitation (NPSH Available vs. NPSH Requirement).
  - Relatively clean fluid.
  - Minimum flow rate as specified by the manufacturer.
  - Protection against high pump discharge pressure.
- Process Control Objectives:
  - Minimum flow control (avoid cavitation and overheating).
  - Control of pump output pressure or flow.

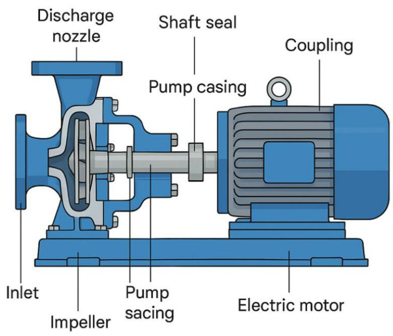


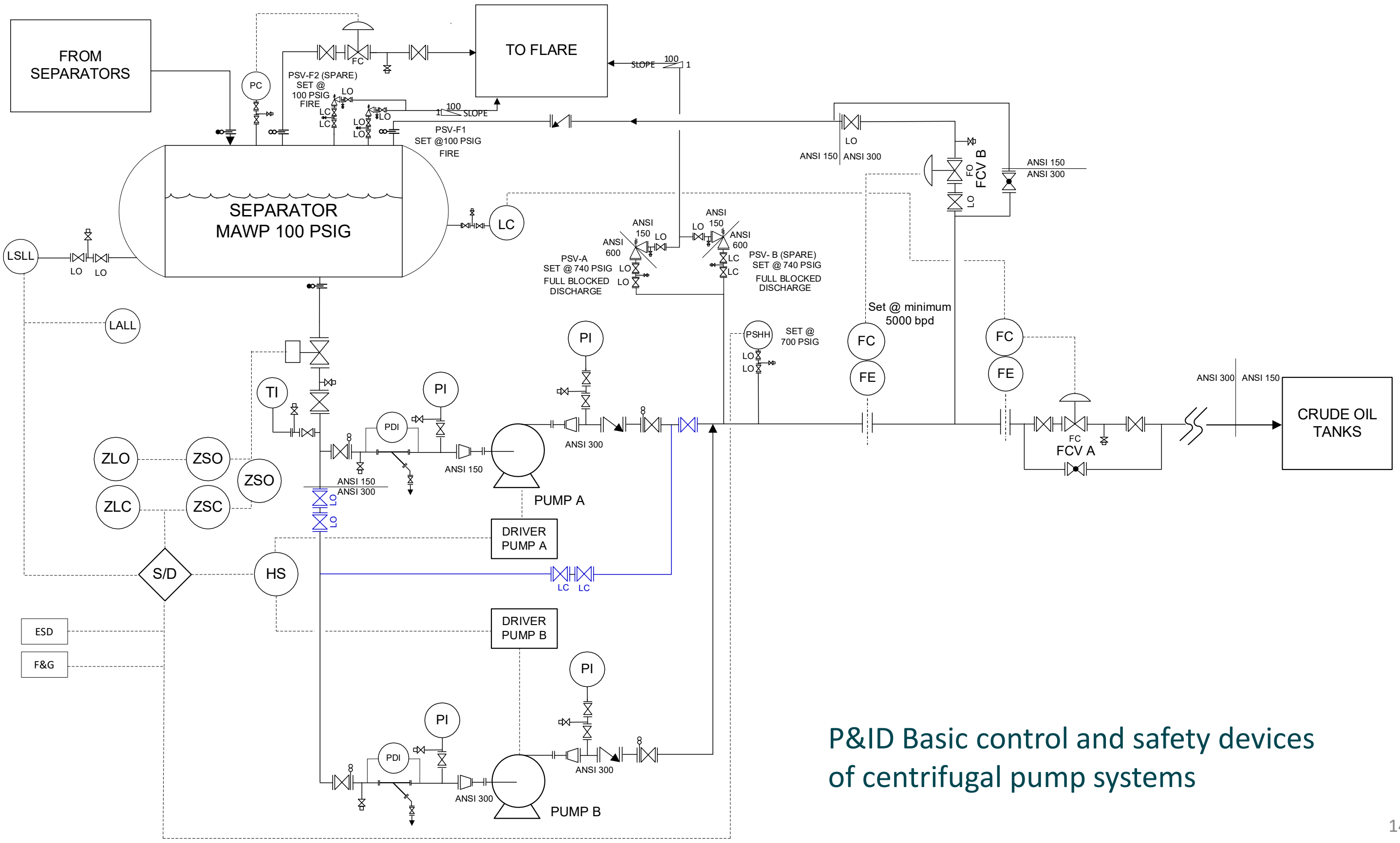
Centrifugal Pump

- The pump discharge pressure is determined by the system downstream of the pump (PCV, pressure in the piping).
- The flow rate follows the discharge pressure.



Pump performance curve - discharge pressure vs flow

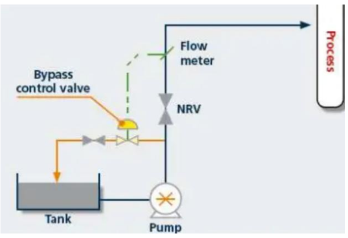


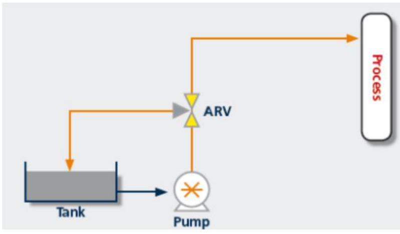


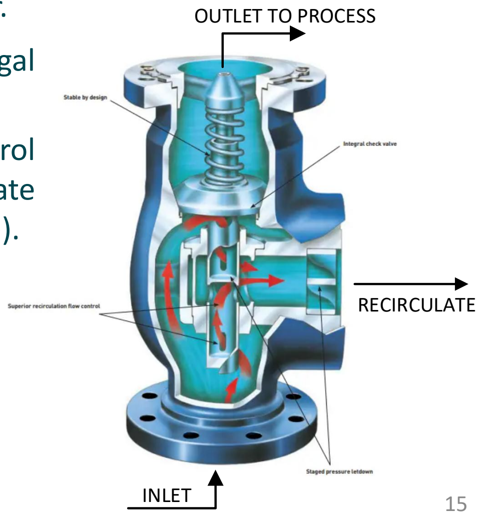
P&ID Basic control and safety devices of centrifugal pump systems

Centrifugal Pump – special duty

- It doesn't shut down even if the power goes out. It has an independent power supply!
- It doesn't shut down even if the ESD or fire signal is active.
- It rarely operates, and will continue operating in the event of a fire, even if the diesel tank level is low, even if the vibration alarm has been sounding from the start.
- The pump is a fire water pump – its job is to supply firefighting water.
- Most are centrifugal-type, so their design follows that of a centrifugal pump.
- Generally, minimum recirculation flow installations do not use control valves and orifices, but instead utilize mechanical principles to regulate flow with a special valve called an ARV (automatic recirculation valve).





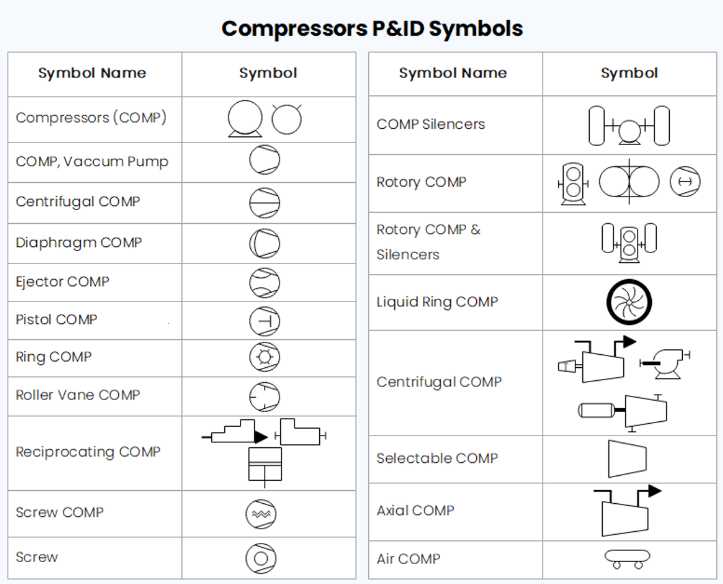


Pompa Sentrifugal – special duty

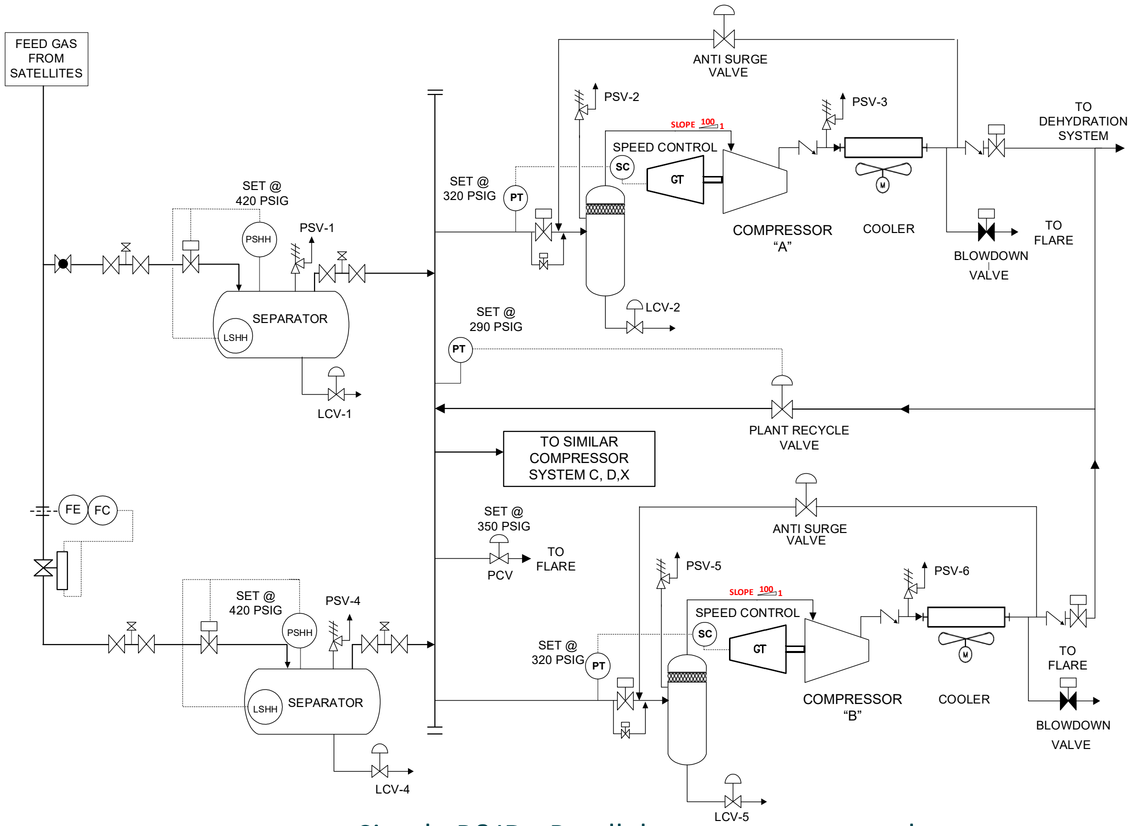
How Equipment Works

Centrifugal Compressor

- A gas fluid transport device.
- Its performance follows a parabolic compressor curve.
- Specificity:
    - Requires a minimum flow rate to prevent ‘surges’.
    - Compressor output pressure must not be too low to prevent ‘stalls’.
    - Requires overpressure protection from high compressor discharge pressures.
- Process Control:
    - Control to avoid surges.
    - Capacity control.
    - Butuh overpressure protection dari tekanan discharge kompresor yang tinggi


Compressors P&ID Symbols

Symbol Name	Symbol	Symbol Name	Symbol
Compressors (COMP)		COMP Silencers	
COMP, Vaccum Pump		Rotory COMP	
Centrifugal COMP		Rotory COMP & Silencers	
Diaphragm COMP		Liquid Ring COMP	
Ejector COMP		Centrifugal COMP	
Pistol COMP		Selectable COMP	
Ring COMP		Axial COMP	
Roller Vane COMP		Air COMP	
Reciprocating COMP			
Screw COMP			
Screw			



Simple P&ID - Parallel compressor control system

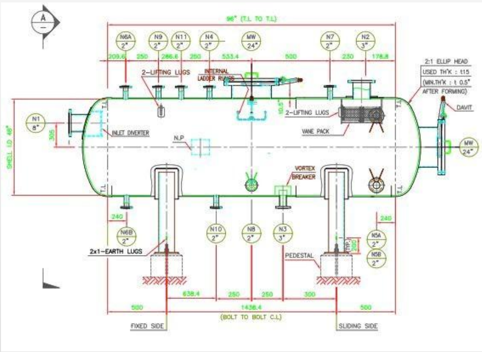
Barrel-type centrifugal compressor

Horizontal-split-type centrifugal compressor

Safety in P&ID

- Maximum Allowable working pressure (MAWP) of pressure vessel, piping, (ANSI rating, API rating)
- Specification Break
- Pressure Safety Valve
- Safety Devices
- Overpressure protection:
  - Equipment (e.g., separators, pumps, compressors)
  - By-pass control valves, flow rate controllers (using RO, limited pipe diameter)
  - Fail-safe conditions (whether an actuated valve should be open, closed, or otherwise)
  - Safe fluid disposal systems (flares, burn pits).

Pressure Vessel

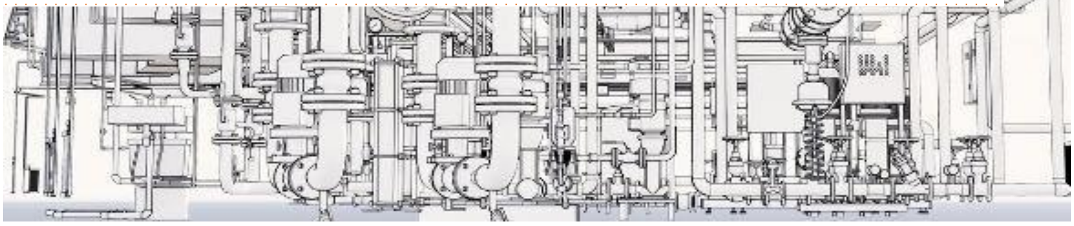


- The P&ID displays the operating unit's Maximum Allowable Working Pressure (MAWP), including its piping and fittings. So, how do you determine whether a system with varying MAWPs can operate safely?
- **The design pressure** of a pressure vessel is also known as its MAWP. In many cases, the MAWP determines the setting pressure of the PSV.
- The operating pressure of a pressure vessel under normal operating conditions is called the **operating pressure**, which is < MAWP. The operating pressure is determined by the process conditions.

Operating Pressure	Minimum difference between operating pressure and MAWP
Less than 50 psig	10 psi
51 s/d 250 psig	25 psi
251 s/d 500 psig	10% of maximum operating pressure
501 s/d 1000 psig	50 psi
> = 1001 psig	5% of maximum operating pressure
Vessels equipped with a pressure switch shutdown must add 5% or 5 psi to the minimum differential between the operating pressure and the MAWP, whichever is greater.	

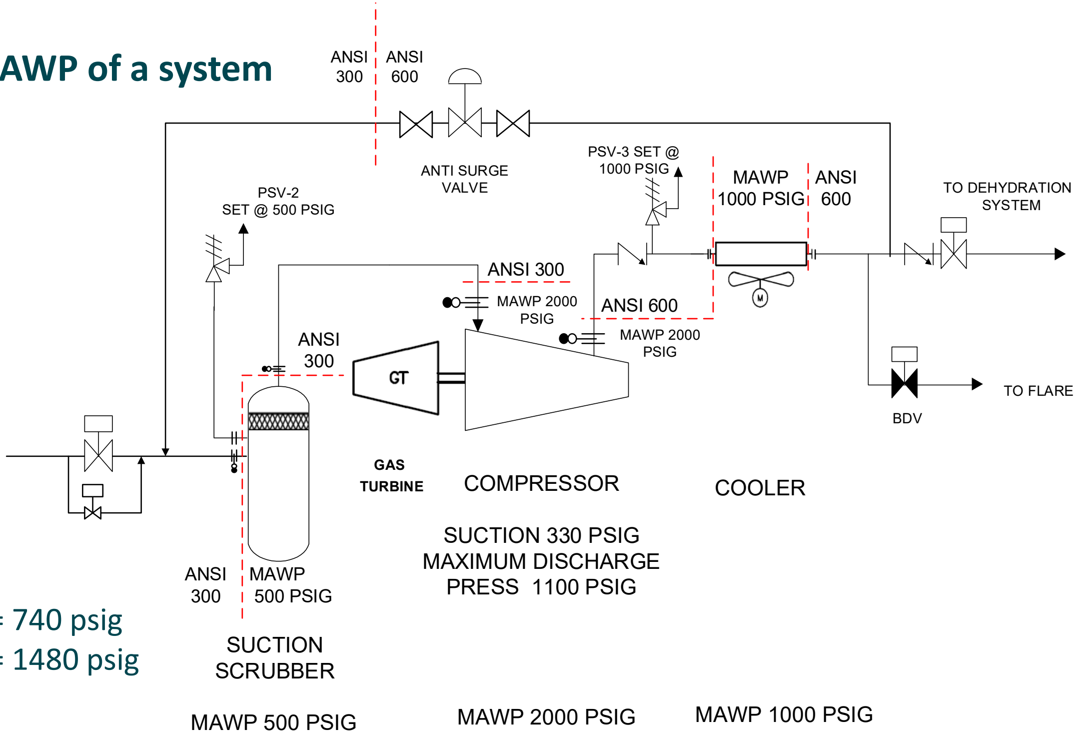
Piping System

- The MAWP in a piping system is determined by the weakest part of the pipe, the connection.
- The MAWP of a flange connection determines the overall MAWP of the piping system.
- Two pipes made of identical material and with the same diameter will have the same MAWP.
- However, because each flange is different, for example, one is ANSI 150 and the other is ANSI 300, the MAWP will depend on the MAWP of those flanges.
- If the piping system is connected to a pressure vessel/process, the overall system MAWP is determined by the MAWP of the weakest piece of equipment/piping in the system.



Note: ANSI is the abbreviation for the American National Standard Institute, an institution from the United States that issues various standards.

Determining the MAWP of a system

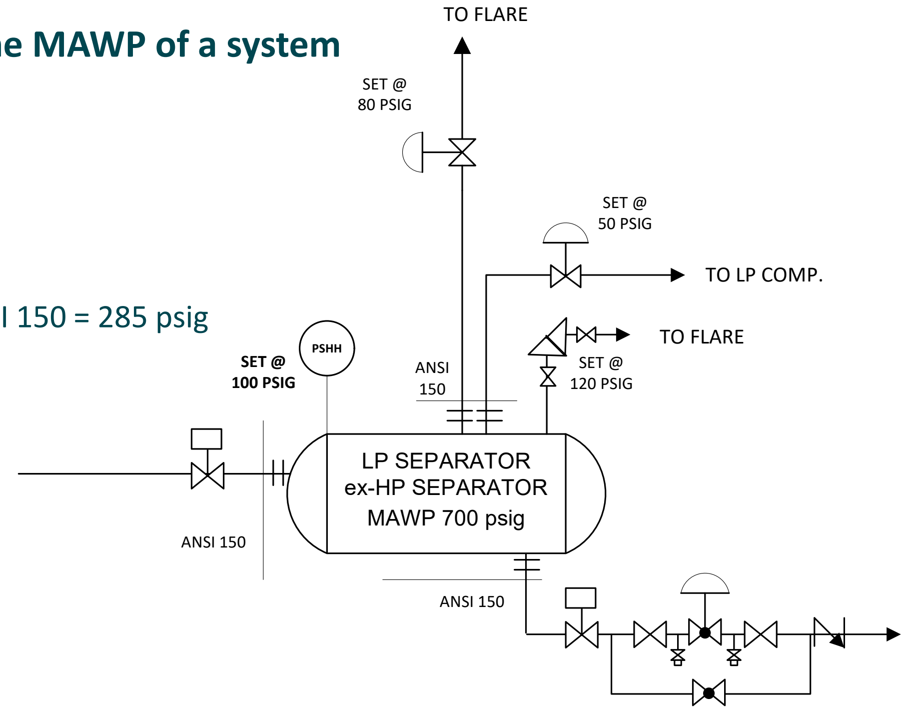


Asumption:
MAWP piping ANSI 300 = 740 psig
MAWP piping ANSI 600 = 1480 psig

From the P&ID above, it can be seen that what determines the weakest MAWP is the MAWP of the suction scrubber (upstream) and aftercooler (downstream) - **and not** the piping system.

Determining the MAWP of a system

Assumption:
MAWP piping ANSI 150 = 285 psig



From the P&ID above, it can be seen that what determines the weakest MAWP is the MAWP of the piping system, and not the separator.

Piping Pressure Rating

ANSI flange rating for material 1.1

MAXIMUM ALLOWABLE NON-SHOCK WORKING PRESSURE MATERIAL GROUP 1.1							
Temp, ^{0}F	MAWP						
	150	300	400	600	900	1500	2500
-20 to 100	285	740	990	1480	2220	3705	6170
200	260	675	900	1350	2025	3375	5625
300	230	655	875	1315	1970	3280	5470
400		635	745	1270	1900	3170	5280
500		600	800	1200		2995	4990
600		550	730	1095			4560
650		535	715	1075			
700			710	1065			
750				1010			
800				825			

Example of Material (Spec-Grade)	
Material Group	**Materials (Spec-Grade), example**
1.1	A105, A181-II, A-216-WCB, A515-70, A-516-70, A-350-LF2
1.2	A216-WCC, A-350-LF3, A203-B
2.2	A-240-317, A-351-CF8M, A182-F316
3.1	B 462, B 463
etc	Etc.

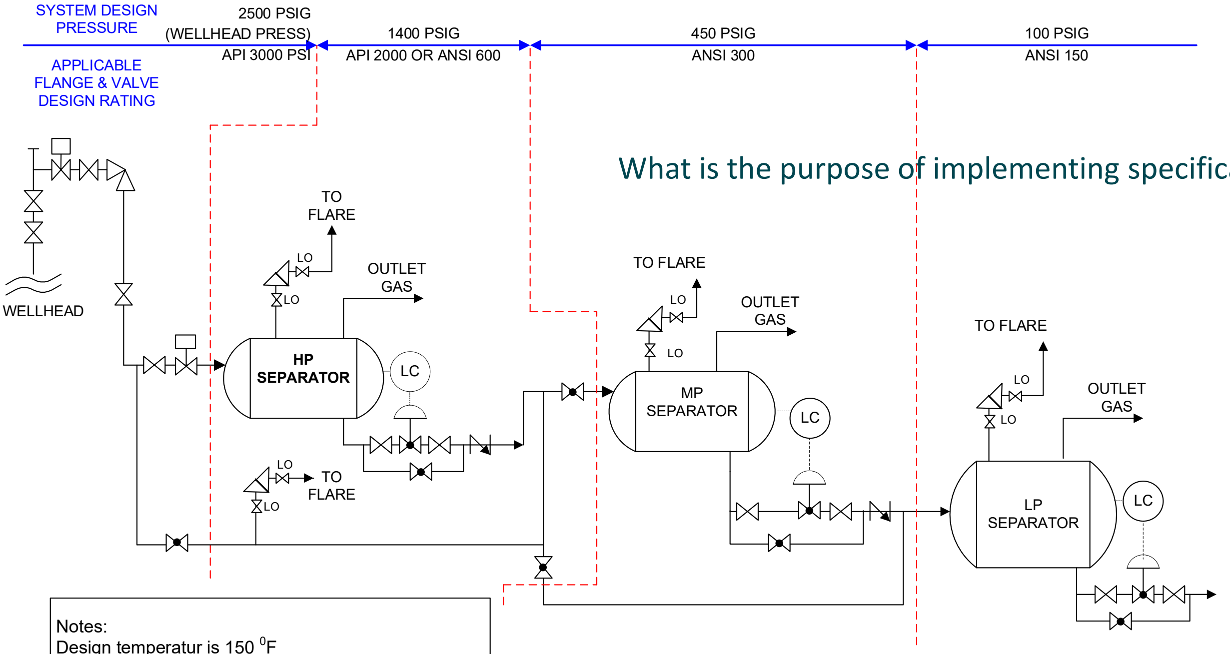
API Flange ratings

PRESSURE – TEMPERATURE RATINGS									
	Temperature , ^{0}F								
	0 – 250	300	350	400	450	500	550	600	650
MAWP	2000	1995	1905	1860	1810	1735	1635	1540	1430
	3000	2930	2860	2785	2715	2505			
	5000	4880	4765	4645					

API and ANSI flange comparison

API Flange	ANSI Flange
Generally used for higher operating pressure	Faster available and cheaper
Used for X'mas tree and surround flowline	Sometimes used for manifold
Sometimes used for manifold	Generally used in production facilities

Specification Break

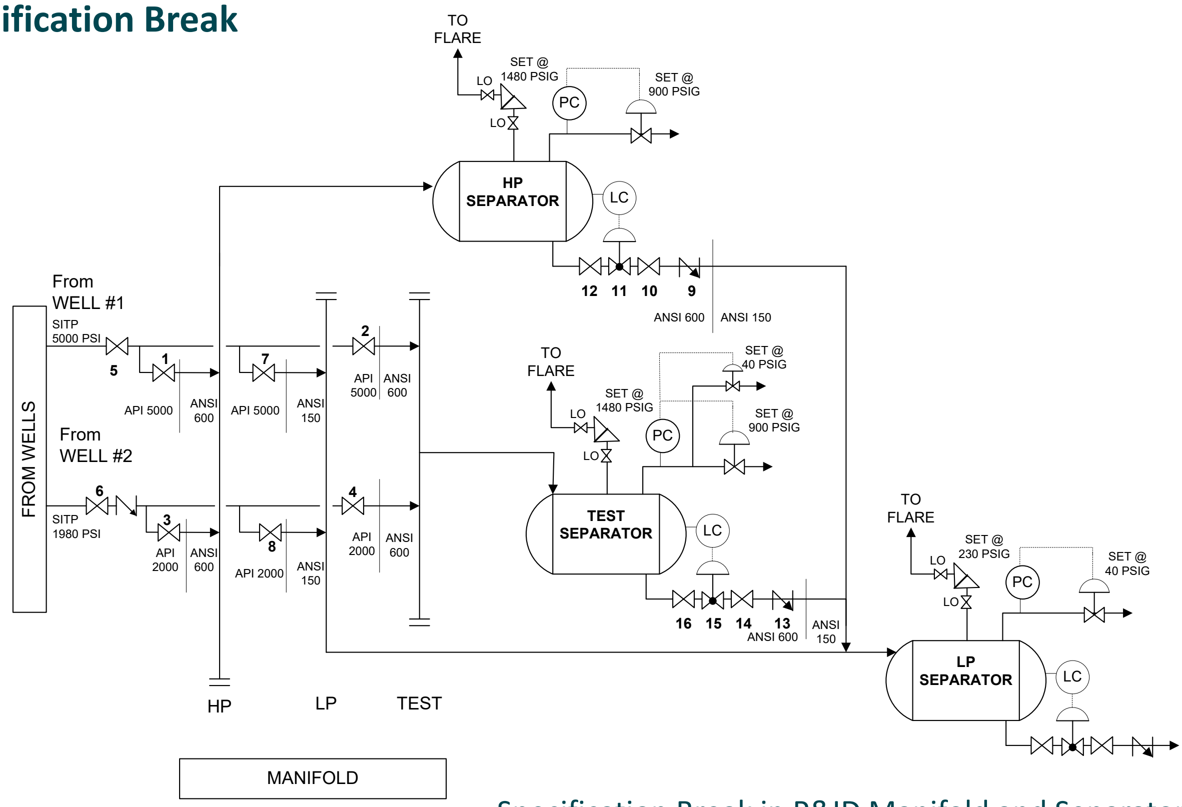


What is the purpose of implementing specification breaks?

Notes:
Design temperatur is 150 ^{0}F
Shutdown system not shown
Flowline & manifold not drawn, assumed its rating design follows wellhead
MAWP of each system is based on the weakest component or equipment
The maximum PSV setting point is MAWP

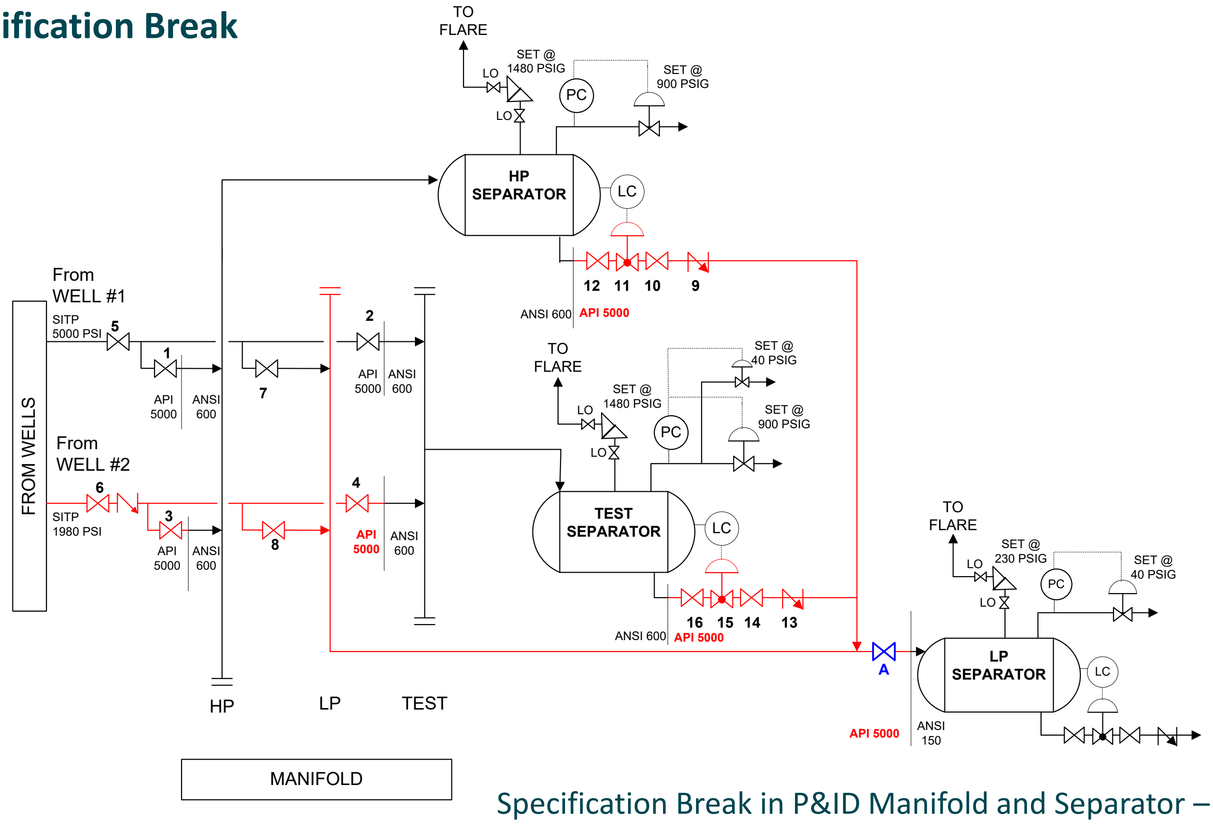
Flange and Valve Pressure Rating Changes

Specification Break



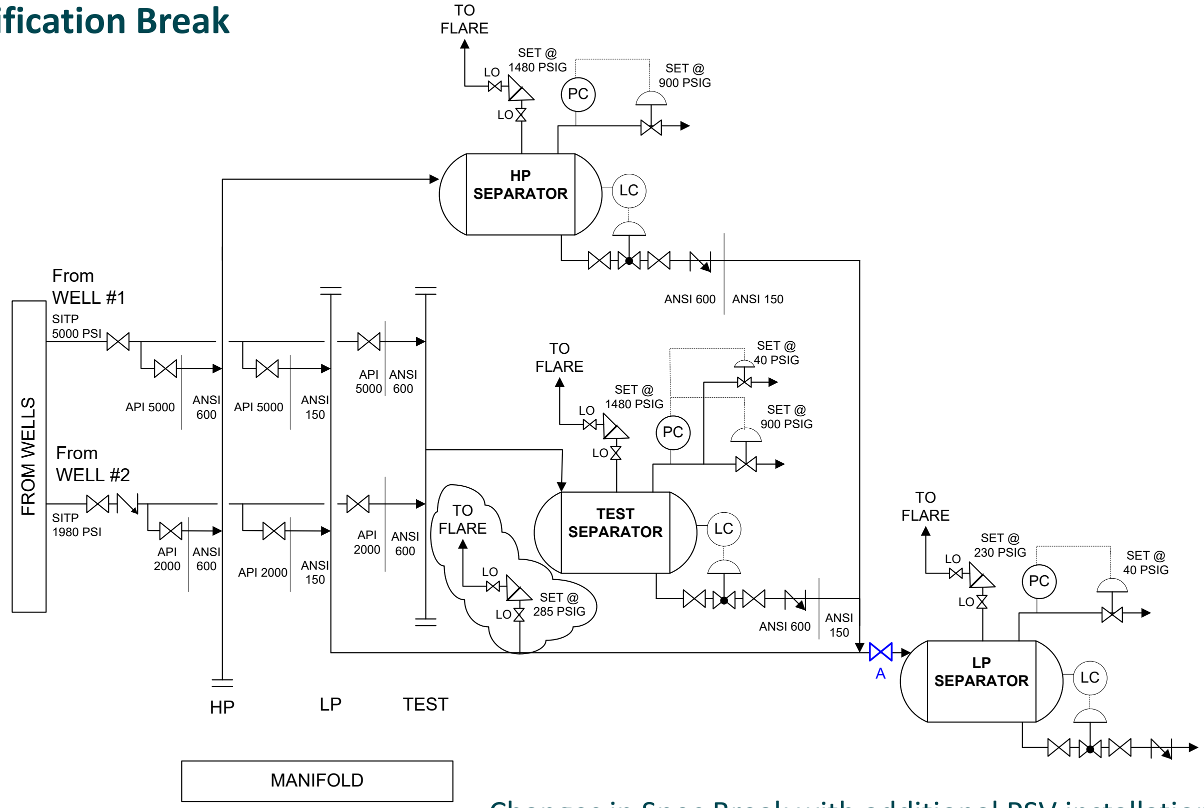
Specification Break in P&ID Manifold and Separator

Specification Break



Specification Break in P&ID Manifold and Separator – Additional valve at upstream of LP Separator

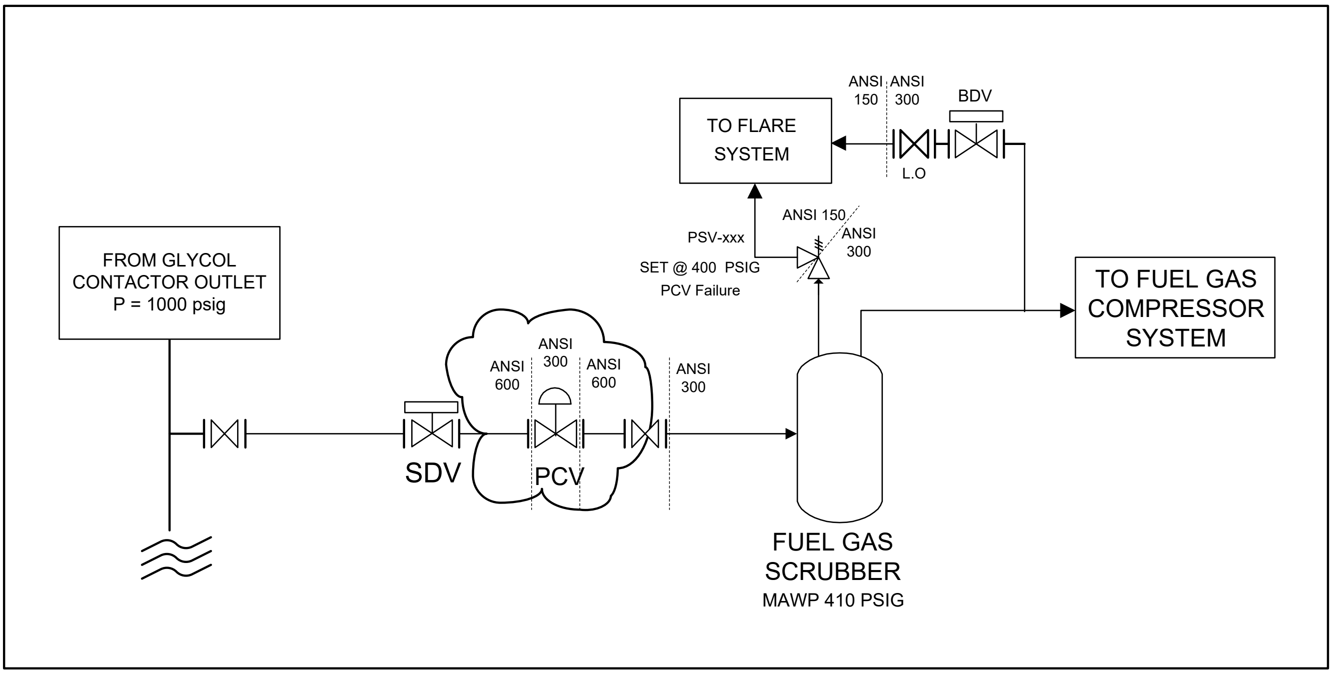
Specification Break



Changes in Spec Break with additional PSV installation

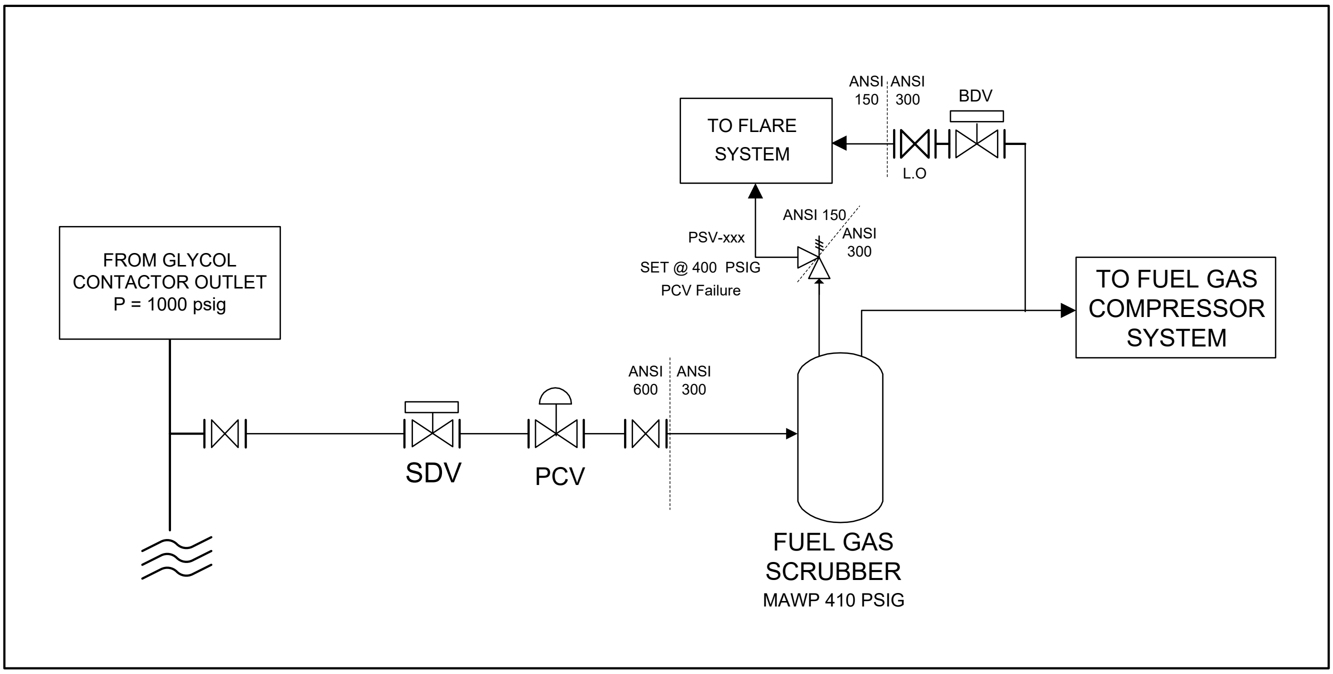
Specification Break

Case Study



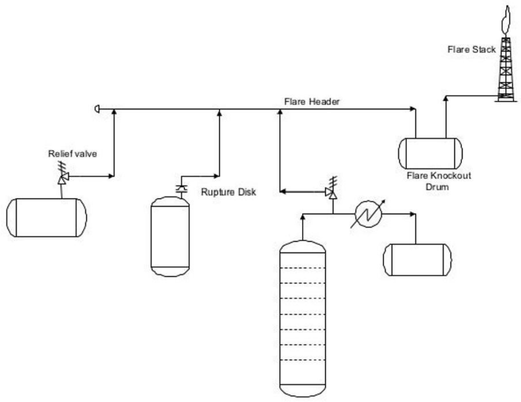
Specification Break

Case Study



Overpressure Protection

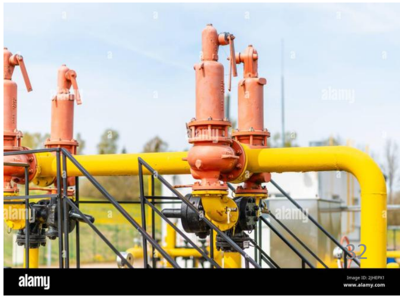
- Excessive pressure in the plant can cause pipes or pressure vessels to burst, which can then cause a fire/explosion.
- Sources of overpressure:
  - Oil and gas wells,Instrumentation failure,
  - Pressure generated by compressors or pumps,
  - Changes in physical properties,
  - hydraulic expansion,
  - Fire.



- There are 2 ways to avoid overpressure:
  1. Cutting off the source, namely:
     - shutting off the gas/oil well or
     - shutting off the injection pressure source (from the gas well or compressor) to the oil well.
  2. Releasing the pressure to a safe system (usually the atmospheric system).
     - The safe system can be a flare, vent, burn pit, or simply venting to the atmosphere.

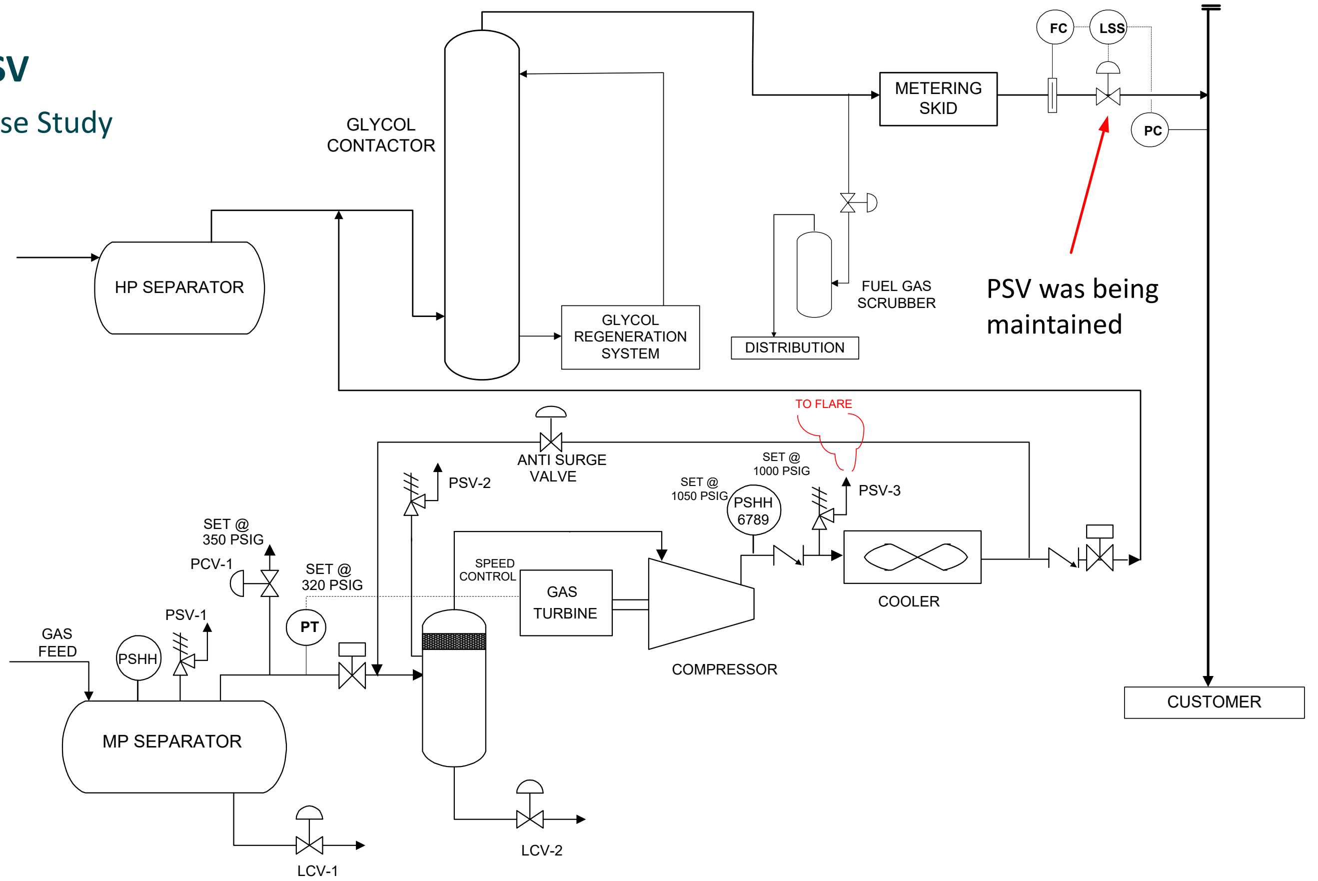
PSV

- A pressure safety valve (PSV) operates according to the second principle: to release excess pressure to a safe system.
- The PSV has a pressure relief setpoint, determined by the engineer. This setpoint must be greater than the PSHH setpoint.
- The PSV capacity is determined by overpressure scenario, which will determine the size and number of PSVs to be installed.
- The PSV design principles **are generally stated in the P&ID**, including: block-discharge PSV, PCV failure PSV, fire PSV, thermal PSV, etc.
- Manual valves are installed upstream and downstream of the PSV for isolation during maintenance.
- Plants typically have identical PSVs (as spares) installed at the facility and operated to back up other PSVs undergoing maintenance.



PSV

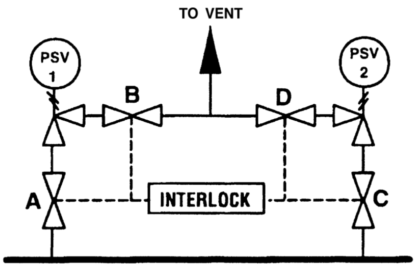
Case Study



PSV

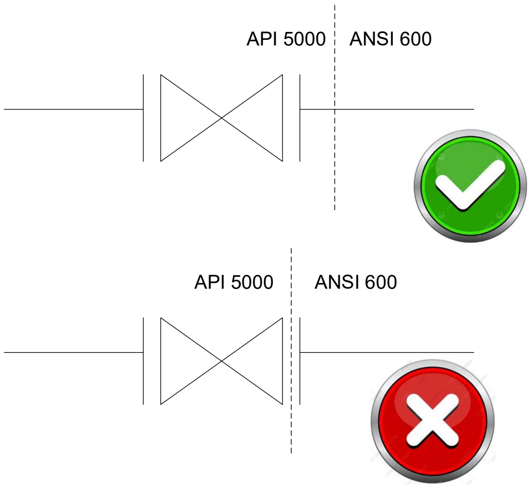
- In facilities/plant where PSVs and their spare parts are installed, an interlock system is often installed.
- Interlocks are installed to ensure that no PSV is accidentally closed while another PSV is isolated (due to maintenance).
- This system ensures that at least one PSV is always operational at all times.
- As an illustration – see the picture:
  - All valves A & B, as well as C & D, are open.
  - Valve A & B are closed **only if** valve C & D are open.
  - Valve C & D are closed **only if** valve A & B are open.

- An interlock system is also installed in the PIG launcher/PIG receiver system, so that the operation of inserting and removing pigs does not endanger workers. *Note: Several accidents resulting in fatalities in pigging operations have been reported.*

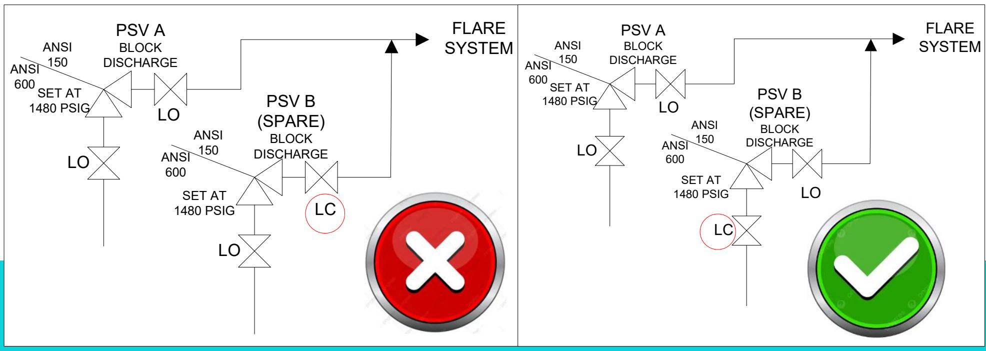


INTERLOCK POSSIBLE STATUS			
A/B	OPEN	CLOSED	OPEN
C/D	OPEN	OPEN	CLOSED

PSV



Proper installation
of specification break

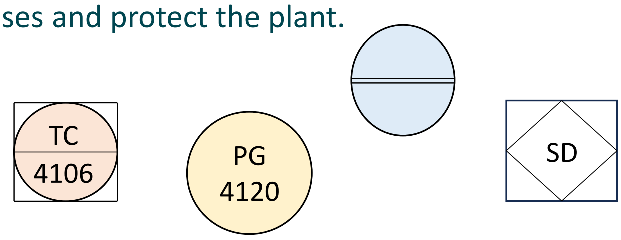


LO/LC principle in the area around the installed PSV

Safety Devices

- Abnormal plant operating conditions can result in personnel injuries, pollution, and loss of assets. Safety devices are installed to address hydrocarbon releases and protect the plant.

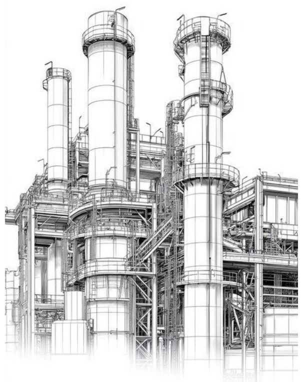
- They consist of three main components:
  - sensing device,
  - a transmission device,
  - and an end device.
- Note: Some devices sense and respond as end devices (check valves, PSVs, etc.).



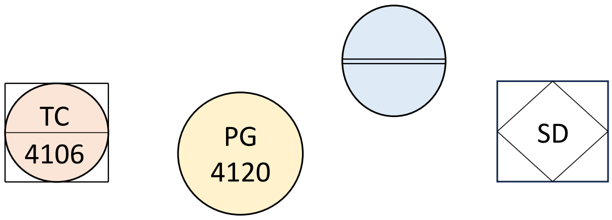
- The primary purpose of a plant safety system:
  - is to prevent the initial release of hydrocarbons (HC)
  - and to shut off the flow of additional HC that has already been released.

- When an abnormal condition is detected:
  - a sensing device sends a signal to an end device.
  - The end device diverts or shuts off the HC flow, sounds an alarm, or takes other corrective action.

Safety Devices

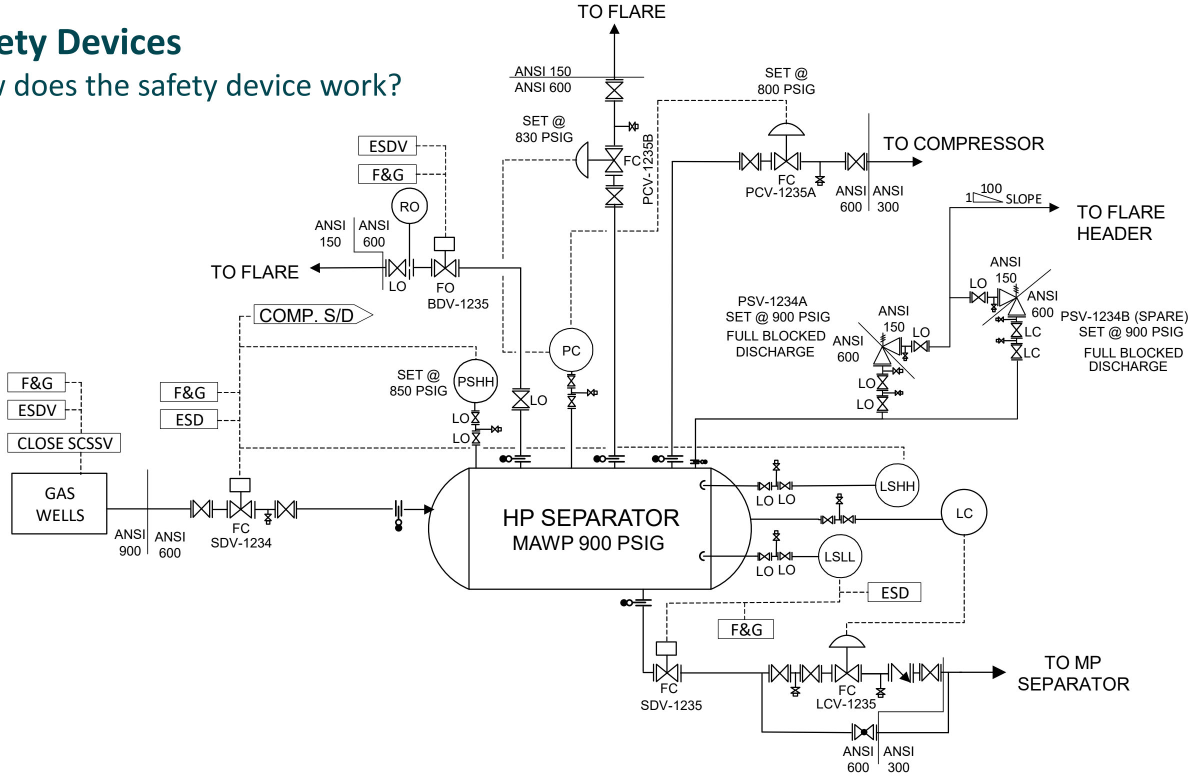


- The previous system is supported by supporting systems, such as:
  - emergency shutdown system (ESD),
  - fire and gas detection system,
  - adequate ventilation,
  - liquid containment system,
  - containment tank,
  - subsurface safety valve (SSSV),
  - pneumatic supply shutdown system,
  - blowdown system.



Safety Devices

How does the safety device work?



P&ID Example - HP Separator with its safety devices

Safety Devices – at HP Separator

Natural gas and condensate flow into the HP separator to separate the gas from the condensate. The gas flows to the compressor, while the condensate enters the MP separator. The PCV-1235A at the separator outlet maintains the separator pressure at 800 psig.

If P ops > or = 830 psig:

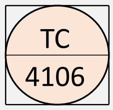
- The PCV-1235B will open to release excess gas to the flare.

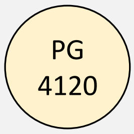
If P ops > the PSHH setpoint (i.e., 850 psig), then:

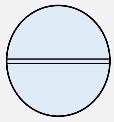
- The switch in the PSHH will send a signal via transmission (either pneumatic or electrical) to the SDV-1234 to shut off the flow to the HP Separator.
- The signal from the PSHH will also be sent to shut down the compressor downstream of the HP Separator.

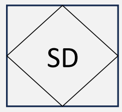
If the level in the separator rises above the LSHH setpoint, then:

- The switch in the LSHH will send a signal to the SDV-1234 to shut off the flow to the HP separator.
- A signal will also be sent to shut down the compressor.





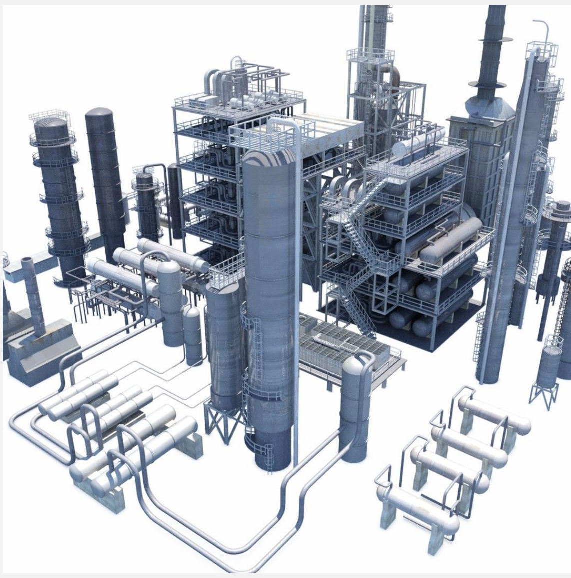




Safety Devices – at HP Separator

If the liquid level drops below the LSLL setpoint, then:

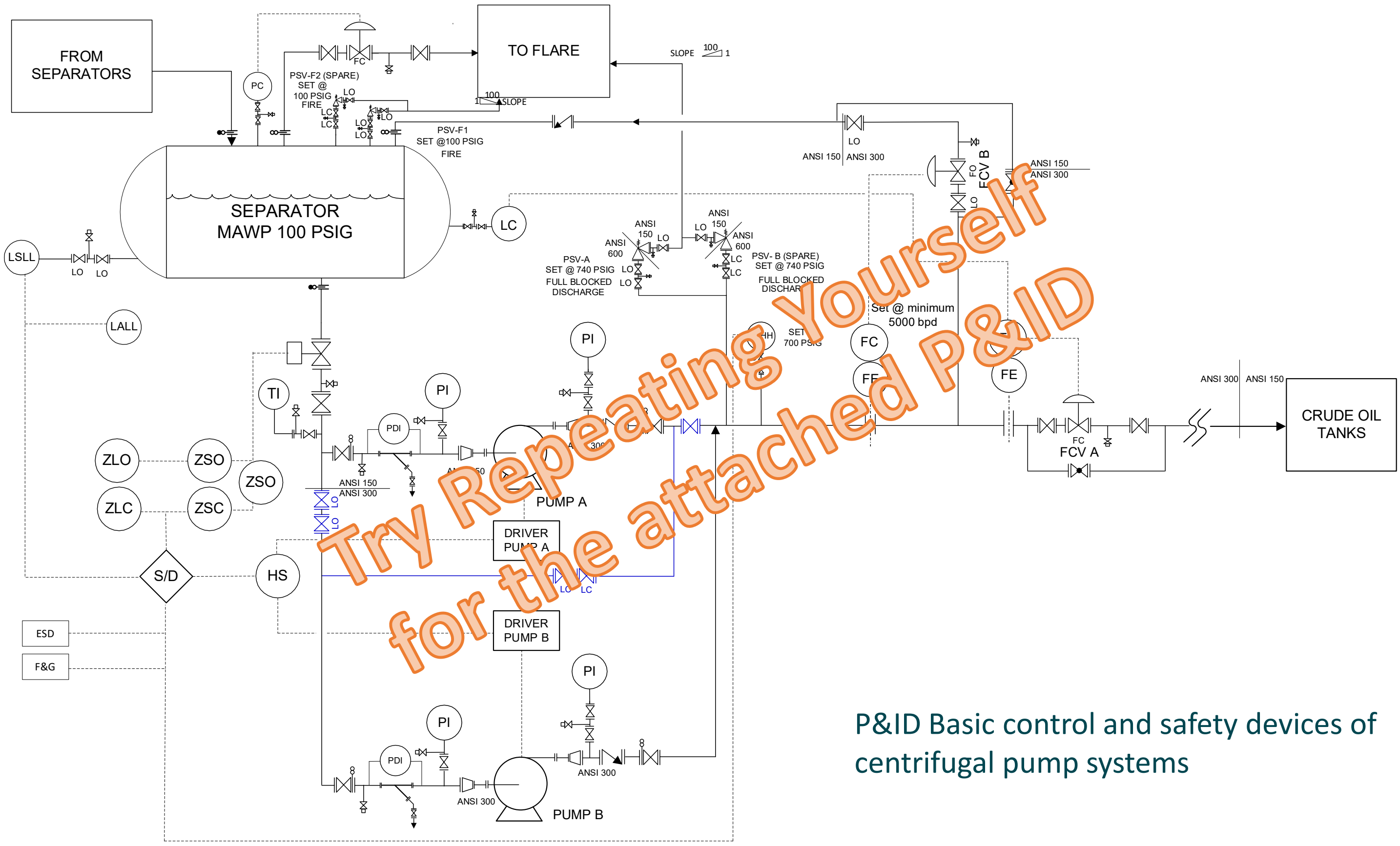
- The switch in the LSLL will send a signal to the SDV-1235 to close, thus preventing further condensate reduction in the separator.
- *Note: The purpose of maintaining the liquid level in the separator in an emergency is to prevent gas from entering the operating unit downstream of the separator, which could cause excessive pressure. It also provides a time delay for temperature increases in the vessel wall thickness (caused by a fire), as the latent heat of vaporization of the liquid requires energy (thermal heat).*



Safety Devices – In case of fire or gas leak

When **a fire and/or flammable gas is detected:**

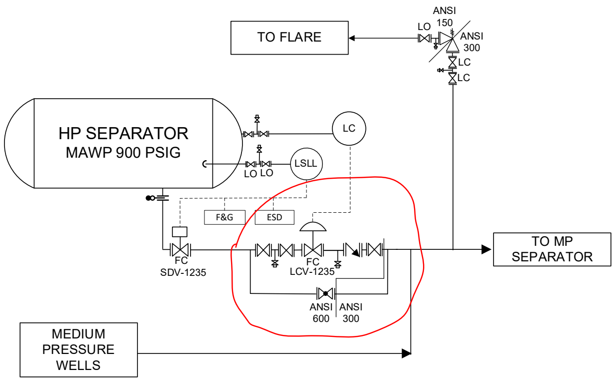
- **The plant will be blown down.**
    - The goal is to reduce the amount of gas/gas inventory within the plant. This action will reduce the gas pressure, which has increased significantly due to the heating effect of the fire.
- **A built-in deluge system sprays water onto** the surface of the separator.
    - Its purpose: to cool the separator surface, which will reduce the impact of a fire or delay the increase in the separator wall temperature. *Note: The deluge system is usually depicted on a separate P&ID from the rest of the process system.*
- **Will activate the F&G (Fire and Gas) system,**
    - The goal: to isolate the separator, namely by:
        - closes SDV-1234,
        - closes SDV-1235 outlet at the HP Separator condensate outlet (to maintain liquid in the separator),
        - shutdown the compressor,
        - and opens and releases gas (blowdown) to the flare through BDV-1235.
- **Will activate** the plant **alarm.**



P&ID Basic control and safety devices of centrifugal pump systems

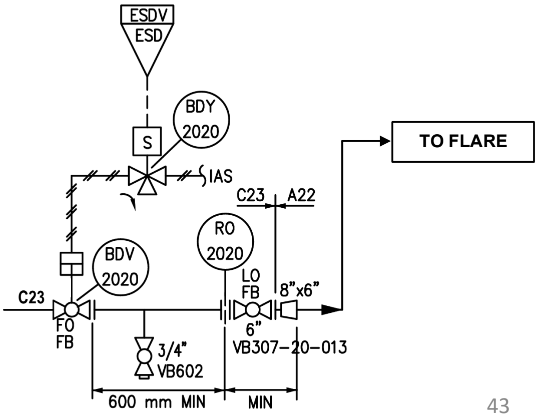
By-pass line

- The bypass line is sized similarly to the main pipe.
- The bypass valve is smaller than the control valve in the main pipe, or the same size but with a smaller trim size.
- Its purpose is to protect against excess pressure downstream, for example, to reduce the rate of gas blowby if it occurs.



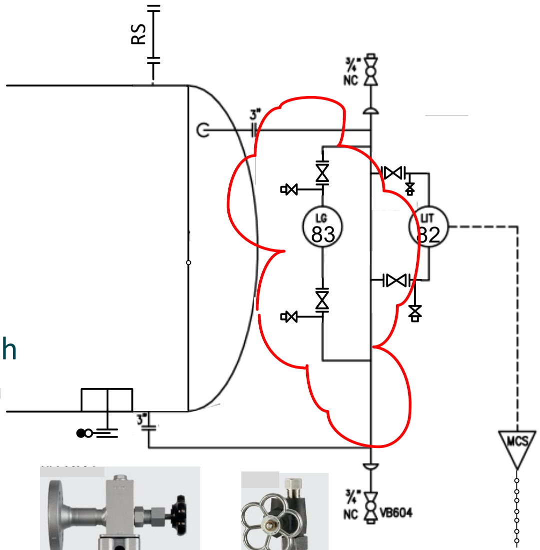
Reducing flow - using RO

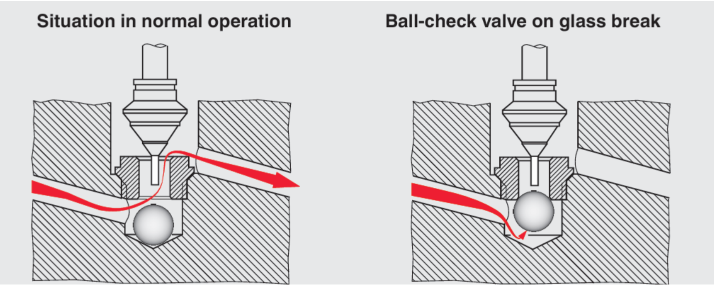
- A restrictive orifice (RO) is typically installed after the blowdown valve (BDV).
- Its purpose is to limit the flow to the flare so it doesn't exceed its capacity.



Sight Glass

- The sight glass is used to monitor the liquid level inside the separator.
- It can also be used to check the interface level or possible contaminants (such as foam).
- If the glass breaks (even if it's relatively strong), two balls at each end of the valve will move, shutting off the flow of hydrocarbon gas/liquid.
- These balls act as check valves to protect against leaks.



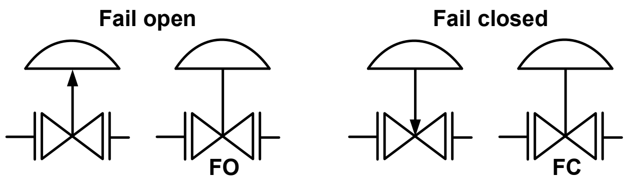


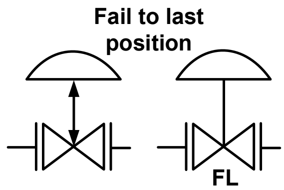
Fail Open or Fail Close or Last Position?

- The P&ID displays the state of the actuated valve when it fails.
- Plant are designed to be failsafe in the event of a failure.
- In engineering, a failsafe system/device will operate in the event of a design feature failure in a manner that will cause minimal or no damage to other equipment, the environment, or people.

Example:

- Elevators have brakes held against the brake pads by the tension of the elevator cables. If the cables break, the tension is lost and the brakes lock the rails in the shaft, preventing the elevator car from falling.
- The isolation valves and control valves used can be designed to close or open when power is lost, for example by using spring force.
- This is known as fail-closed/fail-open when power is lost.

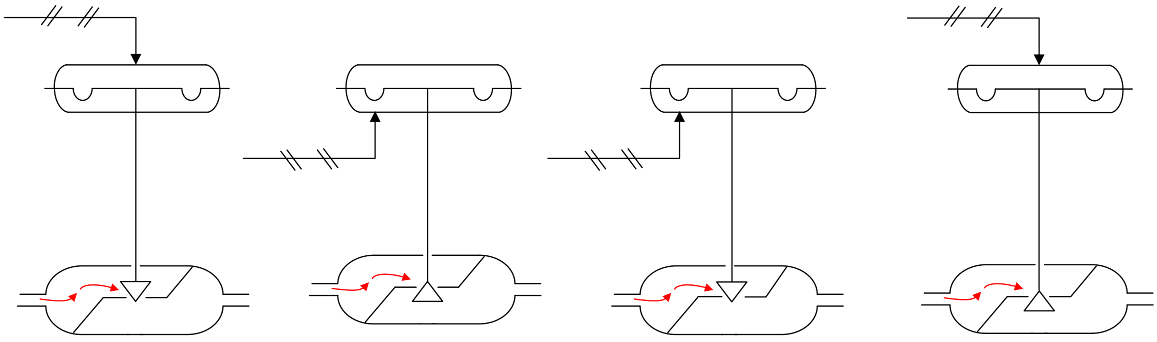




Fail Open or Fail Close or Last Position?

- **The fail-close valve** will automatically close when power is lost.
  - This will stop the flow of media through the system.
  - This design is crucial to prevent potential hazards or contamination
- **A fail-open valve** opens in the event of a power failure.
  - Used to maintain flow or reduce pressure.
  - This design is important for safety reasons, for example, in steam boilers or overpressure protection systems.
- Safety-based selection:
  - A thorough risk assessment of the system's operation is conducted
  - To ensure that in the event of any failure, the system will return to a safe state.
- Based on operational aspects:
  - Process efficiency and
  - Operational continuity (which allows the system to be safely shut down ~~until repairs are made or~~ to continue functioning in fail-safe mode).
- Understanding the nuances between FC and FO valves is crucial for engineers and safety professionals who aim to design systems that not only meet **operational objectives** but also prioritize **safety** and **reliability** under all conditions.

Fail Open or Fail Close or Last Position?



FAIL OPEN (FO)
air to close (A/C)

FAIL OPEN (FO)
air to close (A/C)

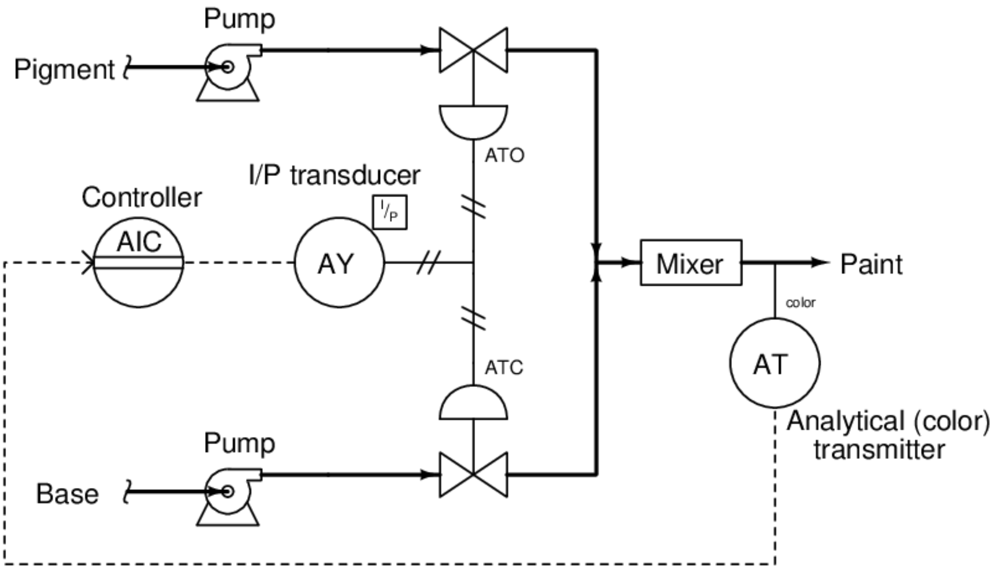
FAIL CLOSE (FC)
air to open (A/O)

FAIL CLOSE (FC)
air to open (A/O)

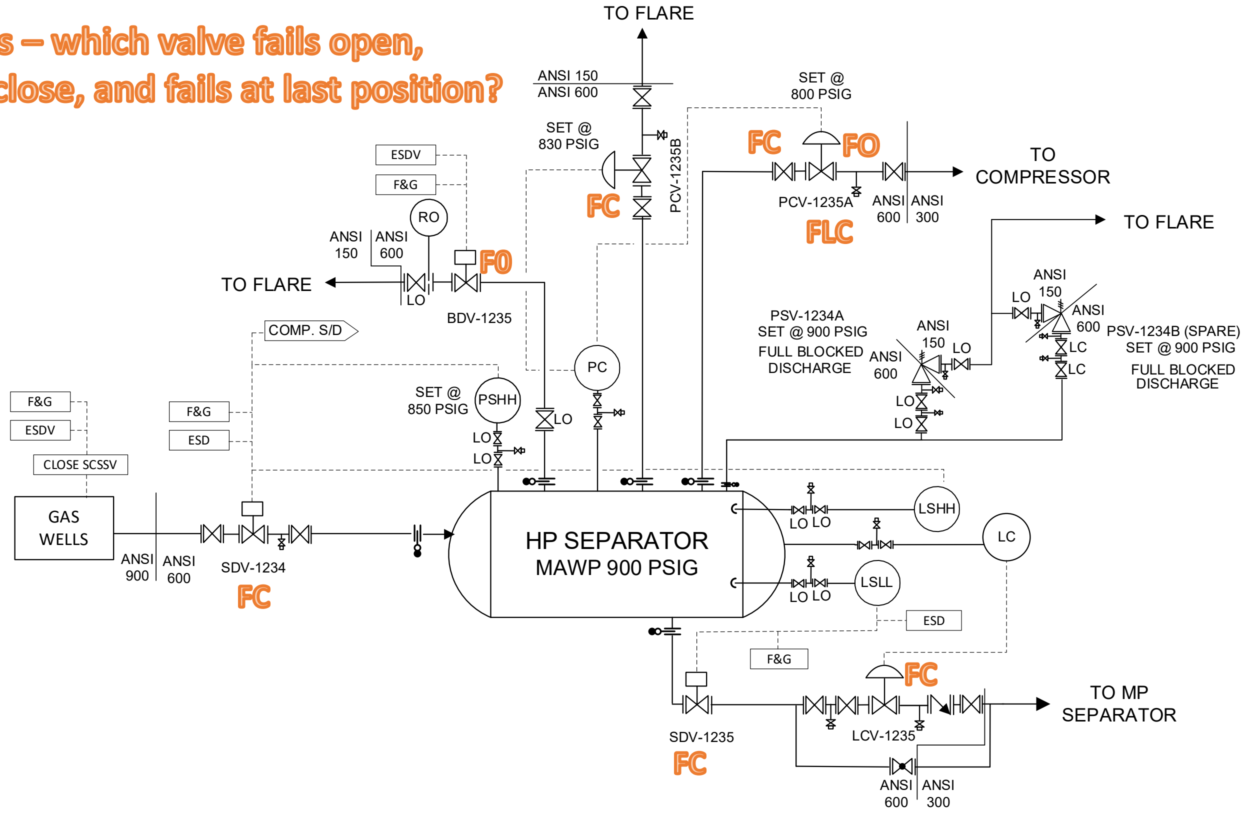
- From the image, it can be seen that a fail-open (FO) valve will air-close (A/C), and a fail-close (FC) valve will air-open (A/O).
- **Can a valve that's FO become FC and vice versa?** Is this possible?

Fail Open or Fail Close or Last Position?

Take a guess!!

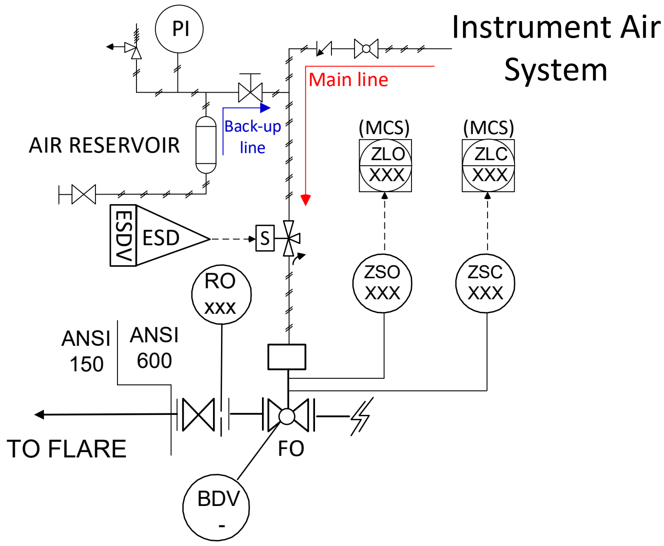


Guess – which valve fails open, fails close, and fails at last position?



Fail Open at Blowdown valve

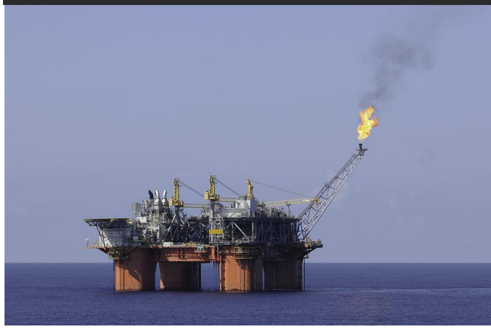
- The blowdown valve, or BDV, is designed to fail open for safety reasons.
- Hydrocarbons in the system it protects will be released/blown down to the flare system, significantly reducing plant pressure.
- However, BDVs can also fail and potentially open suddenly.
- One reason is the loss of instrument air due to a failure of the instrument air generator/Instrument Air System.
- This single-mode failure can cause all BDVs in the plant to open simultaneously



- For this purpose, a small process vessel is installed as a temporary reservoir in the event of instrument air failure.

Flare System

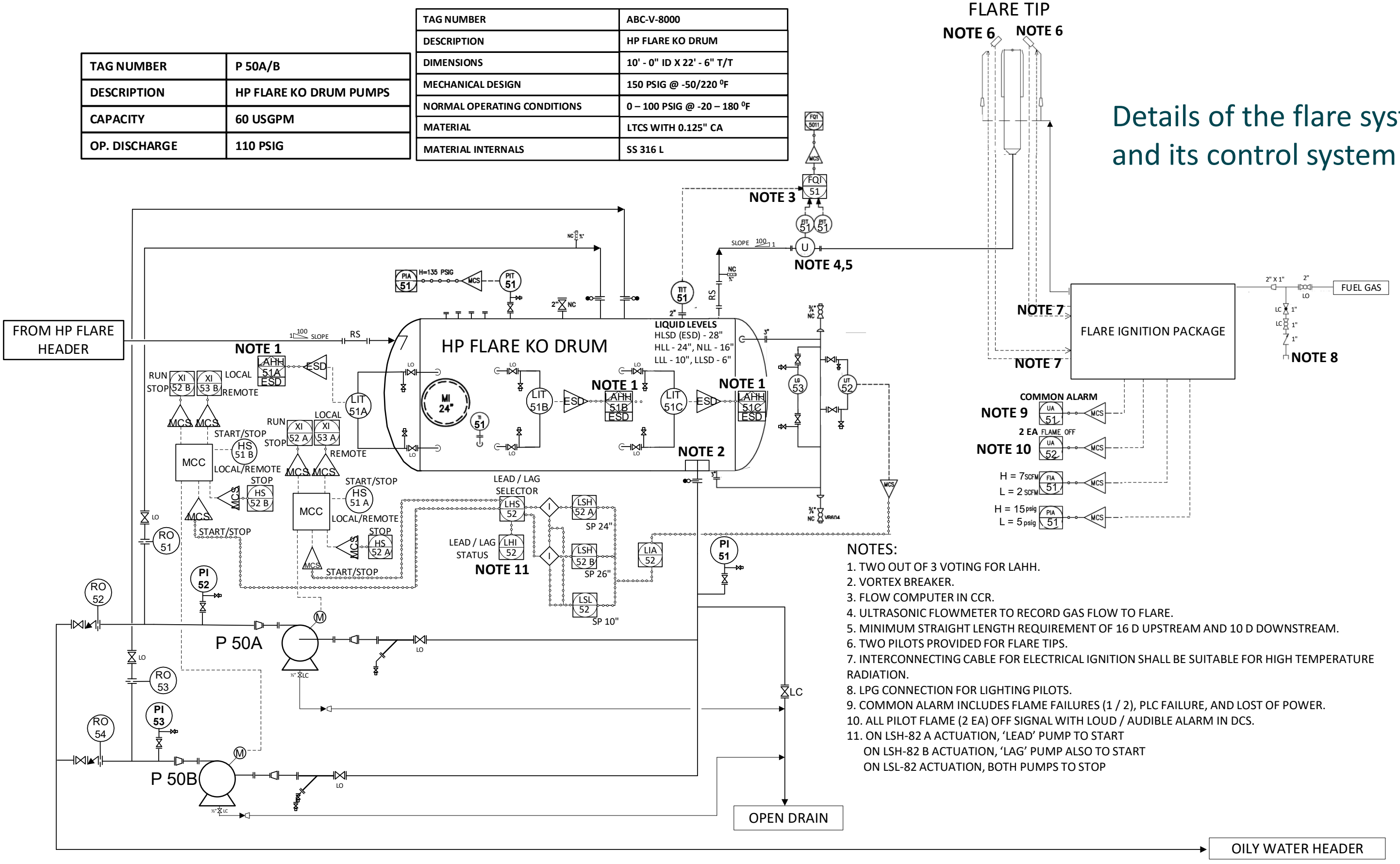
- **Flares are used to burn flammable, toxic, or corrosive vapors** into less hazardous compounds.
- The design of flare systems and the layout of plant equipment **should minimize the need for operator presence.**
- The design of towers or other tall structures exposed to flare radiation **should consider the effects of radiation on the ability to safely evacuate.**
- If personnel exposure to radiant heat is considered hazardous, **protective equipment should be installed.**
- It is often most effective to achieve this by placing ladders and platforms on the side away from the flare.
- **The boundaries of restricted access areas are marked** with signs warning of the potential dangers of thermal radiation exposure.
- **Personnel entry into such restricted areas must be** administratively **controlled** (PTW).
- **It is essential that personnel within restricted areas have immediate access to thermal radiation shielding** or appropriate protective clothing to escape to a safe location.



TAG NUMBER	P 50A/B
DESCRIPTION	HP FLARE KO DRUM PUMPS
CAPACITY	60 USGPM
OP. DISCHARGE	110 PSIG

TAG NUMBER	ABC-V-8000
DESCRIPTION	HP FLARE KO DRUM
DIMENSIONS	10' - 0" ID X 22' - 6" T/T
MECHANICAL DESIGN	150 PSIG @ -50/220 °F
NORMAL OPERATING CONDITIONS	0 – 100 PSIG @ -20 – 180 °F
MATERIAL	LTCS WITH 0.125" CA
MATERIAL INTERNALS	SS 316 L

Details of the flare system and its control system

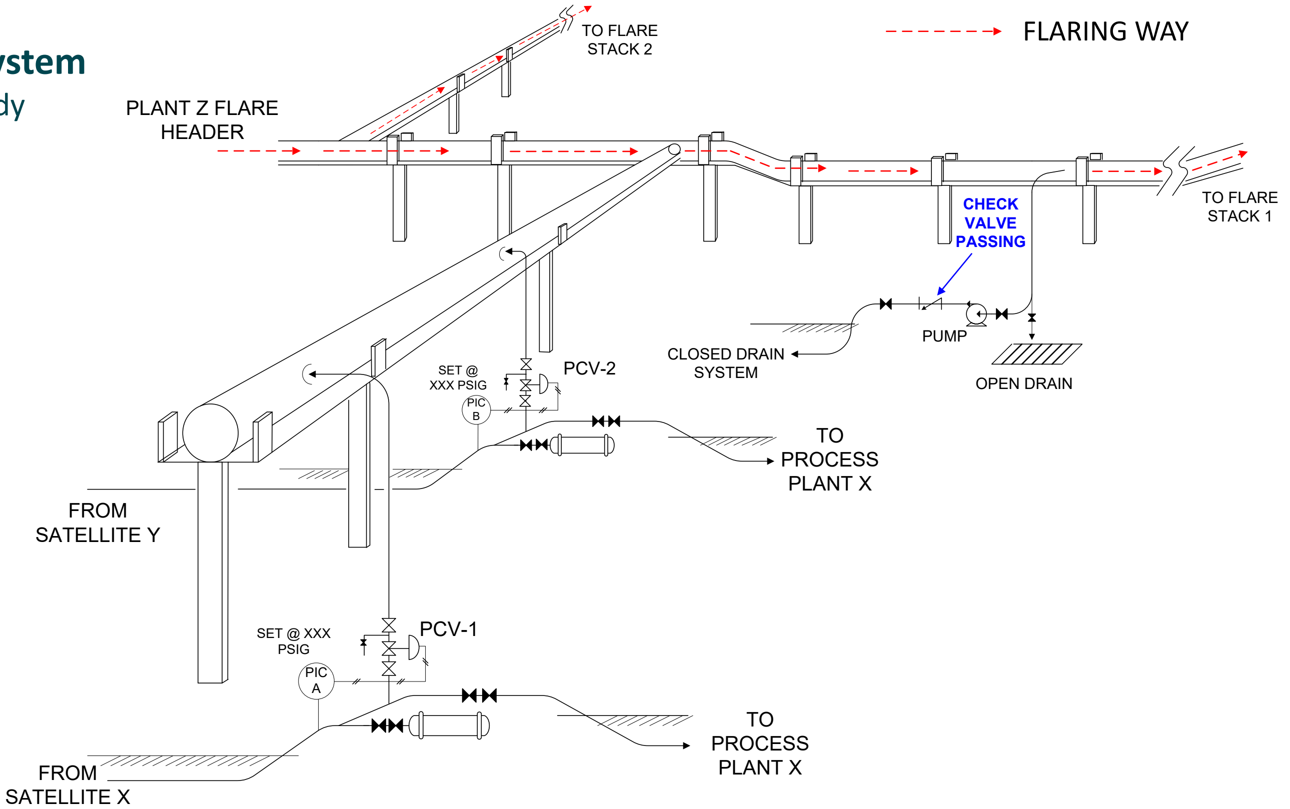


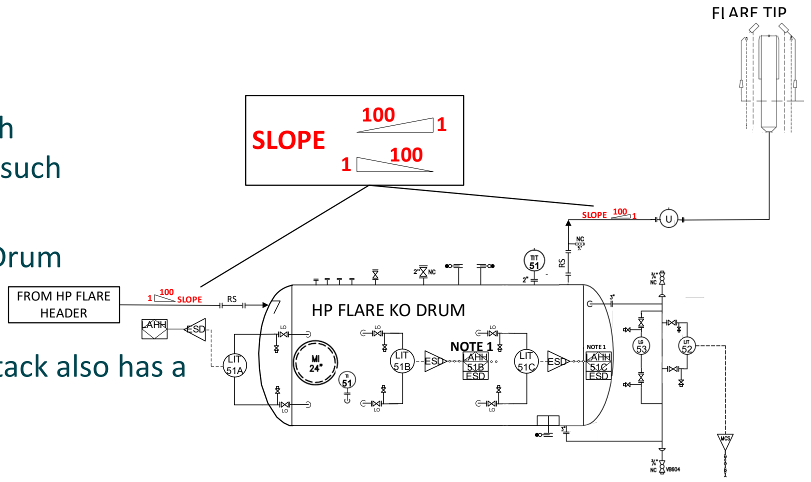
NOTES:

1. TWO OUT OF 3 VOTING FOR LAHH.
2. VORTEX BREAKER.
3. FLOW COMPUTER IN CCR.
4. ULTRASONIC FLOWMETER TO RECORD GAS FLOW TO FLARE.
5. MINIMUM STRAIGHT LENGTH REQUIREMENT OF 16 D UPSTREAM AND 10 D DOWNSTREAM.
6. TWO PILOTS PROVIDED FOR FLARE TIPS.
7. INTERCONNECTING CABLE FOR ELECTRICAL IGNITION SHALL BE SUITABLE FOR HIGH TEMPERATURE RADIATION.
8. LPG CONNECTION FOR LIGHTING PILOTS.
9. COMMON ALARM INCLUDES FLAME FAILURES (1 / 2), PLC FAILURE, AND LOST OF POWER.
10. ALL PILOT FLAME (2 EA) OFF SIGNAL WITH LOUD / AUDIBLE ALARM IN DCS.
11. ON LSH-82 A ACTUATION, 'LEAD' PUMP TO START
    ON LSH-82 B ACTUATION, 'LAG' PUMP ALSO TO START
    ON LSL-82 ACTUATION, BOTH PUMPS TO STOP

Flare System

Case Study



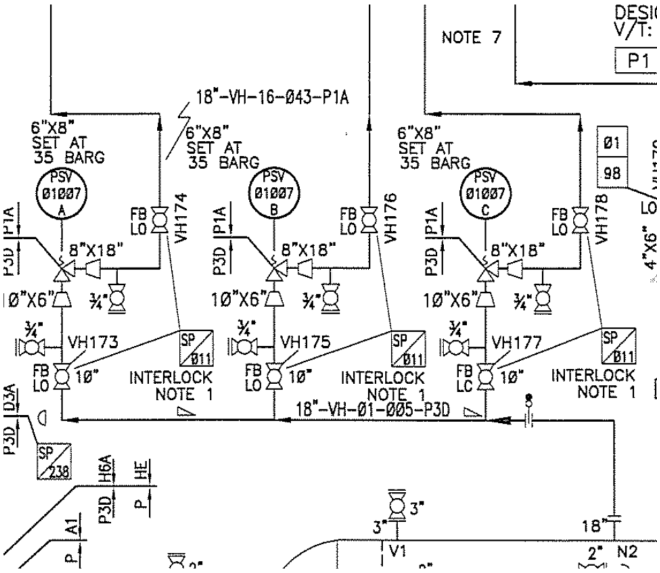


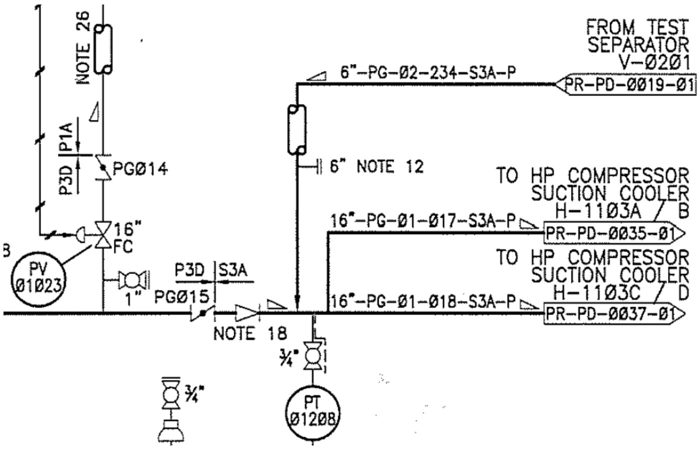
Flare System – Special

- The flare system is designed to ensure a smooth and unobstructed flow of gas from its sources (such as PSVs, BDVs, and manual valves) to the flare.
- All relief lines must slope toward the Flare KO Drum (located upstream of the Flare Stack).
- The flare pipe from the KO Drum to the Flare Stack also has a slope, which is directed towards the KO Drum.
- The purpose:
  - To prevent condensed liquid from pooling in the pipe, which could obstruct the smooth flow of gas to the flare stack.
  - To prevent a reduction in the fluid discharge flow by an active PSV, which could potentially cause overpressure (in the system protected by the PSV).
- Note:
  - Slopes are also common at the suction scrubber outlet to the compressor, PSV header, and main header.
  - Condensed liquid is returned to the suction scrubber so it doesn't accumulate in the piping and get sucked into the compressor (see the compressor P&ID on the previous slide).

Flare System – Slope

Slopes are also common at the suction scrubber outlet to the compressor, PSV header, and main header.



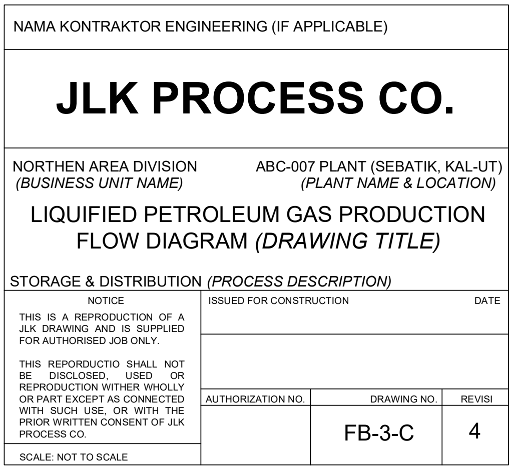


How to read Piping & Instrumentation (P&ID) Diagram

General Information
Symbols

General information in P&ID – Title Block

- P&ID title block contain informasi of:
  - Plant/facility name and its location
  - Process Area
  - Number P&ID
  - Revision status
- Those information must be confirmed so we use the proper and update P&ID. *Use invalid P&ID may cause an error in the job or may cause an accident*. Additional info in P&ID ex: Engineering firm created the P&ID, or one of department in the company who create it.

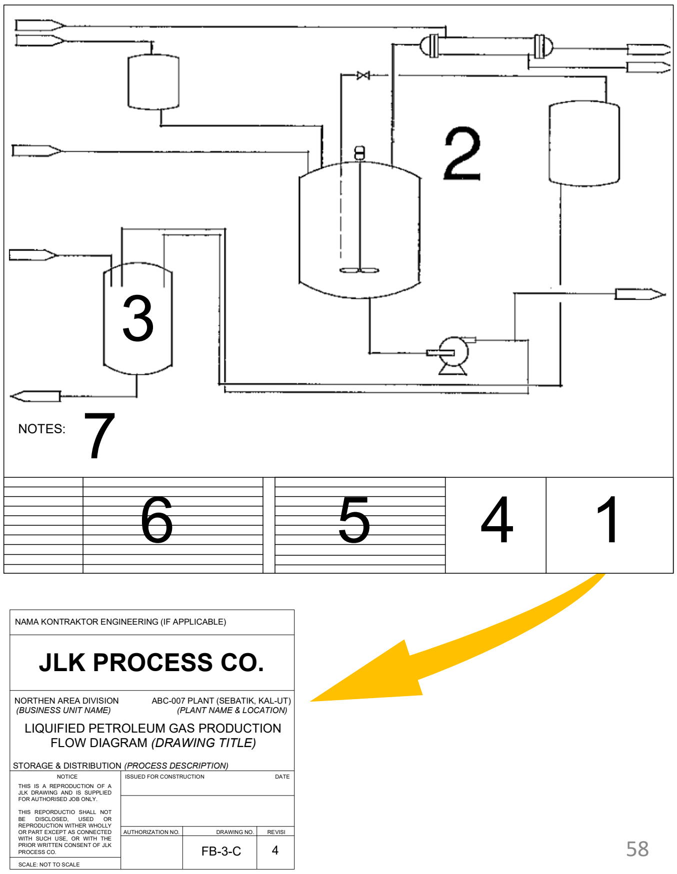


- There are 2 type of information in P&ID:
  - Written information: Title, table label, equipment specification
  - Information via symbol and drawings.

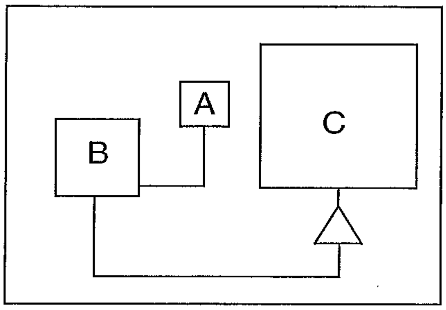
Information in P&ID - Layout

- The layout and amount of information in a P&ID will vary. This slide shows:
    1. Title block
    2. Main diagram
    3. Equipment description
    4. Legal statement (for use/reproduction)
    5. P&ID revision history
    6. Reference drawing
    7. Notes

- The largest section of the P&ID is section 2, which contains symbols and lines. In the P&ID, symbols and lines are used to depict:
    - Equipment
    - Piping connections
    - Instrumentation
    - Lines connecting instruments
    - Instruments control loop



Information in P&ID – Relative Position & Size and Direction of Process Flow

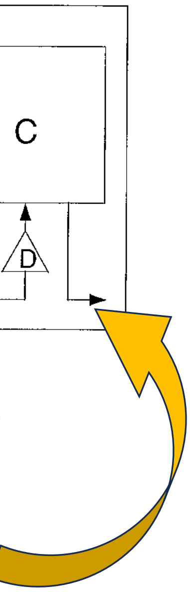


- The relative size of the symbols represents the relative size of the equipment installed in the plant/field.

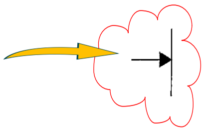
- For example, in the figure, box A is smaller than box B. Therefore, in the field, the vessel depicted as box A is physically smaller than the vessel depicted as box B.

- The relative positions of symbols on a P&ID also represent their actual positions in the field, although the distances are not proportional.
- If the P&ID depicts a pump beneath a vessel, then in the field, the pump is actually located somewhere beneath the vessel.
- The P&ID does not depict the actual distance dimensions between the pump and the vessel.
- Referring to the image above, if the pump is depicted as a triangle, then in the field, the pump is actually located beneath vessel C, and not vessel A or B.

Information in P&ID – Relative Position & Size and Direction of Process Flow



- The P&ID shows the direction of flow in the pipe, depicted by arrows on the lines representing the pipe.
- Each line segment will typically have a flow arrow (created at each intersection or branching of the line).
- The picture on the right shows the arrow directional from the top of box B and exiting from the bottom.
- Sometimes an arrow is made at the point where the lines connect.

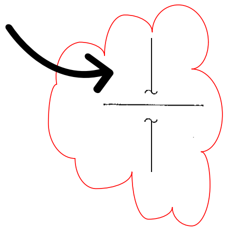


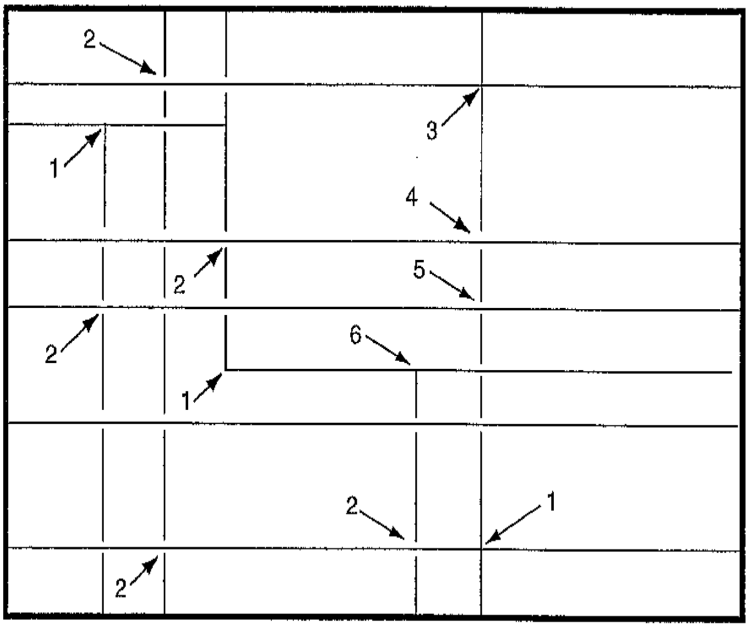
- The P&D also shows the direction of flow leaving the facility or plant or the direction the arrow points to the next P&ID drawing.

Information in P&ID – Connecting Lines

Piping and lines can be installed in various directions, crossover, through walls, or disconnected.

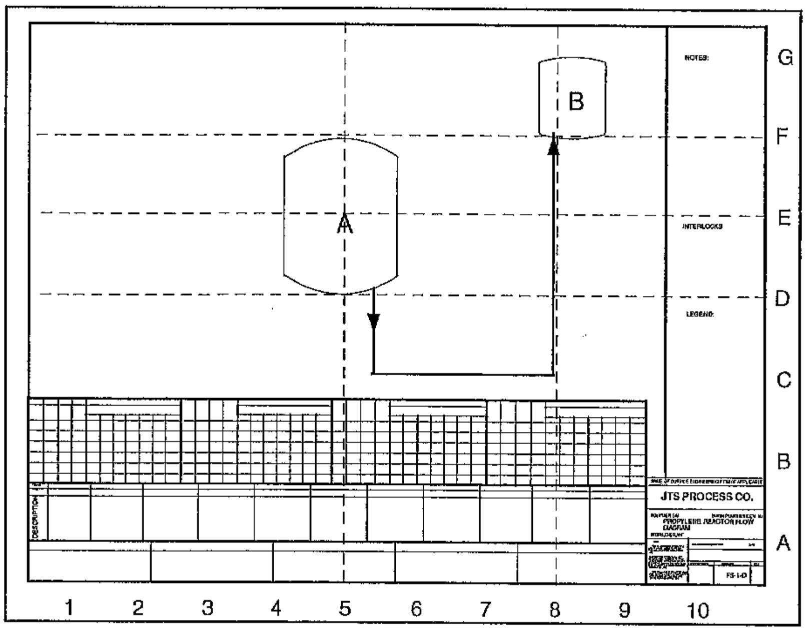
- If two pipes cross over or meet without a break in the P&ID, it means they are physically connected.
- In the picture, the arrow labeled 1 indicates an area where the piping is physically connected, while the arrow labeled 2 indicates a physical disconnect.





Explain pipe with label 3,4,5, and 6 !!

Information in P&ID – Zone Numbers



- Some P&IDs have zone numbering to determine the location of a piece of equipment or the location of two connected pipes.
- There are other P&IDs that don't have a zone number at all.

Information in P&ID – Equipment Description

- The P&ID provides a description of the equipment shown on the P&ID. This description can be written above the equipment, within the drawing, or below the equipment. Regardless of where the description is written, it will generally include:
  - Unique equipment number
  - Capacity
  - Dimensions
  - Materials used
  - Model number
  - Design temperature and pressure
  - Pump power

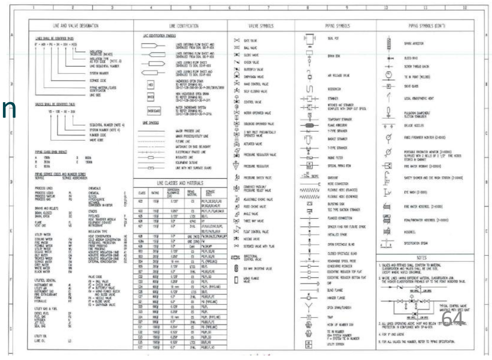
TAG NUMBER	ABC-C-0408
DESCRIPTION	HP COMPRESSOR "A"
OPERATING FLOW	73 – 80 MMSCFD
SUCTION CONDITION	500 – 600 PSIG @ 85 – 95 ^{0}F
DISCHARGE CONDITION	1600 – 1970 PSIG @ 310/330 ^{0}F
MECHANICAL DESIGN	2100 PSIG @ –20–400°F

TAG NUMBER	ABC-V-2004
DESCRIPTION	PRODUCTION SEPARATOR
DIMENSIONS	10'–6" ID x 30'–0" T/T
MECHANICAL DESIGN	800 PSIG @ –20 / 230°F
OPERATING CONDITIONS IN	200- 300 PSIG @ 110°F
OPERATING CONDITION OUT	190- 290 PSIG @ 110°F
MATERIAL	C.S WITH 0.25 CA + INTERNAL EPOXY COATING
MATERIAL INTERNALS	SS 316 L
TRIM	C23

WELL NUMBER	WELL W-001
DESIGN FLOW (MAX/MIN)	70/5 MMSCFD + ASSOCIATED LIQUIDS
PRESSURE LIMITATION	SITHP 1705 PSIA
FLOWING PRESSURE (MAX/MIN)	FWHP 1650/255 PSIA
FLOWING TEMPERATURE (MAX/MIN)	180 / 140 ^{0}F

Legend

- The P&ID legend contains detailed information about the symbols in the P&ID as well as special notes.
- The legend will be frequently referred to when reading and navigating the P&ID, especially for beginners. The more frequently you read a P&ID, the less frequently you will need to consult the legend.
- The information contained in a P&ID legend for one facility/plant location may differ from another, depending on who created it and the processes involved.
- Even within a single company, P&ID legends can differ from facility to facility, unless the company has standardized the information.
- In upstream oil and gas production facilities, the legend generally contains the following information:
  - Line & valve designation
  - Line identification
  - Line class & material
  - Piping specialty items
  - Valve symbol
  - Piping symbol
  - Vessel symbol and its internal
  - Pump symbol
  - Heat Exchanger symbol
  - Compressor symbol
  - Equipment designation
  - System number
  - Instrumentation – abbreviation
  - Instruments designation
  - Instrumentation symbol
  - P&ID/Shutdown definition
  - Special symbol
  - Unit of measurement

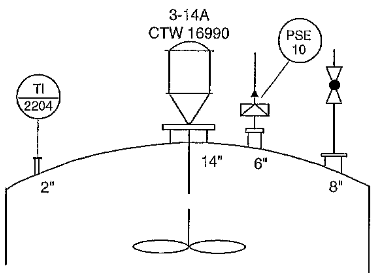


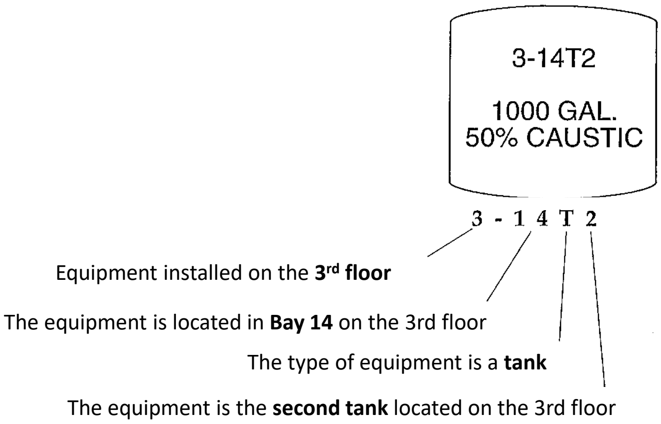
Symbol - Vessel

- The naming of a vessel can be simple (just the first letter followed by the number), or it can also contain information about its location in the field.

- Symbols on the vessel indicate important parts connected to the vessel. For example, the nozzle in the picture below, on the left. The nozzle's shape is circular, following the piping because it serves as the connection point between the pipe and the vessel.

- Nozzle diameters vary. As shown in the picture below, there are four nozzles: 2 inches, 14 inches, 6 inches, and 8 inches in diameter.





Symbol - Vessel

ABC V-2004
Production Separator

ABC = Plant Name

V = Pressure vessel

20 = Gas cooling & liquid separation

04 = the 4th unit

SYSTEM NUMBERS

WELLS/OIL PRODUCTION

10 WELLHEADS
11 MANIFOLD
12
13 DRILLING EQUIPMENT
14
15 INTRA–FIELD IMPORT/EXPORT
16
17 OIL/GAS SEPARATION
18
19 OIL METERING

GAS PRODUCTION

20 GAS COOLING AND LIQUID SEPARATION
21 FUTURE XP COMPRESSION
22 GAS TREATMENT
23
24 DEG PACKAGE
25 ASSOCIATED GAS COMPRESSION
26 EXPORT GAS COMPRESSION
27 TRAIN 2 EGC (FUTURE)
28 GAS METERING
29 GAS EXPORT
30 GAS REINJECTION
31

CONDENSATE PRODUCTION

32 CONDENSATE AND OILY WATER
33
34
35
36

PRODUCED WATER

37 PRODUCED WATER TREATMENT (FIRM FUTURE)
38
39 SLOPS

PROCESS UTILITIES

40 FUEL GAS
41 REFRIGERANT
42 METHANOL
43 SEAWATER/SERVICE WATER
44 COOLING WATER
45
46
47
48 SEAL GAS
49 CHEMICAL INJECTION

FLARE, VENTS AND DRAINS

50 HP FLARE
51 LP FLARE
52 ATMOSPHERIC VENT
53
54
55 CLOSED DRAINS
56 HAZARDOUS OPEN DRAINS
57 NON–HAZARDOUS OPEN DRAINS
58
59

GENERAL UTILITIES

60 POTABLE WATER
61 INSTRUMENT/UTILITY AIR
62 INERT GAS
63 DIESEL FUEL
64 AVIATION FUEL
65 NON–PROCESS OILS
66 WASTE WATER/SEWAGE
67 HYPOCHLORITE
68 N2 GENERATOR
69 MECHANICAL HANDLING/CRANES

BUILDINGS, M & E

70 LIVING QUARTERS
71 HVAC/CHILLED WATER
72 WORKSHOPS

ELECTRICAL POWER

73 MAIN POWER GEN. 1
74 MAIN POWER GEN. 2
75 MAIN POWER GEN. (FUTURE)
76 MAIN POWER GEN. 3
77 SWITCHGEAR/MCC
78 UPS & BATTERY SYSTEMS
79 MISCELLANEOUS ELECTRICAL

LOSS PREVENTION

80 FIREWATER SYSTEM
86 EMERGENCY EQUIPMENT
87 LIFEBOATS AND DAVITS
88 LIFERAFTS

TELECOMMUNICATIONS

90 TELECOMMUNICATIONS AND PA
91 MONITORING

CONTROLS AND SAFETY

96 SAFETY & CONTROL SYSTEM
97 FIRE AND GAS DETECTION

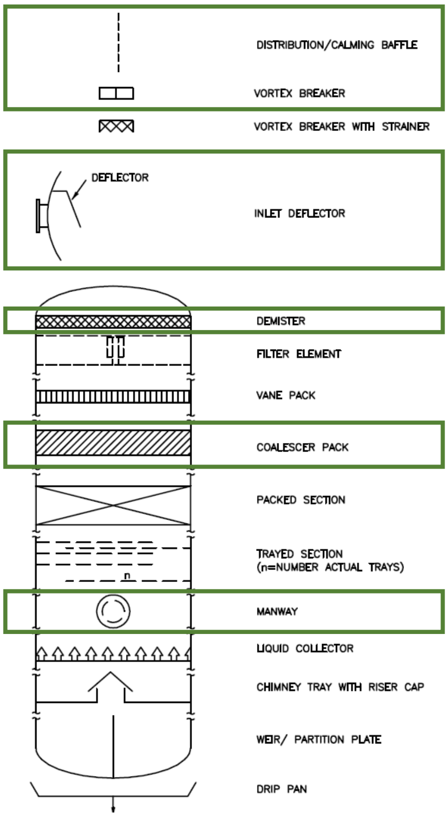
TAG NUMBER	ABC V-2004
DESCRIPTION	PRODUCTION SEPARATOR
DIMENSIONS	10'–6" ID x 30'–0" T/T
MECHANICAL DESIGN	800 PSIG @ –20 / 230F
OPERATING CONDITIONS IN	200- 300 PSIG @ 110F
OPERATING CONDITION OUT	190- 290 PSIG @ 110F
MATERIAL	C.S WITH 0.25 CA + INTERNAL EPOXY COATING
MATERIAL INTERNALS	SS 316 L
TRIM	C23

EQUIPMENT DESIGNATION

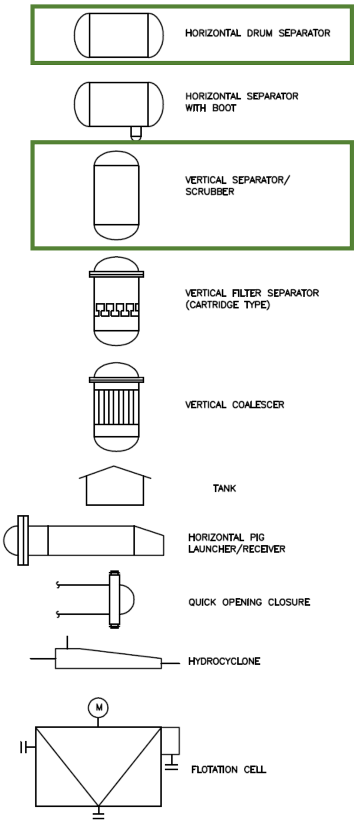
FIRST LETTER		SECOND LETTER
A	PACKAGED UNIT/MISCELLANEOUS EQUIPMENT	
B	CRANE, HOIST	
C	COMPRESSOR/FANS/BLOWERS	
D	DRILLING EQUIPMENT	DIESEL ENGINE
E	UNFIRED HEAT TRANSFER EQUIPMENT	GAS ENGINE
F	FIRED FURNACE/HEATER	FAN
G	GENERATOR	GEARBOX
H	HVAC EQUIPMENT	
I	NOT USED	
J	INSTRUMENTATION AND CONTROL EQUIPMENT	
K	NOT USED	
L	PIG/SCAPER LAUNCHER	
M	MIXER/STIRRER	ELECTRIC MOTOR
N	ELECTRICAL EQUIPMENT	
O	NOT USED	
P	PUMP	
Q	NOT USED	
R	PIG/SCRAPER RECEIVER	
S	STRAINER/FILTER	
T	TANK (ATMOSPHERIC)	GAS TURBINE
U	HYDRAULIC EQUIPMENT	
V	PRESSURE VESSEL	
W	WELLHEAD	
X	MULTIFUNCTION EQUIPMENT	
Y	TELECOMMUNICATIONS EQUIPMENT	
Z	SAFETY EQUIPMENT	

Symbol - Vessel

VESSEL INTERNALS

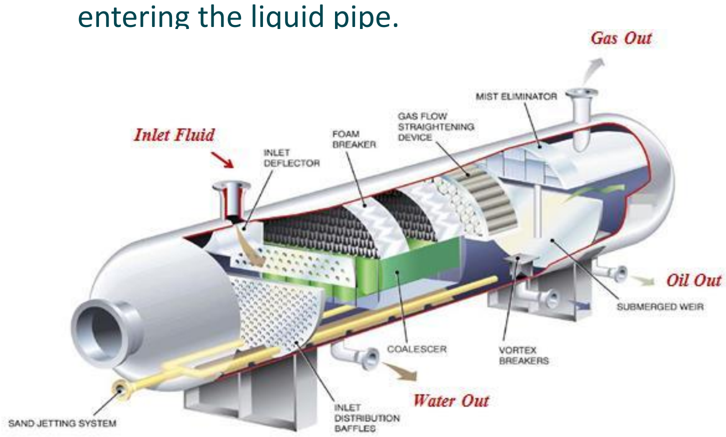


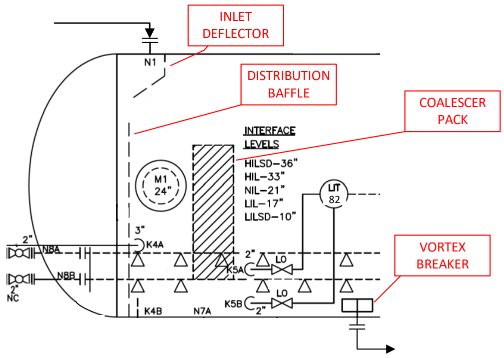
VESSELS



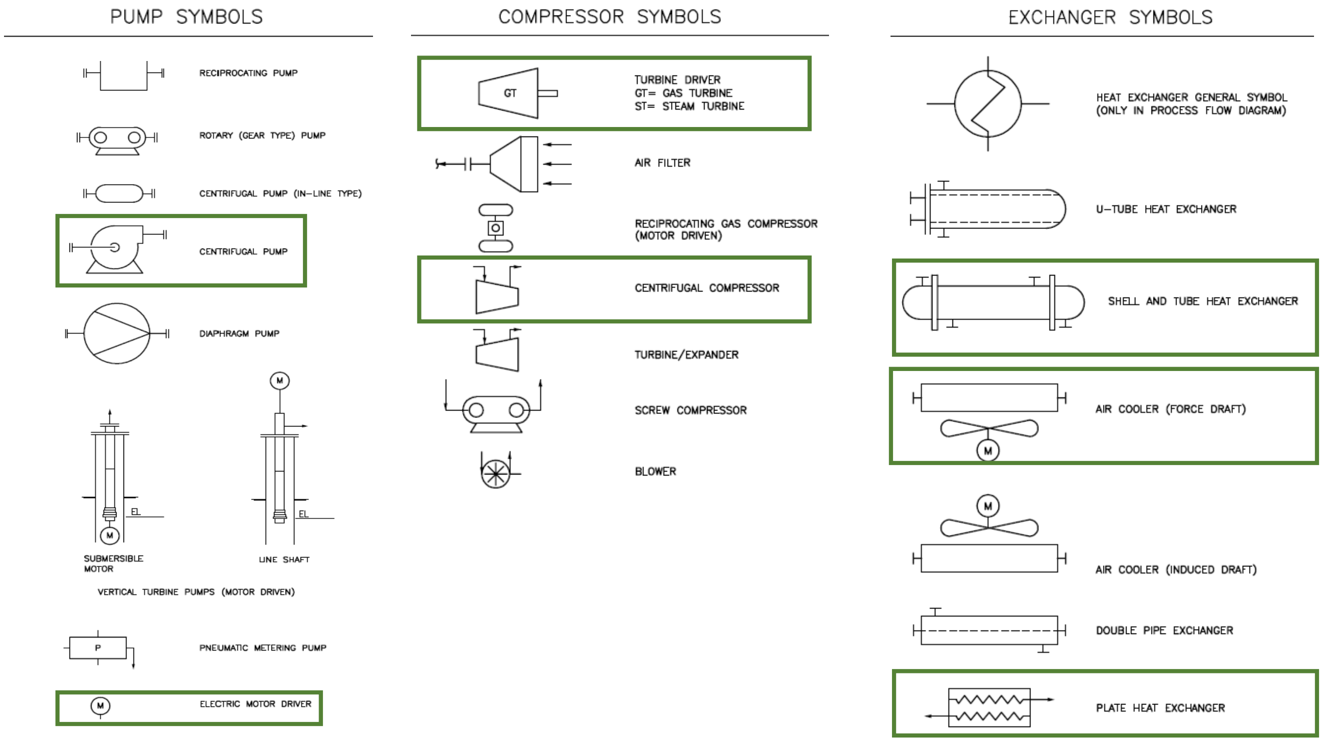
Symbol - Vessel

- Internal vessels vary in content and have specific names and functions.
- The following sectional diagram explains the components:
  - Inlet deflector – functions to separate gas and liquid in the initial stage by utilizing momentum differences, separating large droplets, distributing the flow, and preventing foaming.
  - Distribution baffle - maintains a laminar or calm fluid flow.
  - Coalescer Pack – improves the separation of water from condensate/oil and reduces residence time in the vessel.
  - Vortex breaker – prevents vortex formation at the vessel outlet to prevent gas from entering the liquid pipe.



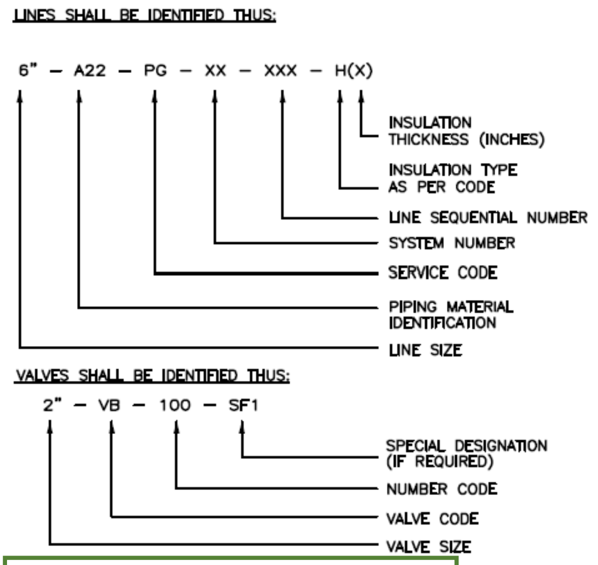


Symbol – Pump, Compressor, Heat Exchanger



Symbol – Line and Valve designation

LINE AND VALVE DESIGNATION



PIPING CLASS (ANSI RATING)

A	150lb	D	900lb
B	300lb	E	1500lb
C	600lb		

PIPING SERVICE CODES AND NUMBER SERIES

SERVICE	SERVICE ABBREVIATION
PROCESS LINES	
PROCESS LIQUID	PL
PROCESS VAPOUR	PV
PROCESS GAS	PG
DRAINS AND RELIEFS	
BLOWDOWN	BD
DRAIN, CLOSED	DC
DRAIN, OPEN	DO
FLARE	FL
SAFETY VALVE DISCHARGE	SV
VENT GAS	VG
UTILITY WATER	
COOLING WATER	CW
FIRE WATER	FW
POTABLE WATER	WP
UTILITY WATER	WU
DELUGE WATER	DE
OIL WATER	OW
TREATED WATER	TW
RAW SEA WATER	SW
INJECTION WATER	WI
FRESH WATER	WF
UTILITIES, GENERAL	
AIR INSTRUMENT	AI
AIR UTILITY	AU
GAS INSTRUMENT	GI
GAS UTILITY	GU
FIRE EXTINGUISHANT	FE
FOAM	FM
HYDRAULIC	HD
UTILITY GAS & FUEL	
BLANKET GAS	BG
DIESEL FUEL	DF
FUEL GAS	FG
NITROGEN	N
JET FUEL	JF
SEAL GAS	SG
UTILITY OIL	
LUBE OIL	LO
CHEMICALS	
CHEMICAL	C
GLYCOL	GL
HYPOCHLORITE	HY
METHANOL	ME
OTHERS	
PIPELINES	P
HEAT TRANSFER MEDIUM	TH
EQUIPMENT EXHAUST	EX
REFRIGERANT	R

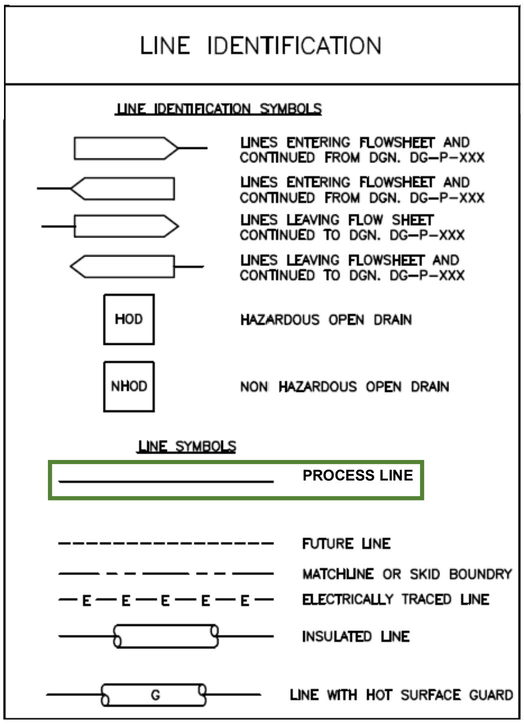
INSULATION TYPE

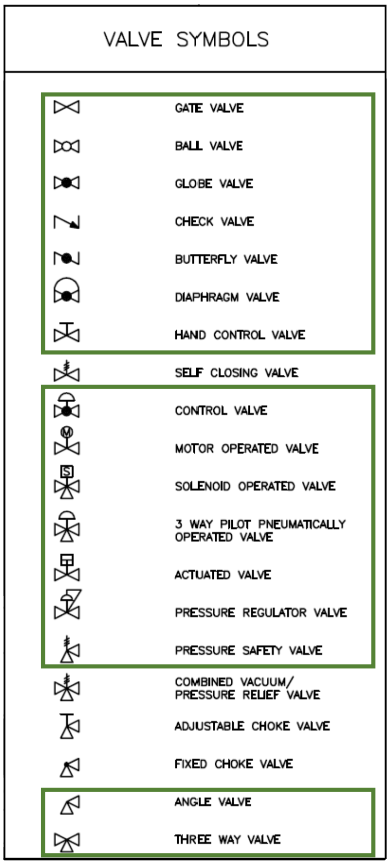
HEAT CONSERVATION	HC
COLD MEDIUM CONSERVATION	CC
PERSONNEL PROTECTION	PP
FROST PROOFING	FR
FIRE PROOFING	FP
ACOUSTIC INSULATION–10dB	A1
ACOUSTIC INSULATION–20dB	A2
ACOUSTIC INSULATION–30dB	A3
EXTERNAL CONDENSATION	EC

VALVE CODE

VB = BALL VALVE
VC = CHECK VALVE
VF = BUTTERFLY VALVE
VM = MONOFLANGE BLOCK AND BLEED VALVE
VN = NEDDLE VALVE
VT = GLOBE VALVE

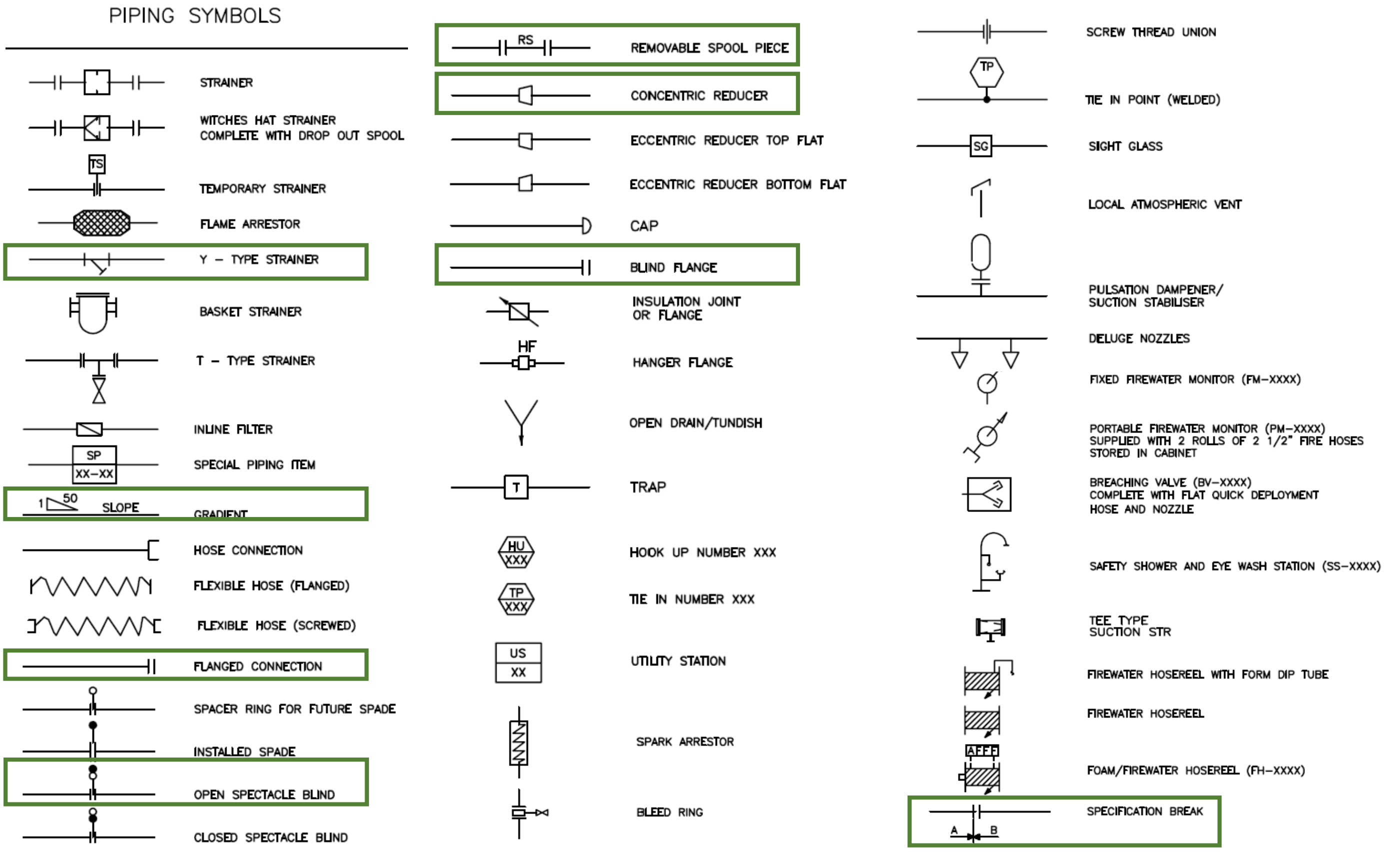
Symbol – Line and Valve designation





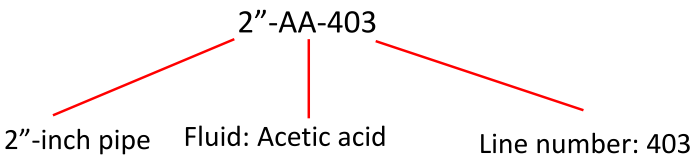
LINE CLASSES AND MATERIALS

CLASS	RATING	CORROSION ALLOWANCE (CA)	PIPING MATERIAL	SERVICE CODE
A22	150#	0.125"	CS	PG, PL, DC, DO, FL, FG, DF, N, BG, R, SG, VG
A23	150#	0.250"	CS	PG, FL, PL, FG, VG
A25	150#	0.125"	LTCS	BD, FL
A26	150#	–	CPVC	HY
A27	150#	0.0	316L SS	PG, BD, JF, FE FL, LO, AI, AU
A28	150#	0.0	GRE	FW, DE, SW, CW, WU, SR, WP, AI, AU, N
A29	150#	0.0	COPPER NICKEL (90%–10%)	FW, SW, WP
B22	300#	0.125"	CS	PG, PL, SG, FG
B23	300#	0.250"	CS	PG, PL, FG, FL
B25	300#	0.125"	LTCS	BD, FL
B27	300#	0.0	316L SS	PG, BD, FG, FE
C22	600#	0.125"	CS	PG, PL, SG
C23	600#	0.250"	CS	PG, PL
C27	600#	0.0	316L SS	PG, BD, FL, FE
D22	900#	0.125"	CS	PG, PL
D23	900#	0.250"	CS	PG, PL
D27	900#	0.0	316L SS	PG, BD, FL, FE
E21	1500#	0.125"	CS	PG
E22	1500#	0.125"	CS	PG, PL, SG
E23	1500#	0.250"	CS	PG, SG
E25	1500#	0.125"	LTCS	BD, FL
E27	1500#	0.0	316L SS	PG, BD, FL
D1	900#	0.05"	CS	BD, GL, P, W, WL

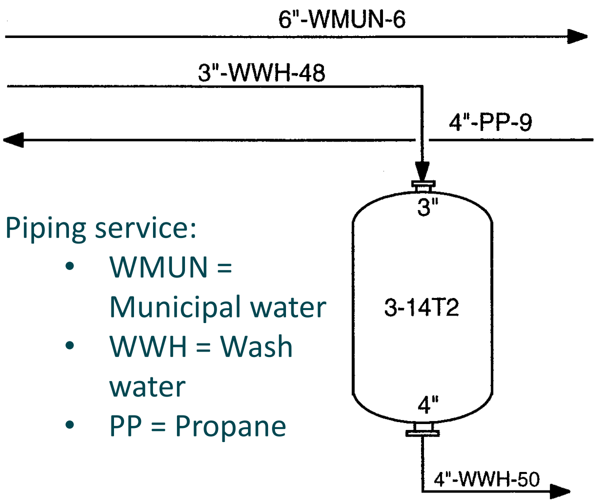

PIPING SYMBOLS
STRAINER
WITCHES HAT STRAINER COMPLETE WITH DROP OUT SPOOL
TS
TEMPORARY STRAINER
FLAME ARRESTOR
Y – TYPE STRAINER
BASKET STRAINER
T – TYPE STRAINER
INLINE FILTER
SP
XX–XX
SPECIAL PIPING ITEM
1
50
SLOPE
GRADIENT
HOSE CONNECTION
FLEXIBLE HOSE (FLANGED)
FLEXIBLE HOSE (SCREWED)
FLANGED CONNECTION
SPACER RING FOR FUTURE SPADE
INSTALLED SPADE
OPEN SPECTACLE BLIND
CLOSED SPECTACLE BLIND
RS
REMOVABLE SPOOL PIECE
CONCENTRIC REDUCER
ECCENTRIC REDUCER TOP FLAT
ECCENTRIC REDUCER BOTTOM FLAT
CAP
BLIND FLANGE
INSULATION JOINT OR FLANGE
HF
HANGER FLANGE
OPEN DRAIN/TUNDISH
T
TRAP
HU
XXX
HOOK UP NUMBER XXX
TP
XXX
TIE IN NUMBER XXX
US
XX
UTILITY STATION
SPARK ARRESTOR
BLEED RING
SCREW THREAD UNION
TP
TIE IN POINT (WELDED)
SG
SIGHT GLASS
LOCAL ATMOSPHERIC VENT
PULSATION DAMPENER/ SUCTION STABILISER
DELUGE NOZZLES
FIXED FIREWATER MONITOR (FM–XXXX)
PORTABLE FIREWATER MONITOR (PM–XXXX) SUPPLIED WITH 2 ROLLS OF 2 1/2" FIRE HOSES STORED IN CABINET
BREACHING VALVE (BV–XXXX) COMPLETE WITH FLAT QUICK DEPLOYMENT HOSE AND NOZZLE
SAFETY SHOWER AND EYE WASH STATION (SS–XXXX)
TEE TYPE SUCTION STR
FIREWATER HOSEREEL WITH FORM DIP TUBE
FIREWATER HOSEREEL
AFFF
FOAM/FIREWATER HOSEREEL (FH–XXXX)
SPECIFICATION BREAK
A
B

Symbol – Line designation

- Line designations in a P&ID include:
  - Pipe size
  - Letters that represent the material/fluid flowing in the pipe
  - Identification numbers, usually related to the origin of the flow.

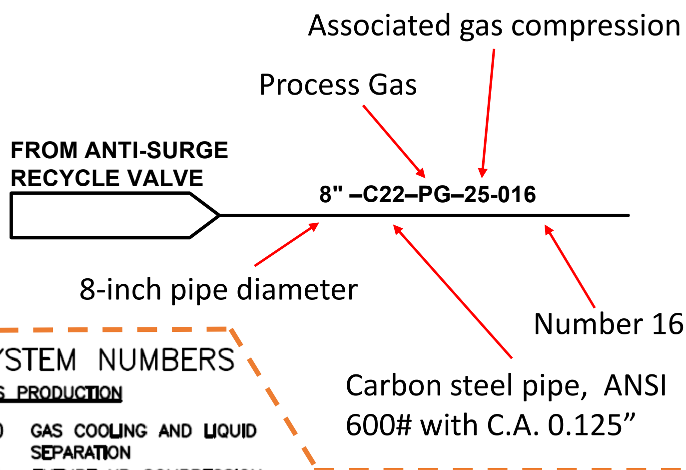


- The following P&ID depicts physically connected and disconnected piping.
- The piping is depicted as a continuous line.
- The 6-inch diameter pipe carries WMUN – Municipal water.
- The upper nozzle of the tank is connected to a 3-inch WWH-48 pipe, and the lower nozzle is a 4-inch in size and connected to a 4-inch WWH-50 pipe.
- The fluid flowing through the 3-inch (tank inlet) and 4-inch (tank outlet) pipes is wash water.
- The fluid flowing through the 4-inch PP-9 pipe is propane.

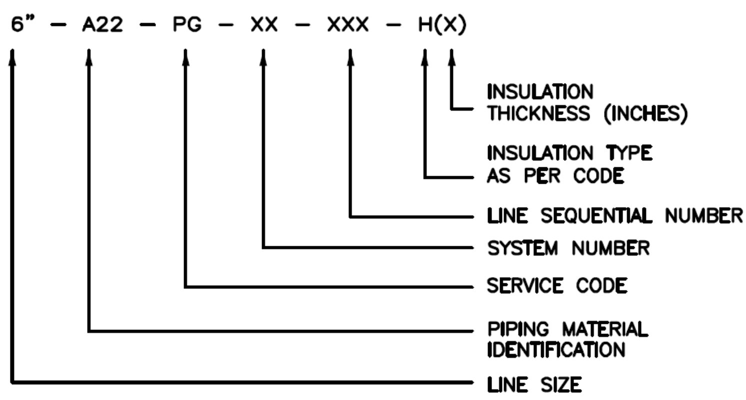


Piping service:
- WMUN = Municipal water
- WWH = Wash water
- PP = Propane

Symbol – Line designation



LINES SHALL BE IDENTIFIED THUS:



SYSTEM NUMBERS

GAS PRODUCTION

20	GAS COOLING AND LIQUID SEPARATION
21	FUTURE XP COMPRESSION
22	GAS TREATMENT
23	
24	DEG PACKAGE
25	ASSOCIATED GAS COMPRESSION
26	EXPORT GAS COMPRESSION
27	TRAIN 2 EGC (FUTURE)
28	GAS METERING
29	GAS EXPORT
30	GAS REINJECTION
31	

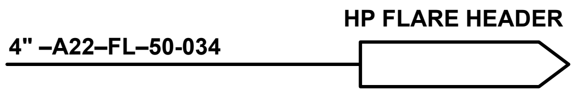
LINE CLASSES AND MATERIALS

CLASS	RATING	CORROSION ALLOWANCE (CA)	PIPING MATERIAL	SERVICE CODE
C22	600#	0.125"	CS	PG, PL, SG

SERVICE	SERVICE ABBREVIATION
PROCESS LINES	
PROCESS LIQUID	PL
PROCESS VAPOUR	PV
PROCESS GAS	PG

Symbol – Line and Valve designation

What kind of this pipe?



SYSTEM NUMBERS

<u>FLARE, VENTS AND DRAINS</u>

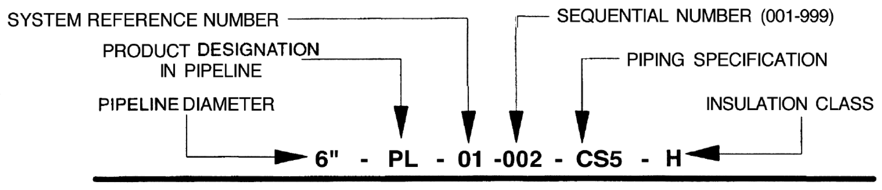
50	HP FLARE
51	LP FLARE
52	ATMOSPHERIC VENT
53	
54	
55	CLOSED DRAINS
56	HAZARDOUS OPEN DRAINS
57	NON–HAZARDOUS OPEN DRAINS
58	
59	

SERVICE	SERVICE ABBREVIATION
BLOWDOWN	BD
DRAIN, CLOSED	DC
DRAIN, OPEN	DO
FLARE	FL
SAFETY VALVE DISCHARGE	SV
VENT GAS	VG

LINE CLASSES AND MATERIALS

CLASS	RATING	CORROSION ALLOWANCE (CA)	PIPING MATERIAL	SERVICE CODE
A22	150#	0.125"	CS	PG, PL, DC, DO, FL, FG, DF, N, BG, R, SG, VG
A23	150#	0.250"	CS	PG, FL, PL, FG VG
A25	150#	0.125"	LTCS	BD, FL
A26	150#	–	CPVC	HY
A27	150#	0.0	316L SS	PG, BD, JF, FE FL, LO, AI, AU
A28	150#	0.0	GRE	FW, DE, SW, CW, WU, SR, WP, AI, AU, N
A29	150#	0.0	COPPER NICKEL (90%–10%)	FW, SW, WP
B22	300#	0.125"	CS	PG, PL, SG, FG
B23	300#	0.250"	CS	PG, PL, FG, FL
B25	300#	0.125"	LTCS	BD, FL
B27	300#	0.0	316L SS	PG, BD, FG, FE
C22	600#	0.125"	CS	PG, PL, SG

Symbol – Line designation - alternate



Illustrates a 6", carbon steel, process liquids pipeline, insulated for full heat conservation.

Piping service:
HC = Hydrocarbons
PL = Process Liquid
PO = Produced Oil
PG = Process Gas
FG = Fuel Gas
IA = Instrument Air
PA = Plant Air
CW = Cooling Water
SW = Seawater
HM = Heating Medium

- Pipe specification identification methods can be quite complex due to the wide variety of pipe materials and pressure ratings.
- A commonly used system is one where a **letter** indicates the pipe material and a specific number indicates the pressure rating.
- Example:
  - CS1 = Carbon Steel - Schedule 40
  - SS2 = Stainless Steel - Schedule 80
  - CU1 = Copper - Schedule 40

Symbol – Valve

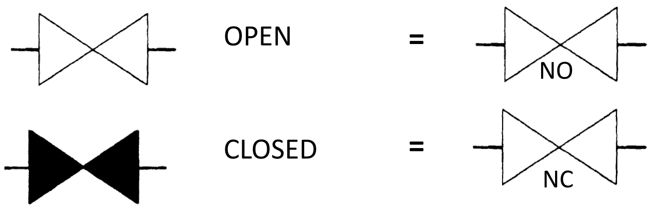
- The three services performed by a valve are:
  - ON/OFF SERVICE: The valve must ensure full flow through it when fully open, and there must be no passing when fully closed.
  - CONTROL SERVICE: The valve must be able to control fluid flow through it according to its design. When the valve is fully closed, it must be free from passing.
  - ONE-WAY SERVICE: Valves are required to ensure that flow is maintained in only one direction. The valve must allow the smooth flow of fluid in the desired direction but block/shut off the flow in the opposite direction - without any passing.
- On most P&IDs, each valve type is assigned a different symbol.
- The valve type selected depends on the operating conditions, product, and service type.
- Other factors such as cost, weight, and maintenance requirements will also be considered.

Symbol – Valve

- In some cases, the various valve types are not indicated on the P&ID. When this occurs, a generic valve symbol, as shown below, is sometimes used.

- A valve that is normally in the OPEN position will generally not be colored.
- Alternatively, the letters "NO" (normally open) are written near the valve.

- A valve that is normally in the CLOSED position will generally be colored (usually black).
- Alternatively, the letters "NC" (normally closed) are written near the valve.

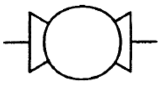


Symbol – Valve – which is used for On/Off service

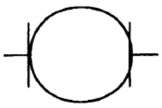
Wedge Gate Valve

Wedge Gate Valve

- The most common valve and used at all pressures.
- Not suitable for dirty flow services because debris can damage the sealing surface or collect on the bottom of the valve, preventing full valve closure.
- Gate valves should not be used for **control services** because flow across the valve will cut the sealing surface.
- Note: The gate valve symbol can be used as a generic symbol for all valve types.

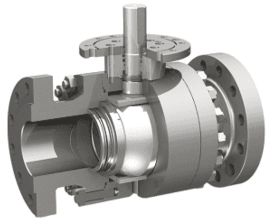


Ball Valve

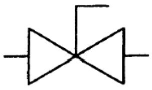


Ball Valve

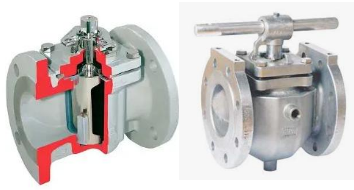
- Used at all operating pressures.
- Features a sealant injection point to improve valve sealing.
- Not suitable for dirty flow services as debris can damage the seal.
- The special internal design allows the valve to be used to control flow with relatively low pressure drop.

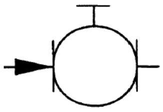


Symbol – Valve – which is used for On/Off service

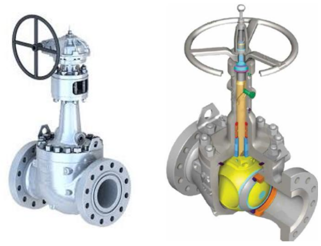

Plug Valve

- Typically used in medium and low pressure operating services.
- It features a sealant injection point to improve valve sealing.
- This valve is not suitable for dirty flow services as debris can damage the seal.




Orbit Valve

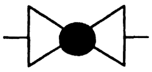
- Typically used in high-pressure service.
- It is a derivative of the ball valve, featuring a rotary/sliding action that forces the ball (inside the valve) against the seal surface.
- It performs well in less clean flow service.




Diaphragm Valve

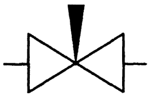
- Typically used in dirty, low-pressure service.
- Care must be taken not to overtighten the valve and damage the flexible diaphragm.

Symbol – Valve – used for control

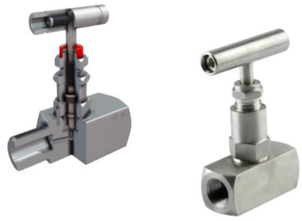

Globe Valve

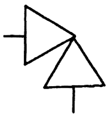
- Used at all operating pressures.
- The most common valve type for control services.
- Suitable for dirty fluid flows.
- Different internal designs can handle all operating and pressure requirements.


Needle Valve

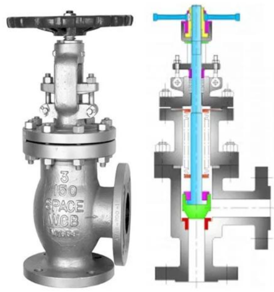

Needle Valve

- Used at all operating pressures.
- A derivative of the globe valve.
- Used for very small flow control (e.g., sample points).
- Not suitable for dirty service.

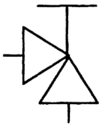


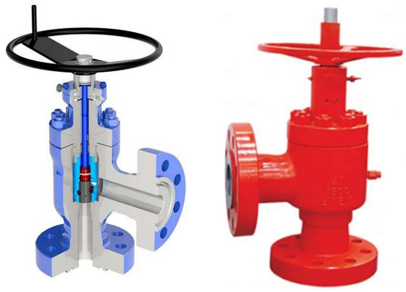

Angle Valve

- Primarily used at high operating pressures.
- A derivative of the globe valve.
- Reduced turbulence within the valve results in better flow than a globe valve.

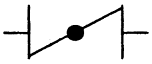


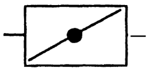
Symbol – Valve – used for control


Choke Valve



- Used for high pressure drop services.
- A derivative of the angle valve.
- Widely used to control flow from the well to the plant/facility.


Butterfly Valve

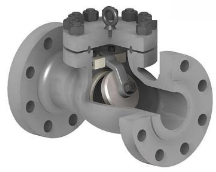

Butterfly Valve

- Primarily used in low-pressure and low-pressure-drop applications.
- Some designs are one-way to improve sealing performance.
- Not reliable for tight closures.
- Commonly found in firewater header applications.

Symbol – Valve – used for one-way/non-return services

- Valves used for one-way operation are called **check valves, non-return valves**, or **one-way valves**.
- These valves ensure free fluid flow in the desired direction and block fluid flow in the opposite direction.
- Types of valves include:

- Swing check valve.
  - This valve contains a hinged, flat circular plate. The plate lifts to allow fluid flow throug plate in one direction.
  - The plate then closes, stopping fluid flow in the opposite direction.
  - This valve is the most common type of check valve.



- Titling plate check valve.
  - Contains a flat circular plate hinged slightly offset from the center position.
  - The offset hinge position causes the valve to open in one direction of flow and close in the opposite direction.
  - Typically used for gas flows with high operating pressures and high flow rates.

Symbol – Valve – used for one-way/non-return services

- Ball check valve.
  - Contains a freely moving ball enclosed in a cage.
  - The ball lifts off the seat to allow fluid flow and falls back into the seat when the flow is reversed.
  - Valves are used for liquid fluids at low flow rates.

- Piston check valve.
  - Contains a piston that moves freely up and down within a cage.
  - The piston lifts off the seat to allow fluid flow and falls back into the seat when the flow is reversed.
  - Used for liquid fluids at low flow rates and high operating pressures.

- All types of valves can be spring-loaded to aid their function.
- Swing check valves can also be equipped with: a device that allows the valve to be screwed down to improve its sealing ability, a hydraulic damper that prevents sudden closing.

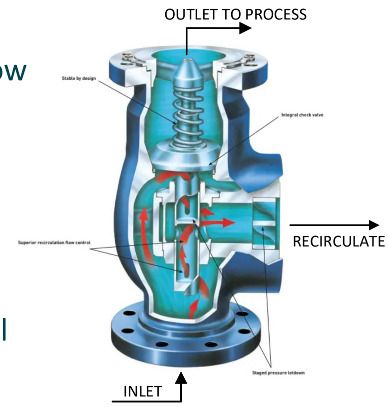
Symbol – Valve – used for one-way/non-return services

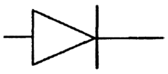
Check Valve

- The direction of the arrow shows the direction of flow from left to right

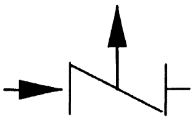
Check Valve

- Alternative symbol. The arrow indicates the direction of flow from left to right.



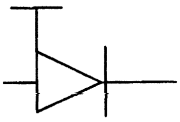


Check Valve



Check Valve with Recycle Facility

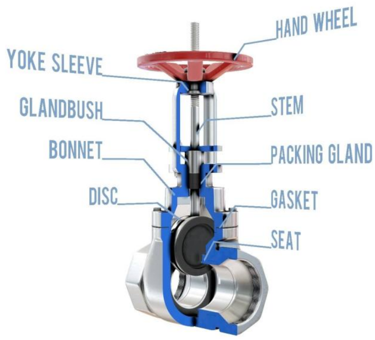
- The main flow is shown from left to right.
- The recycle flow is indicated by a vertical arrow – which is typically used for minimum service flow rates of centrifugal pumps.

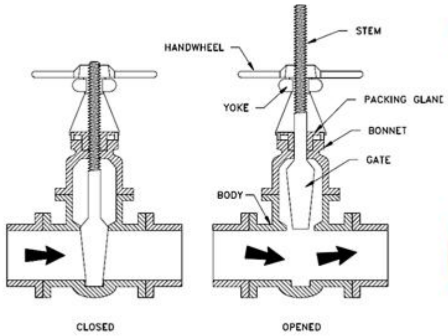


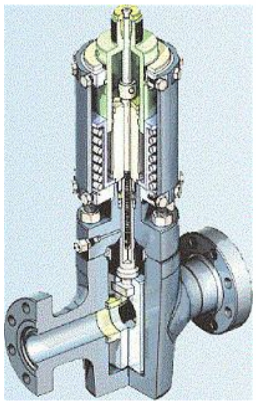
Check Valve with "Screwed Down" Facility

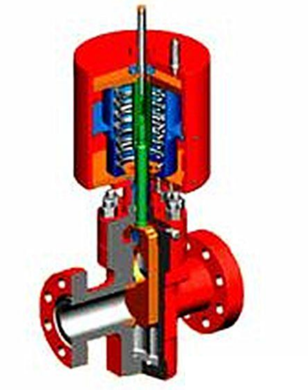
- The check valve can be screwed down to improve sealing.
- Installed on high-pressure pump or compressor installations with a shared header.
- The flow shown is from left to right.

Symbol – Specific Valve – Reverse Gate type



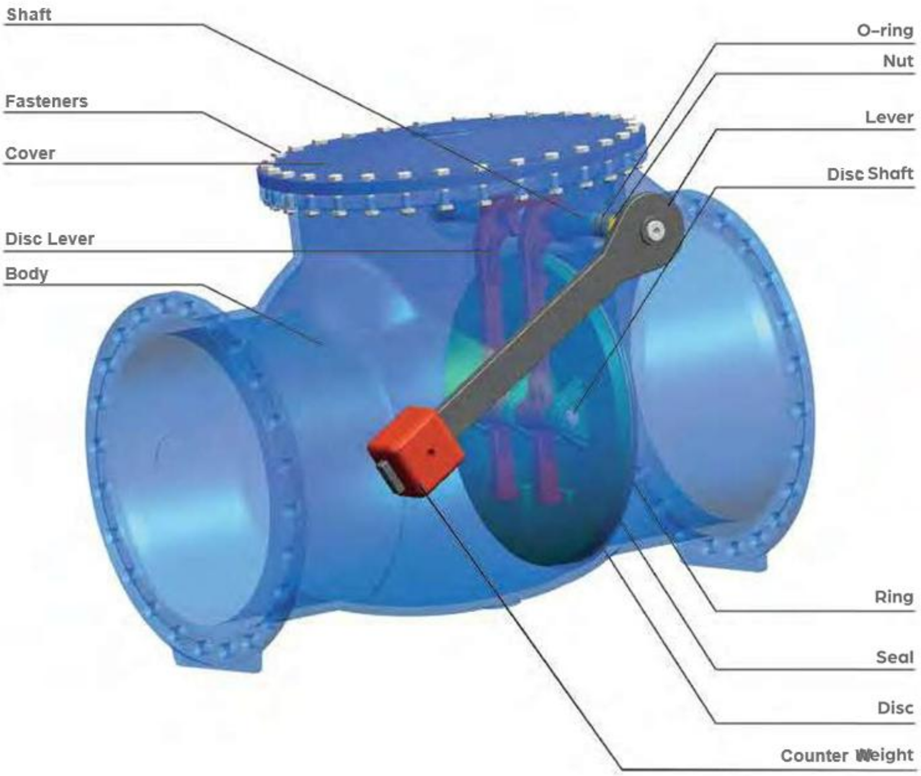






OPENED **CLOSED**

- The image above and to the left shows an actuated 'Reverse Gate Valve.'
- It's called reverse because when closed, the stem protrudes upward (and the stem enters the valve) when open.
- By closing the gate disc, sand cannot fill the space below the disc. This makes it relatively reliable as a reverse gate valve.


Shaft
Fasteners
Cover
Disc Lever
Body
O-ring
Nut
Lever
Disc Shaft
Ring
Seal
Disc
Counter Weight

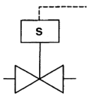
Symbol – Valve

How are valves controlled?

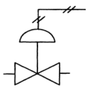
- Controlled by humans/workers – called manual valves
- Valve is also controlled automatically:

Globe valve

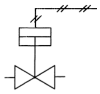
Gate valve



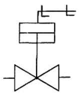
Solenoid actuated valve
The valve will open and close when an electrical signal is applied to the actuator.



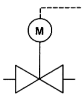
Pneumatic diaphragm valve
A diaphragm inside the actuator is operated by a pneumatic (air) signal.



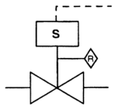
Piston actuated valve
Valves are controlled by pneumatic signals.



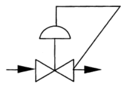
Hydraulic piston actuated valve
The controlling signal is hydraulic fluid.



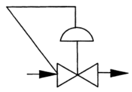
Electric motor actuated valve
The valve is controlled by an electric motor.



Solenoid actuated valve with Reset
Solenoid valve with local reset facility to allow signal to be returned to the valve.



Self acting PCV with downstream tap
PCV uses fluid flow as a signal to the diaphragm to control downstream pressure.



Self acting PCV with upstream tap
Similar to the valve above but to control upstream/upstream pressure

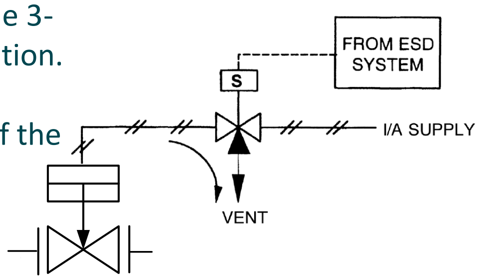
Symbol – Valve

How are valves controlled?

Symbol – Actuated Valve

Under normal operating conditions:

- The ESD system ensures that there is an electrical power supply to the 3-way solenoid valve, and the valve is energized and in the normal position.
- The instrument air supply (I/A) is routed to the diaphragm shutdown valve (SDV) via a 3-way solenoid valve (the normally-closed section of the solenoid will usually be shaded/blacked out and the section showing normal flow through the two open sections will be unshaded).
- SDV is in the open (normal) position.

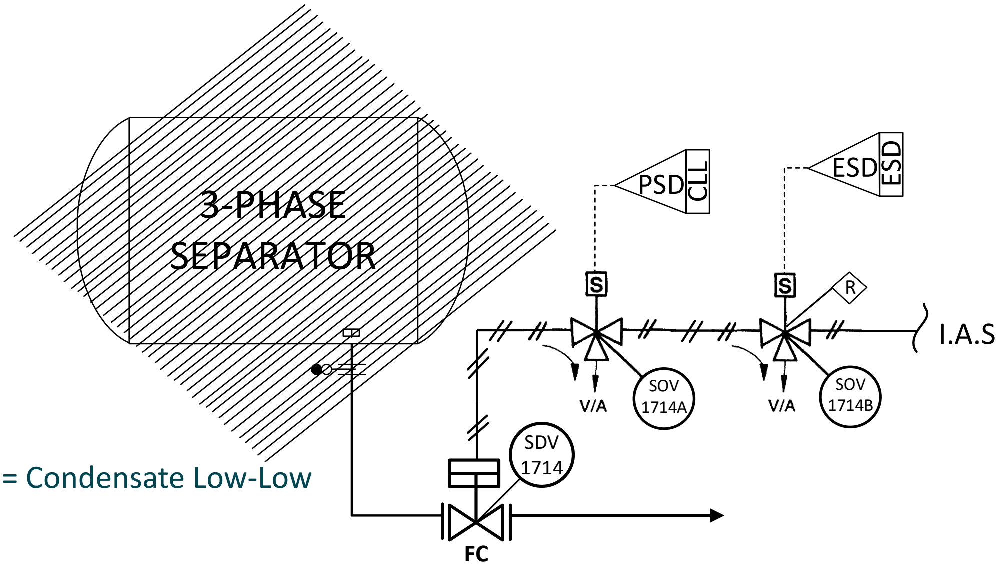


Manual Actuating Element

If the ESD system is activated:

- The ESD system de-energizes/cuts power to the 3-way solenoid valve. The solenoid valve is de-energized and moves to the fail position.
- The solenoid valve changes position to: - shut off the air supply from the instrument air system, and - bleed instrument air from the SDV diaphragm actuator (the curved arrow indicates the air route when the solenoid valve is in the fail position).
- The SDV moves to the fail position (in the example, the SDV will fail to the CLOSED position, as indicated by the downward-pointing arrow – Fail Closed).

Symbol – Actuated Valve

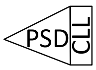


Note:

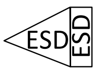
PSD = Process Shutdown

ESD = Emergency Shutdown

CLL = Condensate Low-Low

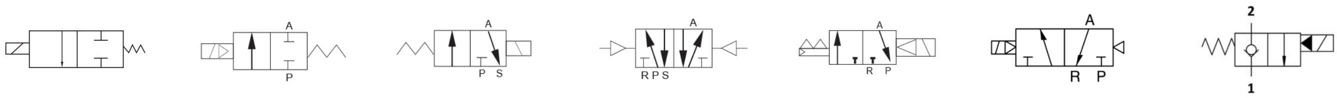


Process Shutdown
Shutdown level = CLL = Condensate Low-Low



Emergency Shutdown
Shutdown level = ESD = Emergency Shutdown

Can anyone explain??



Answer:

Under normal operating conditions:

- The ESD **AND** PSD systems ensure electrical power supply/energizing to the two 3-way solenoid valves (SOV-1741A & 1741B).
- The instrument air supply (I/A) is routed to the SDV-1741 diaphragm valve via the two solenoid valves.

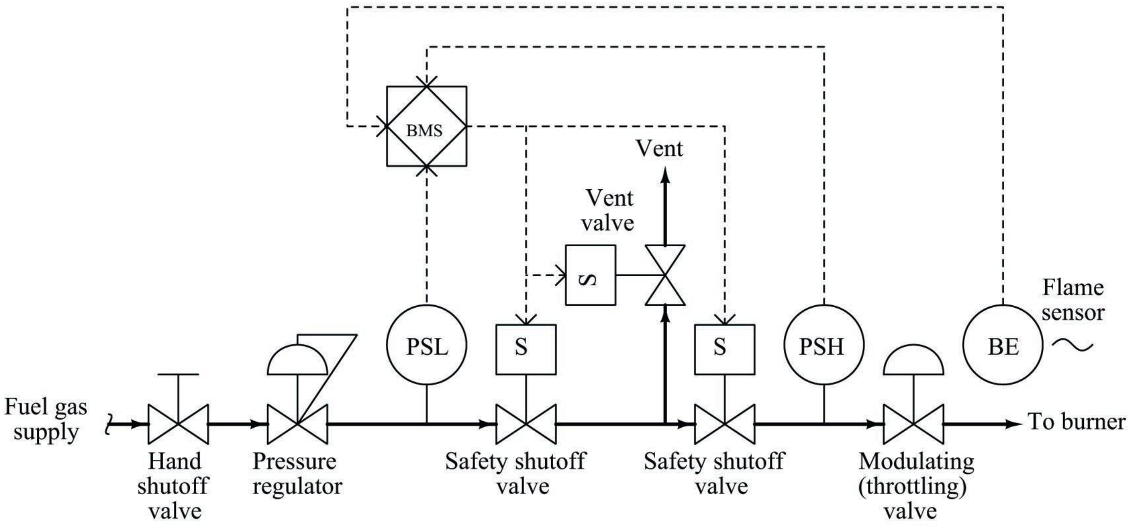
If the ESD system is activated:

- The ESD system de-energizes/cuts power the SOV-1741B and the valve moves to the fail position.
- The SOV-1741B changes position to: - shut off the air supply from the instrument air system, and - bleed instrument air from the SDV-1714 diaphragm actuator.
- The SDV-1714 moves and closes (because the SDV is Fail Closed).
- To reactivate the SDV-1714, the **Reset lever must be pulled.**

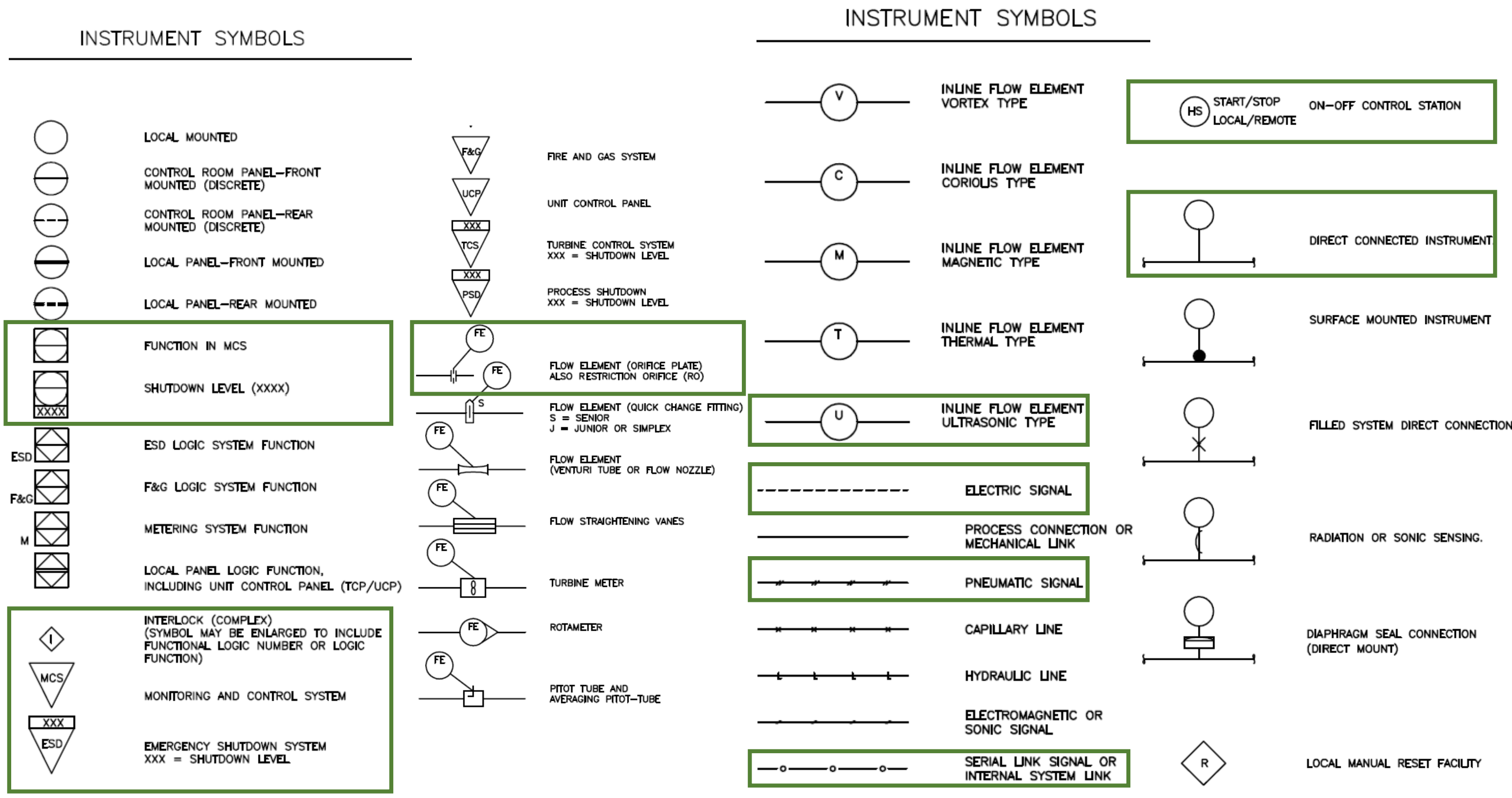
If the PSD system is activated:

- The PSD system cuts power to the SOV-1741A, and the valve moves to the fail position.
- The SOV-1741A changes position to: - shut off the air supply from the instrument air system, and - bleed instrument air from the SDV-1714 diaphragm actuator.
- The SDV-1714 moves and closes.

Symbol – Actuated Valve



Can anyone explain??


INSTRUMENT SYMBOLS
LOCAL MOUNTED
CONTROL ROOM PANEL–FRONT MOUNTED (DISCRETE)
CONTROL ROOM PANEL–REAR MOUNTED (DISCRETE)
LOCAL PANEL–FRONT MOUNTED
LOCAL PANEL–REAR MOUNTED
FUNCTION IN MCS
XXXX
SHUTDOWN LEVEL (XXXX)
ESD
ESD LOGIC SYSTEM FUNCTION
F&G
F&G LOGIC SYSTEM FUNCTION
M
METERING SYSTEM FUNCTION
LOCAL PANEL LOGIC FUNCTION, INCLUDING UNIT CONTROL PANEL (TCP/UCP)
I
INTERLOCK (COMPLEX) (SYMBOL MAY BE ENLARGED TO INCLUDE FUNCTIONAL LOGIC NUMBER OR LOGIC FUNCTION)
MCS
MONITORING AND CONTROL SYSTEM
XXX
ESD
EMERGENCY SHUTDOWN SYSTEM XXX = SHUTDOWN LEVEL
F&G
FIRE AND GAS SYSTEM
UCP
UNIT CONTROL PANEL
XXX
TCS
TURBINE CONTROL SYSTEM XXX = SHUTDOWN LEVEL
XXX
PSD
PROCESS SHUTDOWN XXX = SHUTDOWN LEVEL
FE
FE
FLOW ELEMENT (ORIFICE PLATE) ALSO RESTRICTION ORIFICE (RO)
S
FLOW ELEMENT (QUICK CHANGE FITTING) S = SENIOR J = JUNIOR OR SIMPLEX
FE
FLOW ELEMENT (VENTURI TUBE OR FLOW NOZZLE)
FE
FLOW STRAIGHTENING VANES
FE
TURBINE METER
FE
ROTAMETER
FE
PITOT TUBE AND AVERAGING PITOT–TUBE
INSTRUMENT SYMBOLS
V
INLINE FLOW ELEMENT VORTEX TYPE
C
INLINE FLOW ELEMENT CORIOLIS TYPE
M
INLINE FLOW ELEMENT MAGNETIC TYPE
T
INLINE FLOW ELEMENT THERMAL TYPE
U
INLINE FLOW ELEMENT ULTRASONIC TYPE
ELECTRIC SIGNAL
PROCESS CONNECTION OR MECHANICAL LINK
PNEUMATIC SIGNAL
CAPILLARY LINE
HYDRAULIC LINE
ELECTROMAGNETIC OR SONIC SIGNAL
SERIAL LINK SIGNAL OR INTERNAL SYSTEM LINK
HS
START/STOP LOCAL/REMOTE
ON–OFF CONTROL STATION
DIRECT CONNECTED INSTRUMENT
SURFACE MOUNTED INSTRUMENT
FILLED SYSTEM DIRECT CONNECTION
RADIATION OR SONIC SENSING.
DIAPHRAGM SEAL CONNECTION (DIRECT MOUNT)
R
LOCAL MANUAL RESET FACILITY

ABBREVIATIONS

A/S	INSTRUMENT AIR SUPPLY
ADM	ADMITTANCE PROBE
ARV	AIR RELEASE VALVE
BDV	BLOWDOWN VALVE
CA	CORROSION ALLOWANCE
CC	CORROSION COUPON
CP	CORROSION PROBE
CSO	PRO LOCK TYPE CLOSE
DB & B	DOUBLE BLOCK AND BLEED
DIS	DISPLACER TYPE
DP	DIFFERENTIAL PRESSURE TYPE
ES	EMERGENCY STOP
ESD	EMERGENCY SHUTDOWN
FB	FULL BORE
FC	FAIL CLOSED
F&G	FIRE AND GAS SYSTEM
FL	FAIL LOCKED
FO	FAIL OPEN
FP	FULL PORT
HH	HANDHOLE
HV	HAND VALVE
HPU	HYDRAULIC POWER UNIT
IAS	INSTRUMENT AIR SUPPLY
I/F	INTERFACE
LO	PRO LOCK TUPE OPEN
LC	PRO LOCK TYPE CLOSE
MVG	MANIFOLD GAUGE VALVE
HTR	HEATER
DCV	DIRECTION CONTROL VALVE
CPLG	COUPLING
ABE	ACTUATED BALL VALVE
RD	RUPTURE DISC
MCC	MOTOR CONTROL CENTRE
MCS	MONITORING AND CONTROL SYSTEM
MH	MANWAY
MOV	MOTOR OPERATED VALVE
NC	NORMALLY CLOSED
NNF	NORMALLY NO FLOW
NO	NORMALLY OPEN
PLEM	PIPELINE END MODULE
PN	PANEL NUMBER
PSV	PRESSURE RELIEF/SAFETY VALVE
PVSV	PRESSURE VACUUM SAFETY VALVE
RP	REDUCED PORT/REDUCED BORE
RCU	REMOTE CONTROL UNIT
RO	RESTRICTION ORIFICE
RDR	RADAR PROBE
SB & B	SINGLE BLOCK AND BLEED
SCSSV	SURFACE CONTROLLED SUBSURFACE SAFETY VALVE
SDV	SHUTDOWN VALVE
SSIV	SUBSEA ISOLATION VALVE
SSV	SURFACE SAFETY VALVE
TSS	TOTAL SUSPENDED SOLIDS
TYP	TYPICAL
UA	UNIT ALARM
UCP	UNIT CONTROL PANEL
VF	VENDOR FURNISHED
VSV	VACUUM SAFETY VALVE
WHCP	WELLHEAD CONTROL PANEL
WL	DIFFERENTIAL PRESSURE WITH WET LEG
XMP	AUTOMATIC SAMPLING SYSTEM
XV	MANUAL VALVE WITH LIMIT SWITCH
XDV	TRAIN ISOLATION VALVE

Symbol – Instrumentation

- The P&ID legend for instrumentation includes abbreviations for instrument designations or instrument tags.
- The legend table is usually divided into sections based on the process variables monitored by the instrument.
- Process variables are physical characteristics of the process that can change.
- The process variables we will discuss are temperature, pressure, level, and flow.

The ISA S5.1 standard contains tables and examples of abbreviations used in P&ID.

- The first letter on an instrument's label stands for the **process variable** being measured. For example, "P" stands for pressure, "T" for temperature, "L" for level, and "W" for weight.
- The second letter stands for **function**. For example, "V" stands for valve, so "PV" stands for pressure control valve. Another example: "IT" stands for indicating transmitter, so "FIT" stands for flow indicating transmitter.

Flow

FAL	Flow alarm low
FAH	Flow alarm high
FE	Flow element (orifice, nozzle, etc.)
FFIC	Flow ratio indicating controller
FFRC	Flow ratio recording controller
FG	Flow glass
FI	Flow indicator
FIC	Flow indicating controller
FIS	Flow indicating switch
FIT	Flow indicating transmitter
FO	Flow restriction orifice
FR	Flow recorder
FRC	Flow recording controller
FS	Flow switch (FSL=low, FSH=high, FSHL = high & low)
FQI	Flow totalizing indicator
FQIC	Flow totalizing indicating controller
FQS	Flow totalizer switch
FQT	Flow totalizer transmitter
FQSH	Flow pressure switch
FV or FCV	Flow control valve
FX	Flow strainer and vent
FY	Flow solenoid valve, relay or converter

Level

LA	Level alarm
LAL	Level alarm low
LAH	Level alarm high
LC	Level controller
LCV	Level control valve self-operated
LG	Level glass (gauge glass)
LI	Level indicator
LIC	Level indicating controller
LLL	Level light low
LLH	Level light high
LR	Level recorder
LRC	Level recording controller
LS	Level switch
LSL	Level switch low
LSH	Level switch high
LT	Level transmitter
LIT	Level indicating transmitter
LY	Level solenoid valve, relay or converter

Temperature

TA	Temperature alarm
TAH	Temperature alarm high
TAL	Temperature alarm low
TCV	Temperature control valve self-operated
TE	Temperature element, thermocouple, resistance bulb, or thermopile
TI	Temperature indicator
TIC	Temperature indicating controller
TIS	Temperature indicating switch
TIT	Temperature indicating transmitter
TR	Temperature recorder
TRC	Temperature recording controller
TS	Temperature switch (TSL=low, TSH=high, TSHL = high & low)
TT	Temperature transmitter
TV	Temperature control valve
TW	Temperature well
TY	Temperature solenoid valve, relay or converter

Pressure or Vacuum

P	Pressure test point
PA	Pressure alarm
PAL	Pressure alarm low
PAH	Pressure alarm high
PCV	Pressure control valve self-operated
PV	Pressure control valve
PDI	Pressure differential indicator
PDIC	Pressure differential indicating controller
PDIT	Pressure differential indicating transmitter
DPC	Differential pressure control
PDR	Pressure differential recorder
PI	Pressure indicating gauge or manometer
PIC	Pressure indicating controller
P	Pressure test point
PQIT	Pressure differential indicator
PR	Pressure recorder
PRC	Pressure recording controller
PS	Pressure switch (PSL=low, PSH=high, PSHL = high & low)
PSV	Pressure safety relief valve
PSE	Rupture disc or vacuum breaker or emergency vent valve
PT	Pressure transmitter (blind)
PIT	Pressure indicating transmitter
PY	Pressure solenoid valve, relay or converter

Miscellaneous

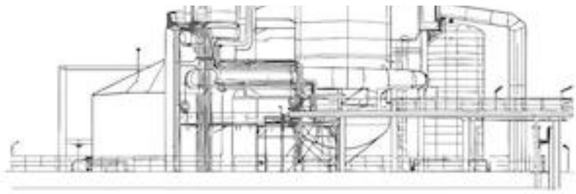
Alarm	Alarm annunciators
BA	Gas burner alarm (no flame)
BC	Burner controller
BV	Burner control valve
BY	Burner solenoid valve
FO	Restriction orifice
HC	Manual controller
HIC	Manual indicating controller
HS	Manual switch
HV	hand valve
II	Current indicator
IT	Current transmitter
JV	Power control valve
PSV	Relief valve, safety valve
PSE	Rupture disc or vacuum breaker
ZI	position indicator

Symbol – Instrumentation

HV	Pressure safety relief valve
PSV	Level interlock solenoid valve
LT	Flow switch
FAL	Manual (hand) control valve
TIC	Temperature indicating controller
LY	Level transmitter
FS	Flow alarm low

Where is the partner?

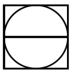
HV	Pressure safety relief valve
PSV	Level interlock solenoid valve
LT	Flow switch
FAL	Manual (hand) control valve
TIC	Temperature indicating controller
LY	Level transmitter
FS	Flow alarm low

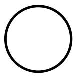


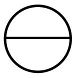
Symbol – Instrumentation

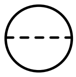
- In general, the letter and number abbreviations that identify an instrument in a P&ID are symbolized by circles.
- The differences between each circle are explained in the legend, often along with its instrument tag.
- Circle symbols typically represent an instrument and also indicate its location within the plant.

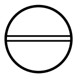
Note:
BPCS –Basic Process Control System
SIS – Safety Instrumented System
CSS – Computer System & Software

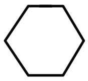
BPCS, shared display shared control, mounted on the central or main panel 

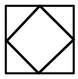
Instruments installed in the field 

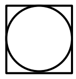
Instruments mounted on a central or main panel 

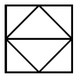
Instruments mounted behind the central or main panel. Usually not accessible to the operator. 

Additional instruments mounted on the local panel. Accessible by the operator. 

Computer controlled function, installed in the field  CSS

PLC, shared display share control, installed in the fied  SIS

BPCS, shared display shared control, installed in the field 

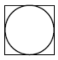
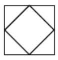
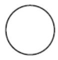
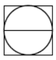
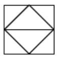
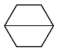
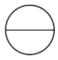
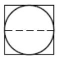
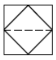
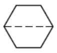
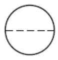
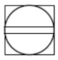
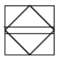
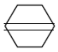
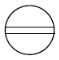
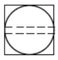
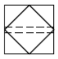
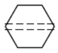
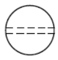
PLC, shared display shared control, mounted on the central or main panel  SIS

Symbol – Instrumentation

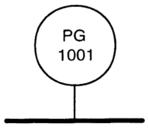
Symbol example – taken from ANSI/ISA-5.1-2009.

Table 5.1.1 — Instrumentation device and function symbols

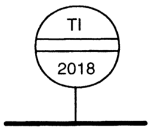
Note: Numbers in parentheses refer to explanatory notes in Clause 5.3.1.

No.	**Shared display, Shared control (1)**		**C**	**D**	**Location & accessibility (6)**
	A	**B**			
	Primary Choice or Basic Process Control System (2)	**Alternate Choice or Safety Instrumented System (3)**	**Computer Systems and Software (4)**	**Discrete (5)**	
1					• Located in field. • Not panel, cabinet, or console mounted. • Visible at field location. • Normally operator accessible.
2					• Located in or on front of central or main panel or console. • Visible on front of panel or on video display. • Normally operator accessible at panel front or console.
3					• Located in rear of central or main panel. • Located in cabinet behind panel. • Not visible on front of panel or on video display. • Not normally operator accessible at panel or console.
4					• Located in or on front of secondary or local panel or console. • Visible on front of panel or on video display. • Normally operator accessible at panel front or console.
5					• Located in rear of secondary or local panel. • Located in field cabinet. • Not visible on front of panel or on video display. • Not normally operator accessible at panel or console.

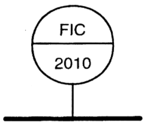
Symbol – Instrumentation – take a guess!!



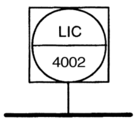
Pressure gauge installed in the field, number 1 on the 10/wellhead system.



Temperature indicator installed on the local panel, number 18 on system 20/ gas cooling & liquid separation.



Flow indicating controller installed on the central/main panel, number 10 on system 20/gas cooling and liquid separation.



Computerized level indicating controller, number 02 on the 40/fuel gas system.

SYSTEM NUMBERS

WELLS/OIL PRODUCTION

10	WELLHEADS
11	MANIFOLD
12	
13	DRILLING EQUIPMENT
14	
15	INTRA–FIELD IMPORT/EX
16	
17	OIL/GAS SEPARATION
18	
19	OIL METERING

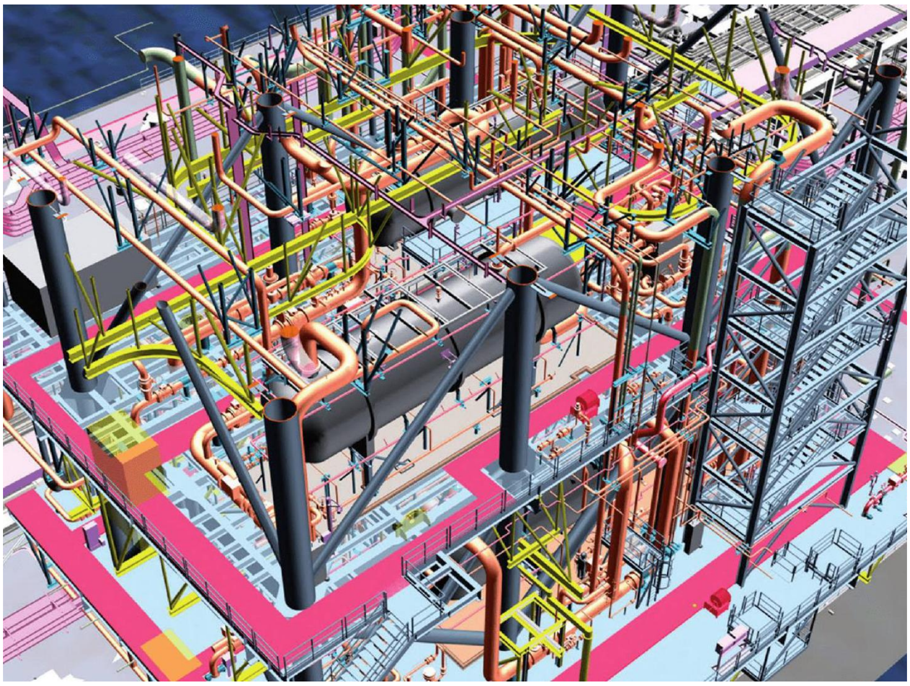
PROCESS UTILITIES

40	FUEL GAS
41	REFRIGERANT
42	METHANOL
43	SEAWATER/SERVICE WATER
44	COOLING WATER
45	
46	
47	
48	SEAL GAS
49	CHEMICAL INJECTION

GAS PRODUCTION

20	GAS COOLING AND LIQUID SEPARATION
21	FUTURE XP COMPRESSION
22	GAS TREATMENT
23	
24	DEG PACKAGE
25	ASSOCIATED GAS COMPRESSION
26	EXPORT GAS COMPRESSION
27	TRAIN 2 EGC (FUTURE)
28	GAS METERING
29	GAS EXPORT
30	GAS REINJECTION
31	

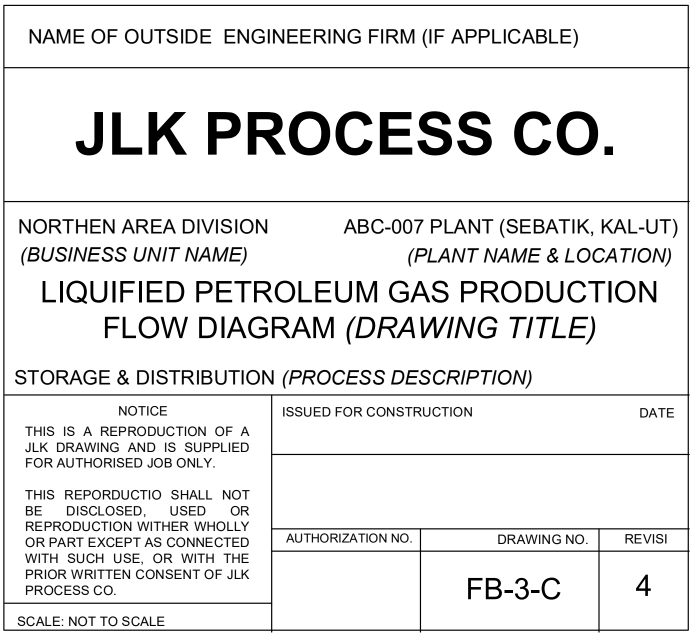
How to Read Piping & Instrumentation (P&ID) Diagram



Tracing the lines
Controlling Process

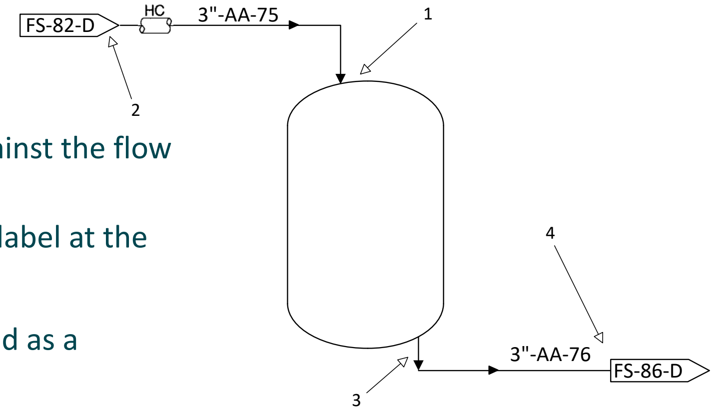
Tracing the line

- The entire process in a plant /production facility typically requires the depiction of more than one P&D.
- The P&IDs used must be correct and up-to-date.
- Each P&ID has an **identification number, area name,** and **issue number** in the Title Block in the lower right corner.
- The following title block shows that:
  - The drawing number is: **FB-3-C Revision 4**
  - The process system depicted is: **Liquified Petroleum Gas Production.**



- The P&ID we're currently reading might depict the beginning of a process flow, but it could also begin elsewhere in the plant (in another P&ID).
- If we need to trace the line/pipeline to find the beginning of the flow, we might need one or more other P&IDs.

Tracing the line



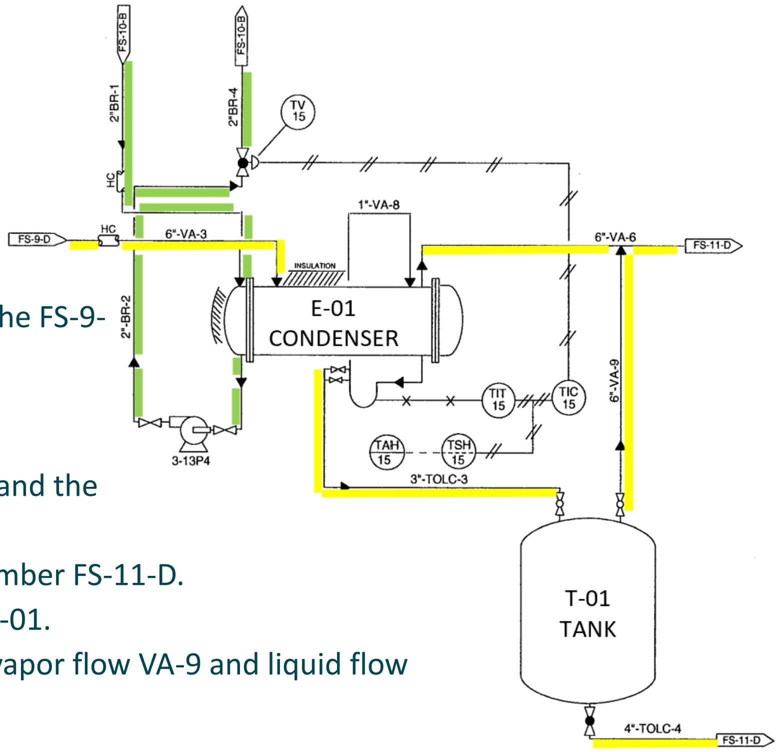
- The trace begins at arrow 1 and works backward against the flow direction of the arrow to the end/edge of the P&ID.
- Each flow/pipeline will have an identification tag or label at the point where it enters the P&ID – see arrow 2.
- The shape of the tag varies, on this slide it is depicted as a rectangle with pointed ends.
- This tag contains the previous P&ID number.
- Note that the incoming pipe is insulated and its number is written near the point where the flow enters the P&ID.
- The trace starts again at arrow 3 and progresses in the direction of the flow arrow. If the flow does not terminate at a device on the current P&ID, it will terminate at the edge of the P&ID.
- The end of the flow will be marked by a tag containing the next P&ID number—where we can continue tracing the flow—see arrow 4.
- Typically, in a P&ID, we will see many process flows and instrumentation. These process flows will enter and exit the P&D around the outer edges of the diagram.

Tracing the line

Note:

VA = Vapour; BR = Brine; TOL = Toluene

TOLC = Toluene-contaminated

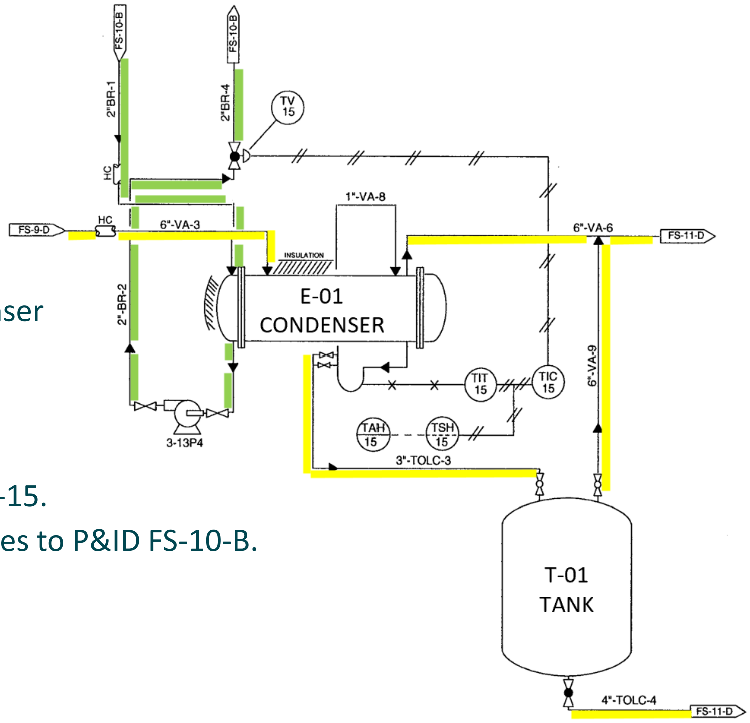


Yellow stream

- The VA-3 vapor stream enters this P&ID from the FS-9-D P&ID.
- This stream enters the E-01 condenser.
- The stream then exits the E-01 condenser and becomes two streams: the VA-6 vapor stream and the TOLC3 liquid stream.
- Flow VA-6 exits this P&ID and goes to P&ID number FS-11-D.
- Another flow (TOLC-3) goes to receiving tank T-01.
- From tank T-01, the flow exits into two parts: vapor flow VA-9 and liquid flow TOLC-4.
- Flow VA-9 joins flow VA-6.
- Flow TOLC-4 leaves the tank and goes to P&ID FS-11-D.

Tracing the line

Note:
VA = Vapour; BR = Brine; TOL = Toluene
TOLC = Toluene-contaminated



Green stream

- The BR-1 flow from P&ID FS-10-B goes to condenser E-01.
- The BR-1 flow from the condenser then goes to pump 3-13P4.
- The BR-2 flow exits pump 3-13P4 and goes to TV-15.
- From TV-15, the BR-4 flow exits this P&ID and goes to P&ID FS-10-B.

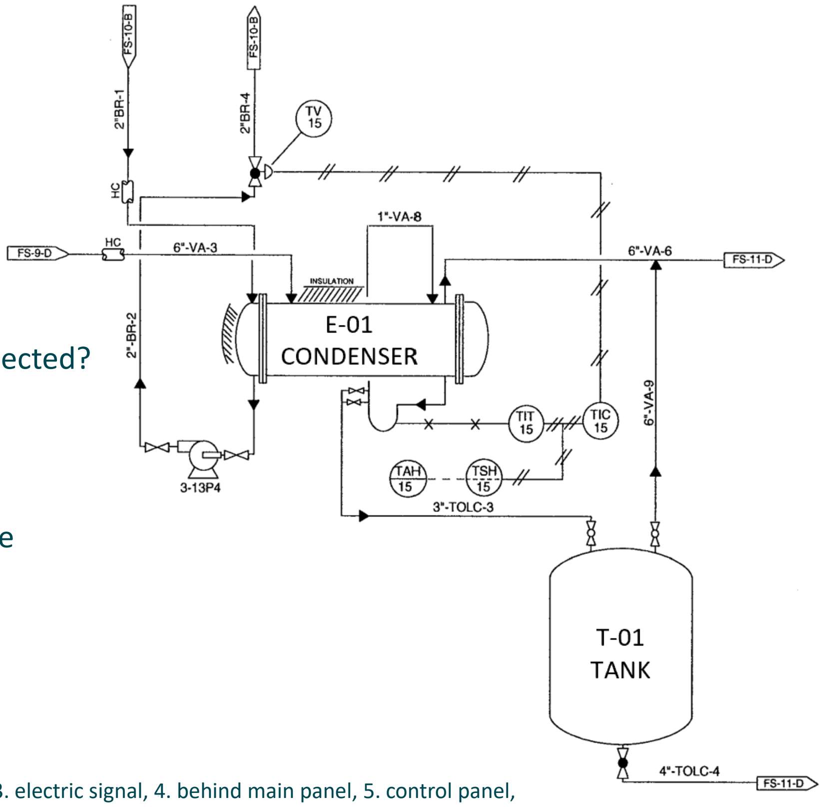
Process Control

- To ensure a process plant remains within a safe operating range, control loops are used to monitor process variables within a system.
- Process variables are physical characteristics of a process that can change, such as flow, pressure, temperature, level, and composition.
- P&ID drawings for an area will show the **lines, instrumentation, and equipment that monitor and control** each process.
- Instruments function to:
  - Monitor process variables.
  - Inform operators about the current condition of a process.
  - Allow process control.
- Instrument control loops are needed to:
  - Maintain process variables within safe ranges
  - Detect potential hazardous situations in progress
  - Provide a means to activate alarms and shut down the process if necessary
  - Maintain product compliance with standard specifications.

Process Control

Take a guess!!

1. What is the variable being monitored?
2. What does TIT stand for?
3. What type of line are TAH and TSH connected?
4. Where is the TSH instrument installed?
5. Where is the TAH instrument installed?
6. What type of line connects the TIT to the toluene-filled piping?



1. Temperature TOLC 3, 2. Temperature indicating transmitter, 3. electric signal, 4. behind main panel, 5. control panel,

6. capillary tubing.

Process Control – function of the control loop parts

Sensors and Transmitters:

- **This is the part of the loop that first detects or measures the process variable.**
- The sensor, labeled TE (temperature element) or TI (temperature indicator) on some P&IDs, can be separated or integrated with a transmitter, which transmits information about the variable.
- In the P&ID, the TIT-15 consists of a sensor and a transmitter.

Controller:

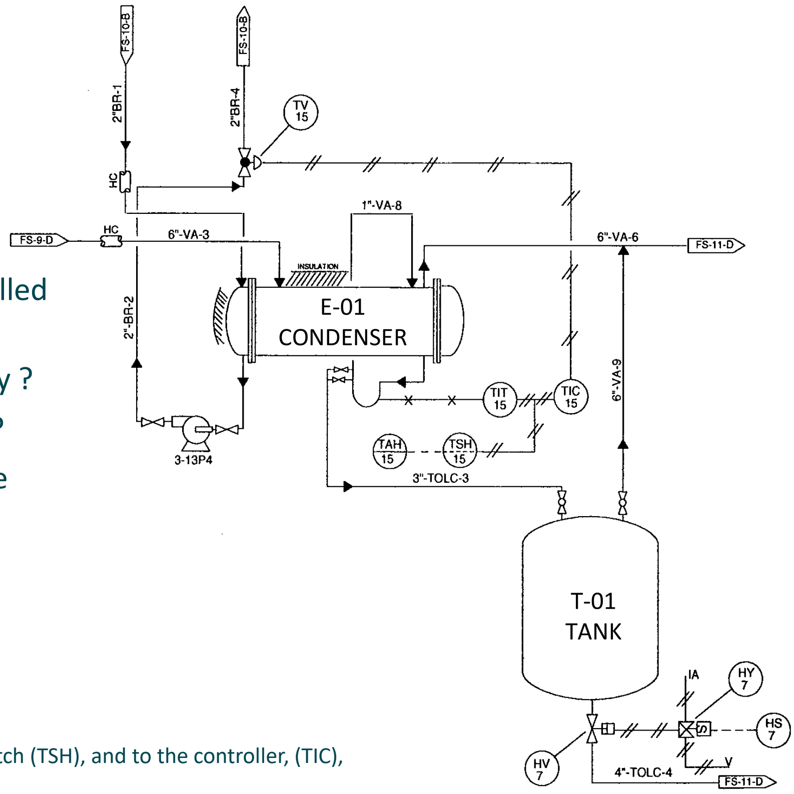
- **Information** (about the variables) **is sent from the TIT-15 to the controller, the TIC-15.**
- **Controller**, then sends a control signal to the **pneumatically operated diaphragm valve (TV-15).**
- **The TV-15 valve controls the brine flow rate** in the pipe.

Switch & Alarm:

- The toluene temperature (TOLC) must be maintained at a certain value.
- A switch, the TSH-15, **is connected to the transmitter. If the toluene temperature exceeds the set point, the switch sends a signal to the alarm, the TAH-15.**
- The alarm can be visual, such as a red light, or audible.
- The TAH-15 alarm symbol indicates that this instrument is located on the panel.

Process Control

Take a guess!!



1. The flowrate of stream TOLC-4 can be controlled by?
2. In control loop 7, the switch HS is actuated by ?
3. In control loop 15, TIT sends signal to where?
4. The valve TV, changes the temperature of the toluene stream, TOLC-3 by controlling the?

1. Piston-operated valve, 2. Hand, 3. To the alarm (TAH), to the switch (TSH), and to the controller, (TIC),

4. Brine flow rate - output from condenser

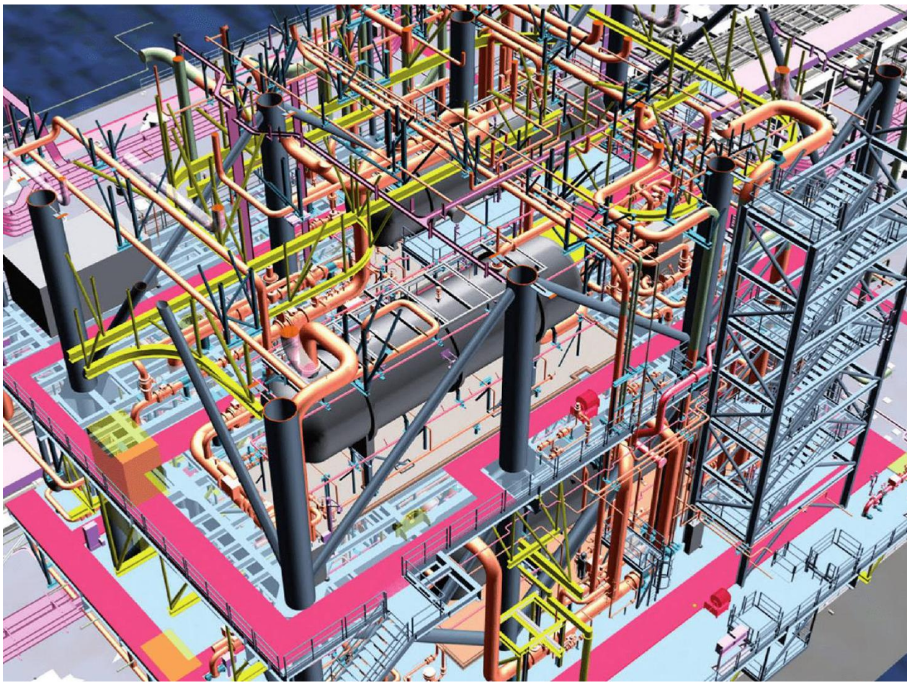
Process Control

Understanding the process controls in a P&ID will make it easier for the reader to understand the process.

This is a key foundation for:

- Process optimization
- Process troubleshooting

How to read Piping & Instrumentation (P&ID) Diagram



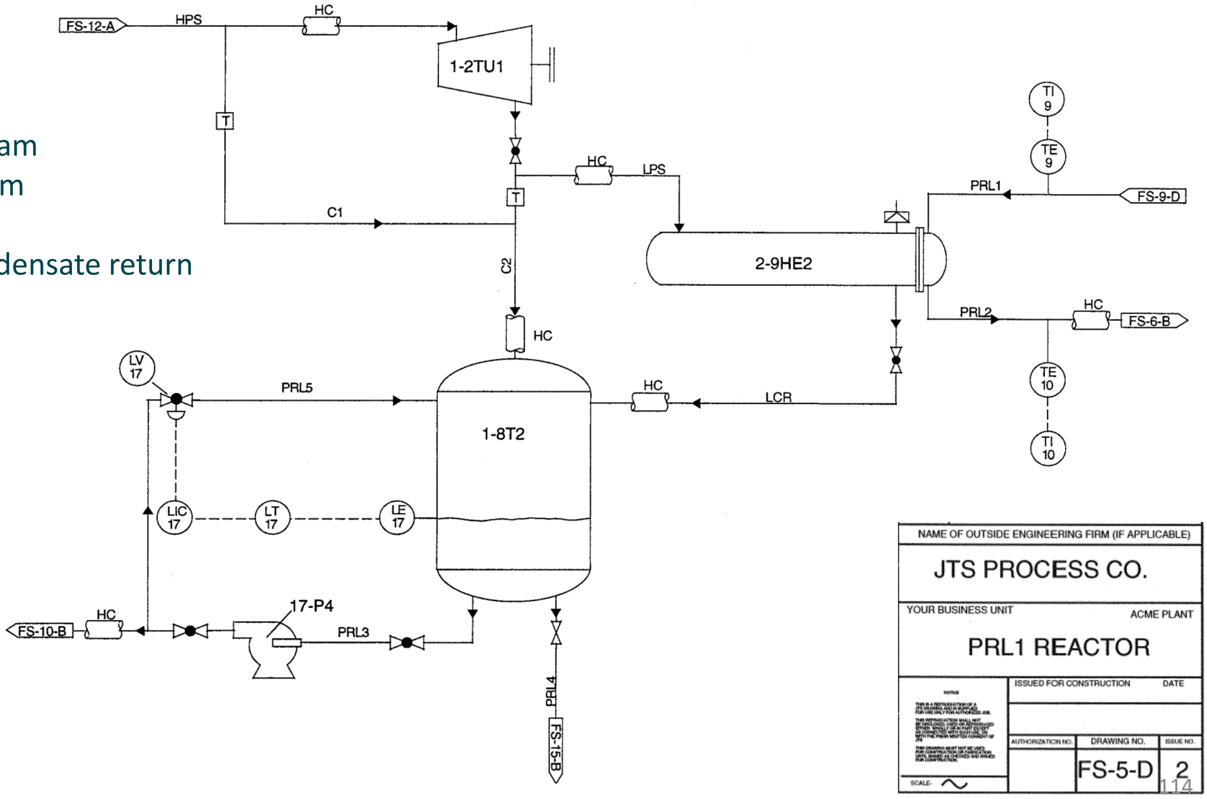
Read P&ID

How to Read Piping & Instrumentation (P&ID) Diagram

P&ID reading exercise - 1

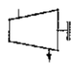
Reading P&ID - Reducing Turbine & Condensate Collection

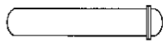
Note:
HPS – High pressure steam
LPS – Low pressure steam
PRL – Process liquid
LCR – Low pressure condensate return

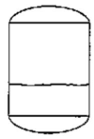


1. The P&ID above shows three large pieces of equipment. Place the letter of the symbol from the right column next to the correct word(s) in the left column. You may use a symbol more than one answer.

___2-9HE2
___Condensate tank
___1-2TU1
___1-8T2
___Turbine
___Heat exchanger

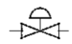
a. 

b. 

c. 

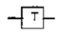
2. Numerous symbols appear on this P&ID. Place the letter of the symbol next to its name.

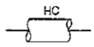
___Insulated line
___Globe valve
___Centrifugal pump
___Gate valve
___Steam trap
___Rupture disc
___Pneumatic valve
___Pneumatic line

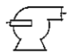
a. 

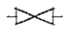
b.

c. 

d. 

e. 

f. 

g. 

h. 

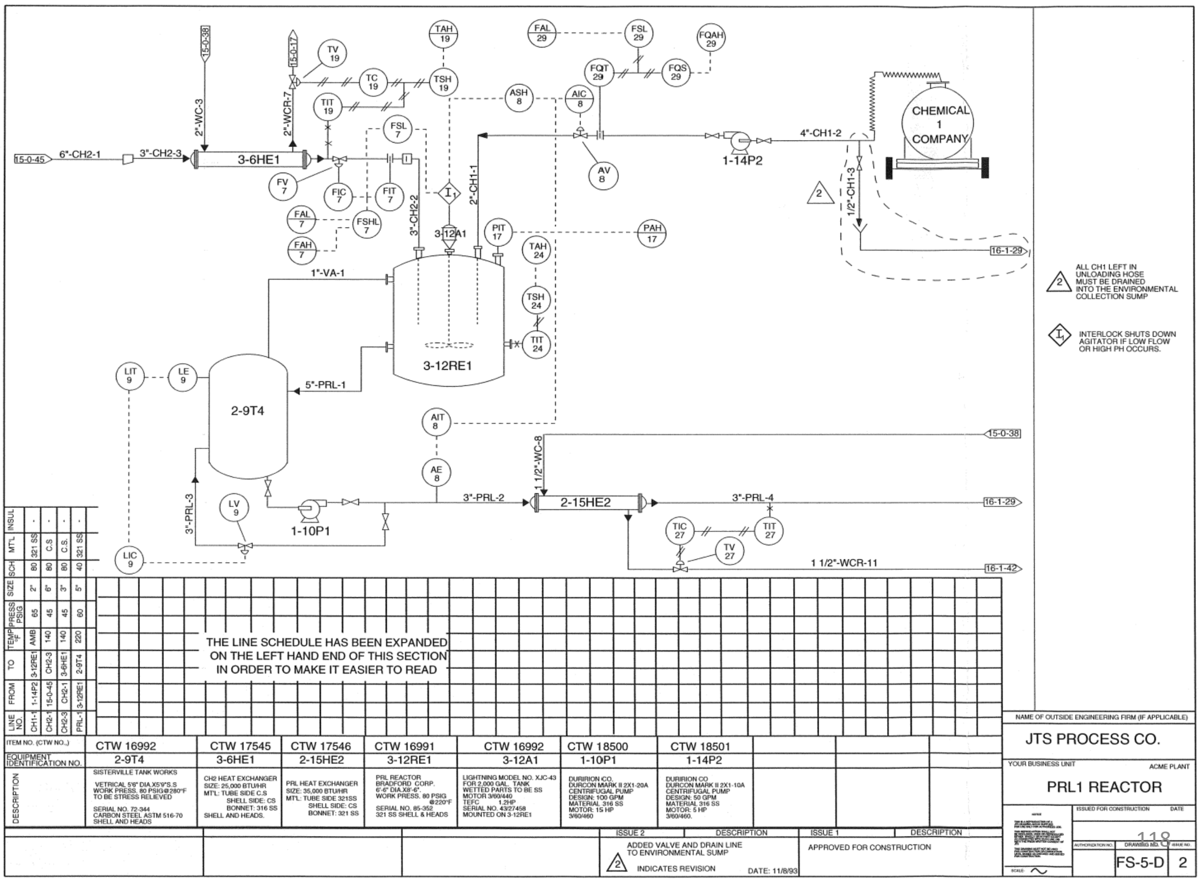
Reading P&ID - Reducing Turbine & Condensate Collection

Choose the correct answers:

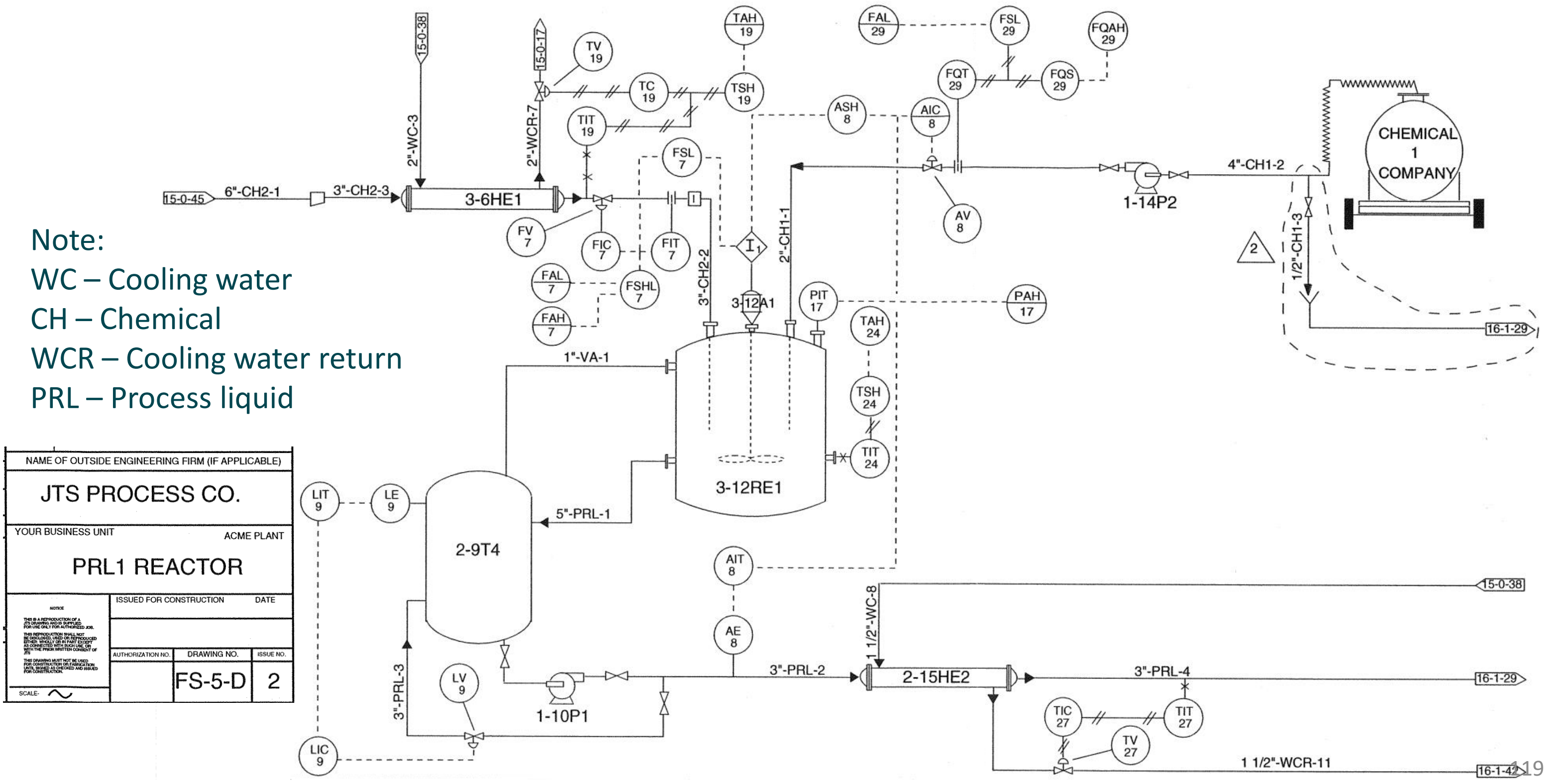
3. _______ (Low pressure steam, High pressure steam) enters the turbine at the top of the equipment and ______ (low pressure steam, high pressure steam) leaves the equipment at the bottom.
4. The steam leaving the turbine is used for in the heat exchanger to heat the ______ (condensate, process liquid).
5. You would expect the material in piping PRL2 to be ________ (warmer, cooler) than the material in piping PRL1.
6. The material leaving heat exchanger and going to the tank is ______ (LPS, PRL, low pressure condensate)
7. The piping LPS and PRL2 are insulated in order to _________ (retain heat of the material, prevent overheating of the material).
8. The level of liquid in the condensate tank is controlled by _______ (pump 17-P4, instrument loop 17).
9. If you wanted to find out what happens to condensate that leaves the tank, you would look on P&ID number ________ and ______ (FS-6-B, FS-10-B, FS-12-A, FS-15-B).
10. LT-17 is a _______ (level transmitter, light indicator).
11. TE-9 and TE-10 measure the temperature of ________ (process liquid, LPS, low pressure condensate/LCR).
12. Opening LV-17 allows some condensate to ___________ (recycle through the turbine, recycle through the tank, stay in the heat exchanger).
13. Steam _________ (can, cannot) be recycled through the turbine. Why?
14. Before servicing the pump 17-P4, close both ______________ (check valve, level element, globe valve) on the line entering and leaving the pump.
15. The pipe that physically join line C2 are ______ (PRL1, C1, PRL3, LPS, HPS).

P&ID reading exercise - 2

Reading P&ID –
Reactor



Reading P&ID – Reactor



Note:
WC – Cooling water
CH – Chemical
WCR – Cooling water return
PRL – Process liquid

Reading P&ID – Reactor

1. Place the letter of the symbols in the right column next to the correct name or label in the left column. You may use a symbol more than once.

____ 3-12RE1

____ Flow element

____ Tank Truck from the CH1 Company

____ 2-9T4

____ Flexible metal hose

____ Nozzle

____ Interlock

____ Flow totalizer switch

____ Hand-operated valve

____ Electric signal line

____ 2-15HE2

____ Pneumatic valve

____ Heat exchanger

____ Holding Tank

____ Reactor

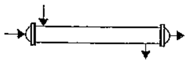
____ Analysis indicating transmitter

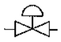
a. 

b.

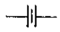
c. 

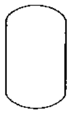
d. 

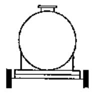
e. 

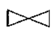
f. 

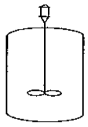
g. 

h. 

i. 

j. 

k. 

l. 

m.

Reading P&ID – Reactor

Choose the correct answer. **Some questions have more than one answer.**

2. CH1 comes into this P&ID from ________ (another reactor, a tank truck, 15-0-45, 16-1-29).
3. The interlock will be activated by the __________ (1-14P2, agitator, low flow of CH2, FSHL-7).
4. CH2 flow that being added to the reaction is controlled by ________ (FV-7, product analysis, flow of CH1, improper function of the agitator).
5. The function of 3-6HE1 is to ________ (heat CH2, cool CH2, heat CH1, cool CH1).
6. If the temperature of stream 3"-CH2-2 is too high, _________ (TV-19, FC-8, TV-27, LV-9) is opened more to allow additional cooling water to flow into the exchanger.
7. 1-10P1 increases the pressure in line _________ (4"-CH-1, 3"-CH2-1, 3"-PRL-2).
8. The function of LV-9 is to __________ (maintain desired level of PRL in the tank, allow vapor to leave the tank)
9. If the temperature of the reaction becomes too high, an alarm will go off __________ (at the reactor, on a computer control panel, near the tank car, TAH-19).
10. If analysis of the process liquid indicates that PRL did not meet specification, ________________ (TV-27, FV-7, AV-8, 1-10P1, PAH-17) changes the flow of ________________ (CH2, CH1, PRL, PRL-3) into the reactor.
11. If PRL is too warm, ___________ (TV-17, TV-27, 1-10P1, AE-8) opens more to allow increased cooling flow through ____________ (3-6HE1, 2-9T4, 3-12RE1, 2-15HE2).
12. If the interlock is activated, a signal is sent to ____________ (FAL-29, 3-12A1, AIC-8, FIT-7, FIC-7, 3-12RE1).
13. The title of this P&ID is ______________ (JTS Process Co., Engineering Drawing/Reactor, PRL-1 Reactor); this diagram covers a process occurring at the _______________ (Acme Plant, Engineering Firm, coordinate FS-5-D).

Reading P&ID – Reactor

Choose the correct answer. **Some questions have more than one answer.**

14. Specific information about the line 5”-PRL-1 can be found in the _____________ (Title Block, Line Schedule, Issue Description).
15. Line 2”-CH1-1 is made of ____________ (carbon steel, copper, stainless steel) and carries the material ____________ (cooling water, CH1, CH2, PRL) at a pressure of _____________ (60 psig, 65 psig, 45 psig).
16. Any excess liquid left in the tank truck hose after unloading is completed, must go to _____________ (the reactor, the holding tank, the environmental collection sump, the sewer drain).
17. The changes in the process system that was made with this issue was ______________ (addition of an environmental collection sump, a new source of CH1, addition of line ½”-CH1-3 and drain, additional CH1).
18. The serial number of the holding tank is ___________ (2-9T4, T4, T2-344, FS-5-D).
19. The diameter of the nozzle where CH1 enters the reactor is __________ (4 inches, 3 inches, 5 inches, 2 inches).
20. To locate the changes made on main diagram of issue 2 of P&ID number FS-5-D, look for ___________ (a cloud-like sketch around specific areas on the P&ID, an arrow with FS-5-D, a triangle with a 2 in it, two changes).
21. Material in line CH2 enters the heat exchanger at approximately ____________________ degrees Fahrenheit (45, 220, 140, 65).
22. During the reaction, the material ______________ (PRL, cooling water, vapor) is allowed to recycle between the holding tank and the reactor.
23. 3-12RE1 has a working pressure of ______________ (45, 80, 65) psig and a shell made of ___________ (stainless steel, carbon steel).

Choose the correct answer. **Some questions have more than one answer**

24. Trace the flow of PRL from where it is created to where it leaves the current P&ID by arranging the following pieces of equipment in the proper order. Place a “1” next to where PRL is created, “2” where it goes next, and so on..

    ___2-15HE2 _____3-6HE1 _____16-1-29 _____TV 27 _____2-9T4

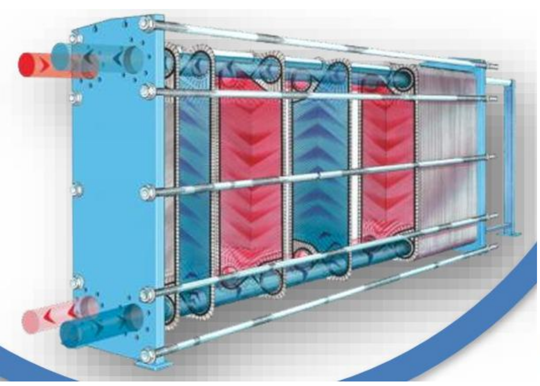
    ___3-12RE1 _____16-1-42 _____1-10P1 _____1-14P2 _____15-0-45

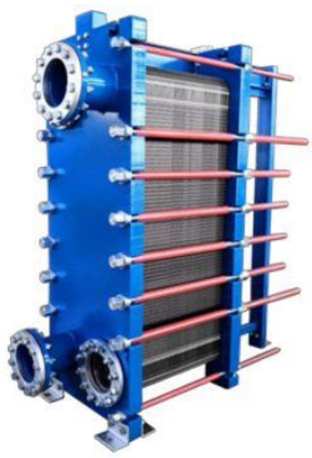
How to Read Piping & Instrumentation (P&ID) Diagram

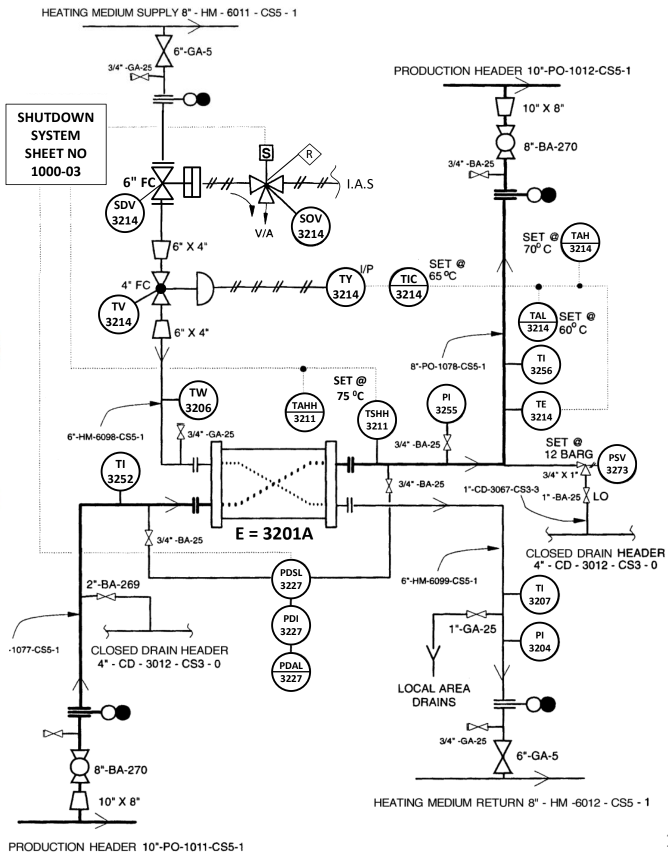
P&ID reading exercise - 3

Reading P&ID

Try to explain the following P&ID:







E-3201A – Plate-Plate Heat Exchanger

Finished…

Note:
If you want to improve yourself, it is recommended that you:

- Practice reading P&IDs extensively under the guidance of a process engineer or someone who is an expert in interpreting P&IDs.
- Go on site to compare the P&ID on paper with the actual installation.

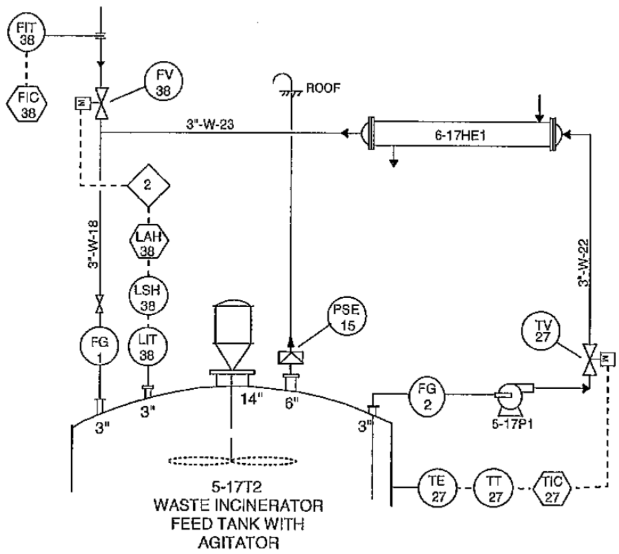
Next level:

- Able to read control types: closed vs. open control loops, feedback, feedforward, low vs. high switch selectors, cascade control.
- Identify errors in a P&ID.
- Improve processes in P&IDs.
- Review P&ID.


THANK
YOU
Merci
Grazie
Спасибо
Gracias
Paldies
Takk
Kiitos

How to Read Piping & Instrumentation Diagram for Beginners

Answer to the questions

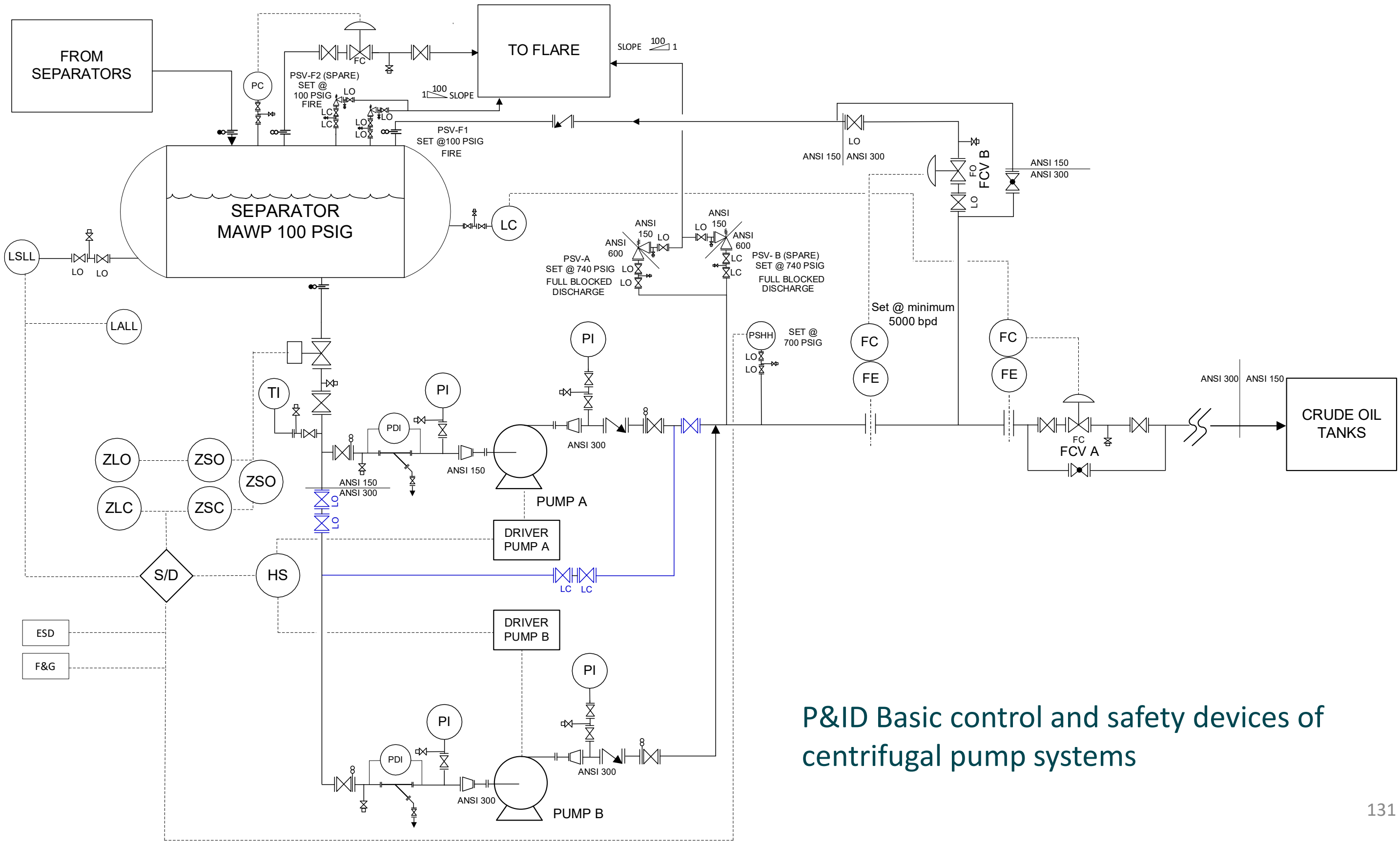


By Cahyo Hardo, B.Eng., M.OHS.

It's a good idea to try answering the following questions before opening them. This will improve your memory and test your understanding.

Happy Learning!

Answer to slide 42



P&ID Basic control and safety devices of centrifugal pump systems

Safety Devices – in centrifugal pump system

Crude oil from various separators flows to the final separator to separate the gas. The gas flows to the flare, while the oil flows to two centrifugal pumps. The control concept is to maintain the level in the separator by regulating the pump output flow rate (cascade control).

The oil flow rate at the pump output is controlled by FCV A. FCV A receives input from the FC, which receives flow rate measurement data from the FE and the set point from the LC.

The minimum pump flow rate is controlled by the FC (set point 5000 bpd) which receives input from the FE which then forwards an electronic signal to FCV B to recycle low to the tank. If the pump output flow rate is still or below the set point, then FCV B will remain open. If the flow rate is above the set point, FCV B will start to close until fully closed and the LCV-FCV instrument will control the separator level and pump flow rate.

The two installed pumps can be operated individually or simultaneously in parallel operation by setting them on the HS (hand switch) in the field. The two pumps can also be operated in series by opening two valves, normally set to LC, and closing two valves at the pump inlet, normally set to LO.

Safety Devices – in centrifugal pump system

When the pump discharge pressure reaches 700 psig, the PSHH setpoint, the switch sends an electronic signal to the shutdown system, shutting down the operating pump, or both pumps in parallel or series operation.

If the pump discharge pressure continues to rise to 740 psig, PSV A or PSV B (depending on which is operating) will discharge crude oil into the flare system to prevent overpressure.

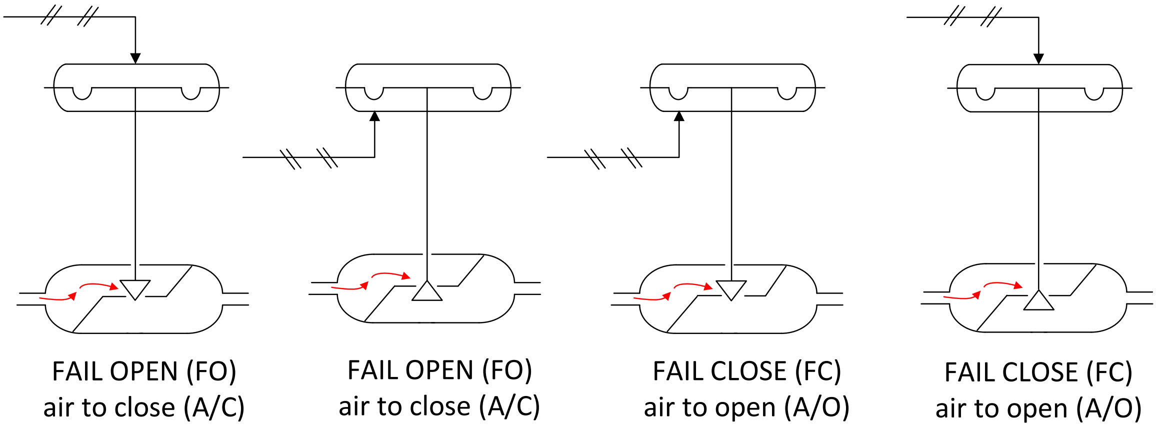
If the level in the separator drops to the LSLL setpoint, the switch also sends an electronic signal to the shutdown system, shutting down the pumps.

Safety Devices – In case of fire or gas leak

When **fire and/or flammable gas is detected:**

- **Will the plant be blown down??**
  - It turns out the separator isn't blown down when the Fire or Fire & Gas Signal is activated. This is because the operating pressure in the separator is already relatively low, which doesn't require blowdown.
- **The installed deluge system sprays water** onto the separator surface.
  - The purpose: to cool the separator surface, which will reduce the impact of a fire or delay the increase in separator wall temperature.
- **Will activate F&G (fire and gas) system.**
  - Purpose: to isolate the separator, by:
    - Turning off Pump A and Pump B.
    - Closing the SDV-XXXX outlet from the separator in the upstream system (not pictured).
- **Will activate** plant **alarm.**

Answers to slides 47 - 48



Can a valve that is FO become FC and vice versa?

Is this permissible?

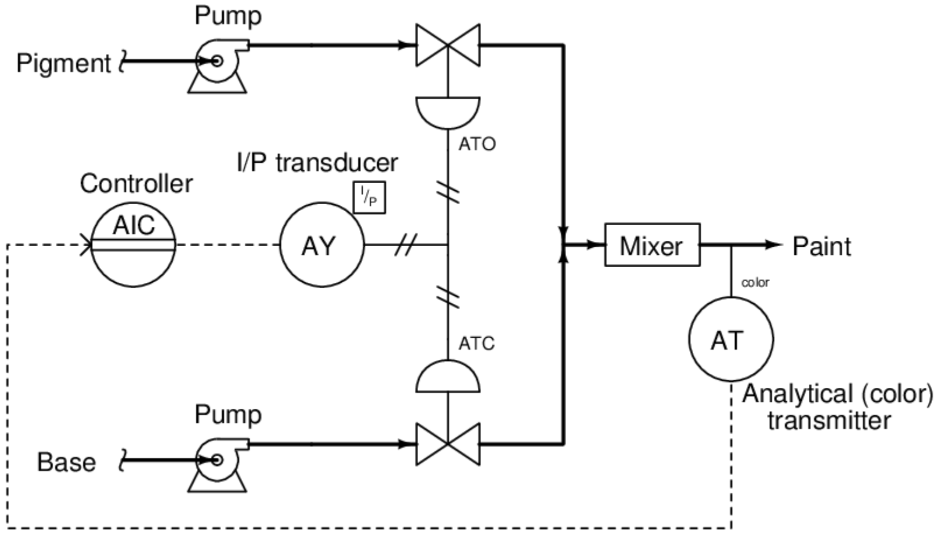
Answer:

Yes, as long as the trim valve model is the same. In the example above, the valve with the leftmost actuator can be changed to the valve and actuator positioned third from the left.

This is permissible as long as a prior risk assessment is conducted.

Fail Open or Fail Close or Last Position?

Take a guess!!



Answer:

ATO = Air to Open = Fail Close

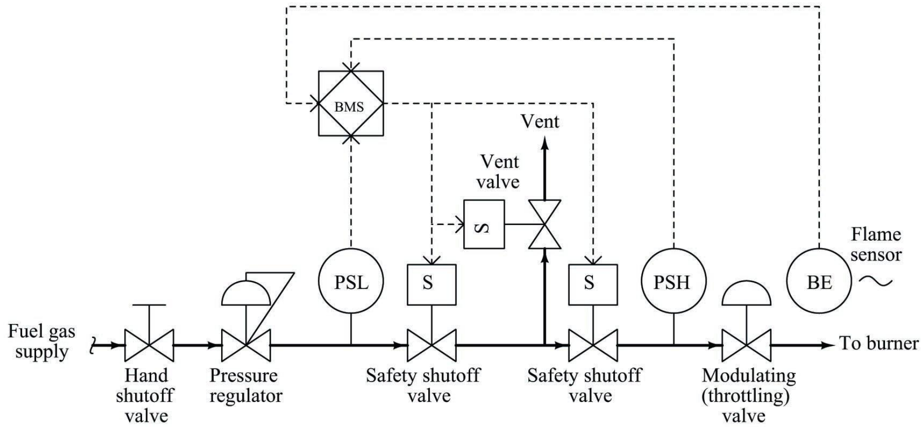
ATC = Air to Close = Fail Open

Answer to slide 61

- Pipe 3 is physically connected
- Pipe 4 is not physically connected
- Pipe 5 is not physically connected
- Pipe 6 is physically connected

Answer to slide 93

Symbol – Actuated Valve



Can anyone explain????
Answer:

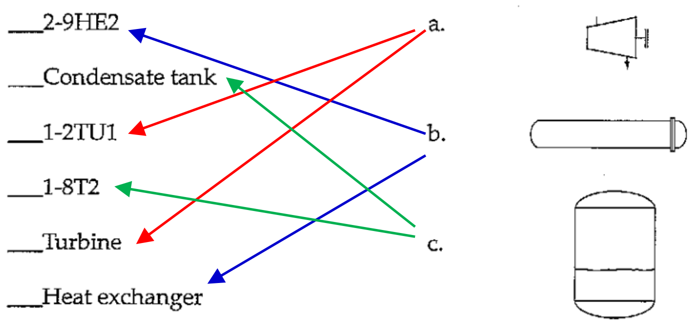
The P&ID above is the P&ID regarding BMS

The BMS is a burner management system installed to manage fuel flow to the combustion system in the burner. It consists of a pressure and flow/throttling control system, as well as two shut-off valves and one vent valve. When the flame-off signal from the burner is sent to the BMS, the two shut-off valves close, and the vent valve opens, releasing fuel gas to the atmosphere.

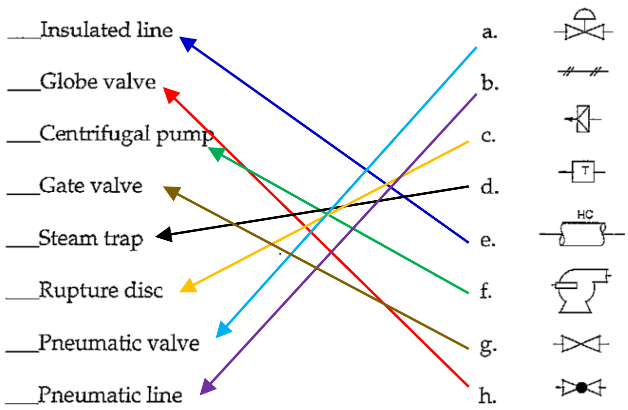
Reading practice answer P&ID - 1

Reading P&ID - Reducing Turbine & Condensate Collection

1. The P&ID above shows three large pieces of equipment. Show the three drawings of the equipment on the right to match them to the choices on the left. Each piece of equipment may have more than one answer.



2. Numerous symbols appear on this P&ID. Place the letter of the symbol next to its name.



Reading P&ID - Reducing Turbine & Condensate Collection

Choose the correct answers:

3. _______ (~~Low pressure steam~~, High pressure steam) enters the turbine at the top of the equipment and ______ (low pressure steam, ~~high pressure steam~~) leaves the equipment at the bottom.
4. The steam leaving the turbine is used for in the heat exchanger to heat the ______ (~~condensate~~, process liquid).
5. You would expect the material in piping PRL2 to be ________ (warmer, ~~cooler~~) than the material in piping PRL1.
6. The material leaving heat exchanger and going to the tank is ______ (~~LPS, PRL~~, low pressure condensate)
7. The piping LPS and PRL2 are insulated in order to _________ (retain heat of the material, ~~prevent overheating of the material~~).
8. The level of liquid in the condensate tank is controlled by _______ (~~pump 17-P4~~, instrument loop 17).
9. If you wanted to find out what happens to condensate that leaves the tank, you would look on P&ID number ________ and ______ (~~FS-6-B,~~ FS-10-B, ~~FS-12-A~~, FS-15-B).
10. LT-17 is a _______ (level transmitter, ~~light indicator~~).
11. TE-9 dan TE-10 measure the temperature of ________ (process liquid, ~~LPS, low pressure condensate/LCR~~).
12. Opening LV-17 allows some condensate to ___________ (~~recycle through the turbine~~, recycle through the tank, ~~stay in the heat exchanger~~).
13. Steam _________ (~~can,~~ cannot) be recycled through the turbine. Why?
14. Before servicing the pump 17-P4, close both ______________ (~~check valve, level element~~, globe valve) on the line entering and leaving the pump.
15. The pipe that physically join line C2 are ______ (~~PRL1,~~ C1, ~~PRL3, LPS, HPS~~).

How to Read Piping & Instrumentation (P&ID) Diagram

Reading practice answer P&ID - 2

Reading P&ID – Reactor

1. Place the letter of the symbols in the right column next to the correct name or label in the left column. You may use a symbol more than once!

____ 3-12RE1

____ Flow element

____ Tank Truck from the CH1 Company

____ 2-9T4

____ Flexible metal hose

____ Nozzle

____ Interlock

____ Flow totalizer switch

____ Hand-operated valve

____ Electric signal line

____ 2-15HE2

____ Pneumatic valve

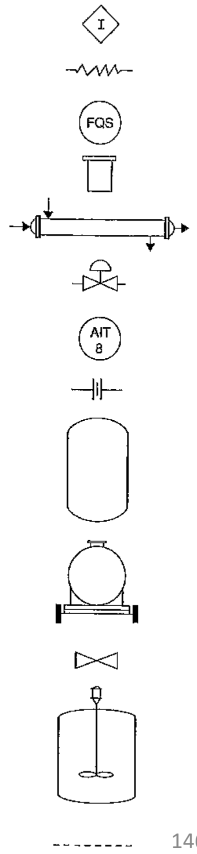
____ Heat exchanger

____ Holding Tank

____ Reactor

____ Analysis indicating transmitter

a.
b.
c.
d.
e.
f.
g.
h.
i.
j.
k.
l.
m.



Reading P&ID – Reactor

Choose the correct answer. **Some questions have more than one answer.**

2. CH1 comes into this P&ID from ________ (~~another reactor~~, a tank truck, ~~15-0-45, 16-1-29~~).
3. The interlock will be activated by the __________ (~~1-14P2, agitator, low flow of CH2~~, FSHL-7).
4. CH2 flow that being added to the reaction is controlled by ________ (FV-7, ~~product analysis, flow of CH1, improper function of the agitator~~).
5. The function of 3-6HE1 is to ________ (~~heat CH2~~, cool CH2, ~~heat CH1, cool CH1~~).
6. If the temperature of stream 3"-CH2-2 is too high, _________ (TV-19, ~~FC-8, TV-27, LV-9~~) is opened more to allow additional cooling water to flow into the exchanger.
7. 1-10P1 increases the pressure in line _________ (~~4" CH-1, 3" CH2-1~~, 3"-PRL-2).
8. The function of LV-9 is to __________ (maintain desired level of PRL in the tank, ~~allow vapor to leave the tank~~)
9. If the temperature of the reaction becomes too high, an alarm will go off __________ (at the reactor, ~~on a computer control panel, near the tank car, TAH-19~~).
10. If analysis of the process liquid indicates that PRL did not meet specification, ________________ (~~TV-27, FV-7~~, AV-8, ~~1-10P1, PAH-17~~) changes the flow of ________________ (~~CH2~~, CH1, ~~PRL, PRL-3~~) into the reactor.
11. If PRL is too warm, ___________ (~~TV-17~~, TV-27, ~~1-10P1, AE-8~~) opens more to allow increased cooling flow through ____________ (~~3-6HE1, 2-9T4, 3-12RE1~~, 2-15HE2).
12. If the interlock is activated, a signal is sent to ____________ (~~FAL-29,~~ 3-12A1, ~~AIC-8, FIT-7, FIC-7, 3-12RE1~~).
13. The title of this P&ID is ______________ (~~JTS Process Co., Engineering Drawing/Reactor~~, PRL-1 Reactor); this diagram covers a process occurring at the _______________ (Acme Plant, ~~Engineering Firm, coordinate FS-5-D~~).

Reading P&ID – Reactor

Choose the correct answer. **Some questions have more than one answer.**

14. Specific information about the line 5"-PRL-1 can be found in the ____________ (~~Title Block~~, Line Schedule, ~~Issue Description~~).
15. Line 2"-CH1-1 is made of ___________ (~~carbon steel~~, copper, stainless steel) and carries the material ___________ (~~cooling water~~, CH1, ~~CH2, PRL~~) at a pressure of ____________ (~~60 psig~~, 65 psig, ~~45 psig~~).
16. Any excess liquid left in the tank truck hose after unloading is completed, must go to ____________ (~~the reactor, the holding tank,~~ the environmental collection sump, ~~the sewer drain~~).
17. The changes in the process system that was made with this issue was _____________ (addition of an environmental collection sump, ~~a new source of CH1~~, addition of line ½"-CH1-3 and drain, ~~additional CH1~~).
18. The serial number of the holding tank is __________ (2-9T4, ~~T4, T2-344, FS-5-D~~).
19. The diameter of the nozzle where CH1 enters the reactor is _________ (~~4 inches, 3 inches, 5 inches~~, 2 inches).
20. To locate the changes made on main diagram of issue 2 of P&ID number FS-5-D, look for __________ (a cloud-like sketch around specific areas on the P&ID, ~~an arrow with FS-5-D~~, a triangle with a 2 in it, ~~two changes~~).
21. Material in line CH2 enters the heat exchanger at approximately __________________ degrees Fahrenheit (~~45, 220,~~ 140, ~~65~~).
22. During the reaction, the material _____________ (~~PRL, cooling water~~, vapor) is allowed to recycle between the holding tank and the reactor.
23. 3-12RE1 has a working pressure of _____________ (~~45~~, 80, ~~65~~) psig and a shell made of __________ (stainless steel, ~~carbon steel~~).

Reading P&ID – Reactor

Choose the correct answer. **Some questions have more than one answer**

24. Trace the flow of PRL from where it is created to where it leaves the current P&ID by arranging the following pieces of equipment in the proper order. Place a "1" next to where PRL is created, "2" where it goes next, and so on..

_4__2-15HE2 _____3-6HE1 _5___16-1-29 _____TV 27 _2__2-9T4

_1_3-12RE1 _____16-1-42 _3__1-10P1 _____1-14P2 _____15-0-45

Reading practice answer P&ID - 3

Reading P&ID

Flowchart tracing shows:

- A 6-inch heating medium (HM) line, made of carbon steel (CS), 6"-HM-6098-CS5-1, flows from an 8-inch diameter heating medium supply header made of the same material (8"-HM-6011-CS5-1) into and out of a plate-type heat exchanger, E-3201 A, to exchange heat with the Produced Oil (PO) stream. (Note: The name "heating medium" suggests this.)
- After passing through E-3201A, the heating medium, 6"-HM-6099-CS5-1, flows to the 8-inch heating medium return header (8"-HM-6012-CS5-1).
- The letter E in E-3201A in the P&ID Legend indicates the equipment designation, including the Unfired Heat Transfer Equipment (UHE) heat exchanger.
- The number 32 in E-3201A in the system numbering in the Legend indicates the Condensate and Oily Water System area.
- The heated stream, an 8-inch produced oil (PO) stream, also made of CS (8"-PO-1077-CS5-1), flows from the 10"-PO-1011-CS5-1 production header line into E-3201A, then exits as the 8"-PO-1078-CS5-1 stream to the 10"-PO-1012-CS5-1 production header line.
- The 6-inch HM stream has a 1-inch drain to the open drain at the outlet of E-3201A, while the drain from the 8-inch PO stream located at inlet to E-3201A, and its drain line goes to the closed drain header.

Reading P&ID

Instrumentation tracing shows:

- The temperature of the Produced Oil leaving the E-3201 A is measured by a locally installed temperature element (TE-3214). The electronic signal from the TE-3214 is forwarded to a temperature indicator controller installed in the control room panel (TIC-3214). The TIC-3214 will control the temperature of the Produced Oil at 65°C. The electronic signal coming out of the TIC-3214 is forwarded to a locally installed temperature relay (TY-3214). The temperature relay converts the electronic signal into a pneumatic signal (indicated by the letters "I/P"). The pneumatic signal flows to the control valve (TV-3214) to regulate the amount of heating medium entering the E-3201A.
- The electronic signal from the TE-3214 is also used to generate a high temperature alarm (TAH-3214) set at 70°C and a low temperature alarm (TAL-3214) set at 60°C. A locally installed high-high temperature switch (TSHH-3211) is another over-temperature protection device. The TSHH-3211 is set at 75°C.
- If the temperature of the oil produced reaches 75°C, the TSHH-3211 will activate an alarm (TAHH-3211) on the control room panel and also send a signal to the Shutdown System.

Reading P&ID

Instrumentation tracing shows:

- Another input to the Shutdown System is generated by a locally installed low-pressure differential switch (PDSL-3227). Low differential pressure may indicate a leak within the E-3201A heat exchanger. If low differential pressure occurs, the PDSL-3227 will activate an alarm (PDAL-3227) on the control room panel and also send a signal to the Shutdown System.
- If the Shutdown System is activated by the TSHH-3211 or the PDSL-3227, power to the SOV-1014 will be disconnected. Instrumentation air will be vented by the SOV-3214 (as indicated by the small curved arrow). The SDV-3214 will then close as indicated by the letters "FC."
- Other features include:
  - Spectacle blinds installed on the E-3201A inlet and outlet lines to isolate the equipment during maintenance.
  - The PSV-3273 is set to release pressure at 12 barg. This device protects against overpressure in the produced oil side of the E-3201A.
- Note: The system doesn't have a spec break.
  - **The E-3201A doesn't have a MAWP value. Can you guess what it might be?**
  - Answer: The MAWP of the E-3201A can be estimated from its PSV setting, which is 12 barg. Therefore, **the maximum MAWP of the vessel is 12 barg.**